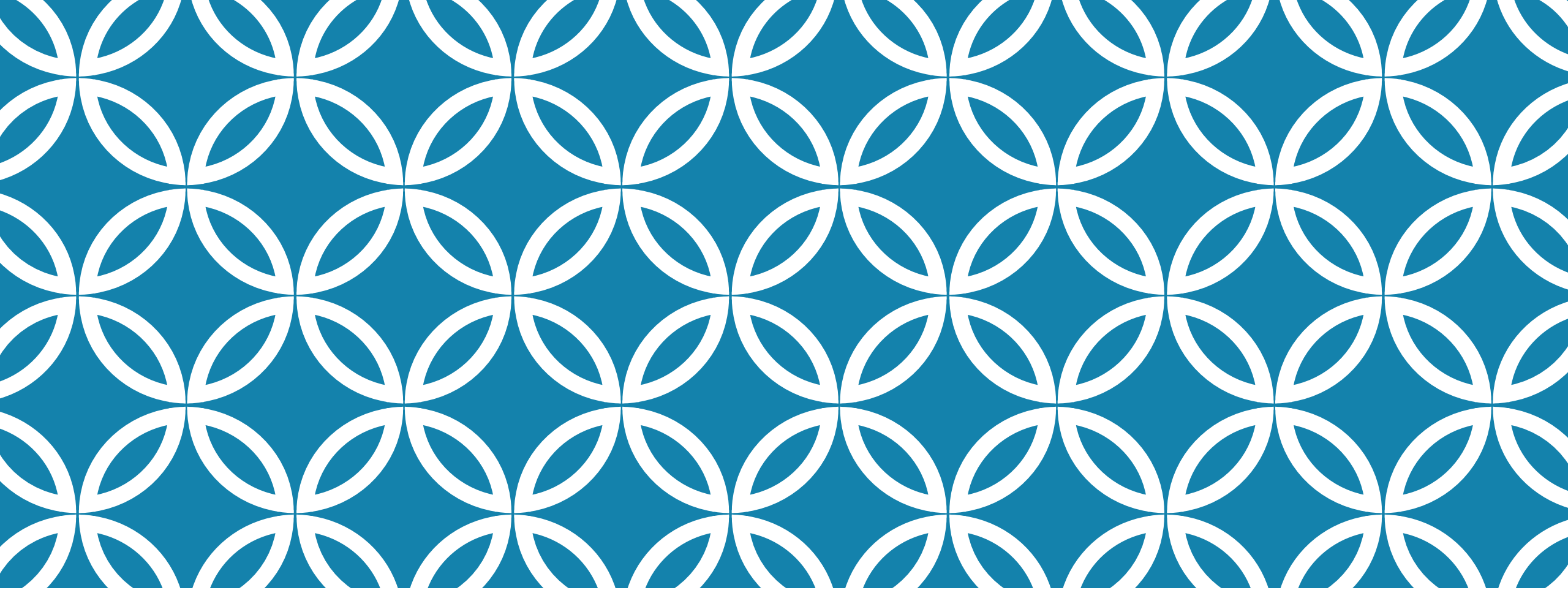
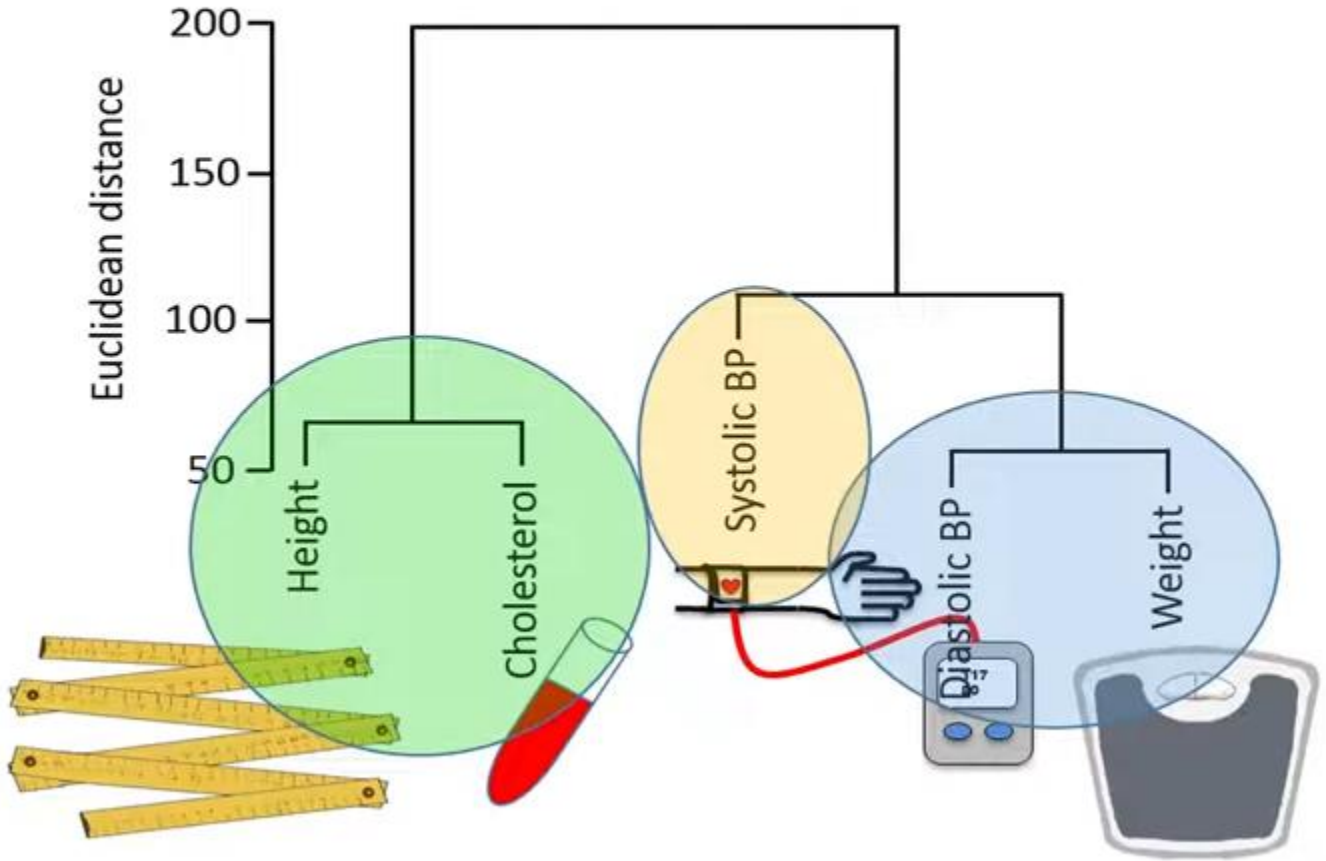


HIERARCHICAL CLUSTERING

by Dr. Hafeez Anwar

Hierarchical clustering



- How to build a dendrogram based on a simple data set
- How to identify clusters
- Different linkage functions
- How to select an appropriate distance metric
- How to transpose a matrix

Clustering

Cluster analysis, or clustering, is an unsupervised machine learning method where the aim is to identify a set of similar objects.

Clustering

Cluster analysis, or clustering, is an unsupervised machine learning method where the aim is to identify a set of similar objects.

A group that includes similar objects is called a cluster.

Clustering



How would you group the following objects in a natural way?

Clustering







Maybe you would group the cigarette with the car because they can both produce smoke,

Clustering







the money with the die because they can be associated with gambling,

Clustering







and you might have put the plant and the flower in one group.

Clustering









Or maybe you grouped based on the color? Before we start to cluster, we must therefore first decide how we define similarity.

Clustering

- Hierarchical clustering
- k-means clustering

There are various clustering methods but the most commonly used methods are hierarchical clustering and k-means clustering.

Clustering

- Hierarchical clustering
- k-means clustering

In hierarchical clustering, we identify the number of clusters after we have computed the clustering.

Clustering

- Hierarchical clustering
- k-means clustering

In comparison, the k-means clustering requires that we first specify how many clusters the method should generate before we run the actual analysis.

Hierarchical clustering



	A	B	C	D	E
Diastolic BP (mmHg)	70	86	78	60	75
Systolic BP (mmHg)	130	147	137	121	134
Weight (kg)	100	90	100	90	106
Height (cm)	180	170	181	171	185
Total cholesterol (mg/dl)	190	215	192	215	200

To explain how hierarchical clustering works, we will use the following data set that comes from measurements on 5 different individuals, A, B, C, D and E.

Hierarchical clustering

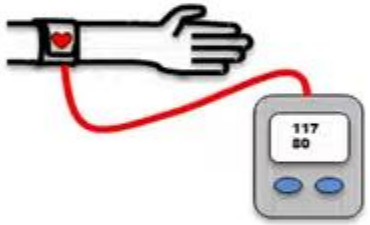




	A	B	C	D	E
Diastolic BP (mmHg)	70	86	78	60	75
Systolic BP (mmHg)	130	147	137	121	134
Weight (kg)	100	90	100	90	106
Height (cm)	180	170	181	171	185
Total cholesterol (mg/dl)	190	215	192	215	200

This row shows the diastolic blood pressure of the five individuals,

Hierarchical clustering





	A	B	C	D	E
Diastolic BP (mmHg)	70	86	78	60	75
Systolic BP (mmHg)	130	147	137	121	134
Weight (kg)	100	90	100	90	106
Height (cm)	180	170	181	171	185
Total cholesterol (mg/dl)	190	215	192	215	200

whereas this row shows the systolic blood pressure.

Hierarchical clustering





	A	B	C	D	E
Diastolic BP (mmHg)	70	86	78	60	75
Systolic BP (mmHg)	130	147	137	121	134
Weight (kg)	100	90	100	90	106
Height (cm)	180	170	181	171	185
Total cholesterol (mg/dl)	190	215	192	215	200

We also have data on the body weights, in kg, and body heights, in cm,

Hierarchical clustering



	A	B	C	D	E
Diastolic BP (mmHg)	70	86	78	60	75
Systolic BP (mmHg)	130	147	137	121	134
Weight (kg)	100	90	100	90	106
Height (cm)	180	170	181	171	185
Total cholesterol (mg/dl)	190	215	192	215	200



as well as the total cholesterol level in the blood of the five individuals.

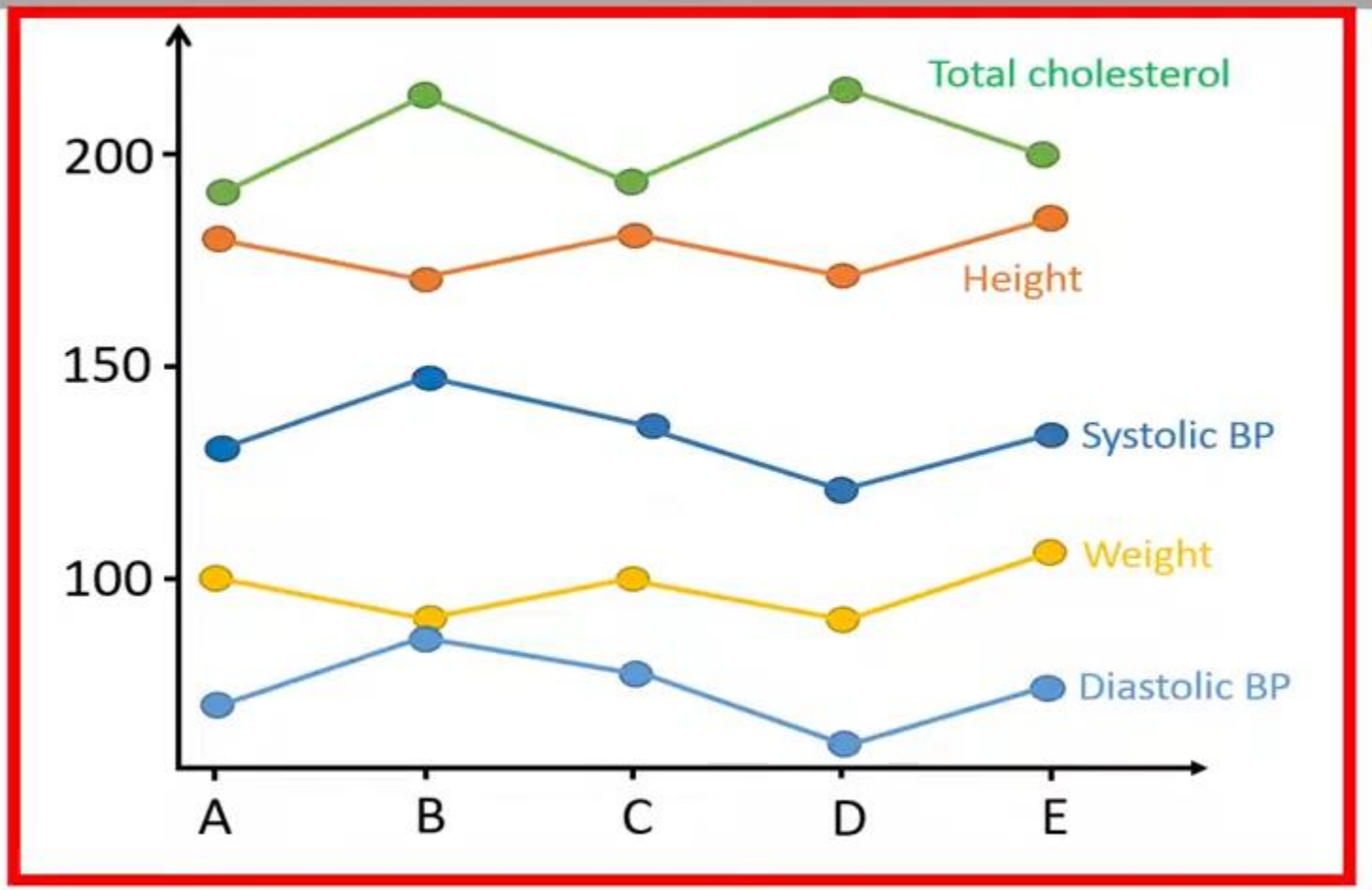
Hierarchical clustering

	A	B	C	D	E
Diastolic BP (mmHg)	70	86	78	60	75
Systolic BP (mmHg)	130	147	137	121	134
Weight (kg)	100	90	100	90	106
Height (cm)	180	170	181	171	185
Total cholesterol (mg/dl)	190	215	192	215	200

Let’s place the data table up here.

Hierarchical clustering

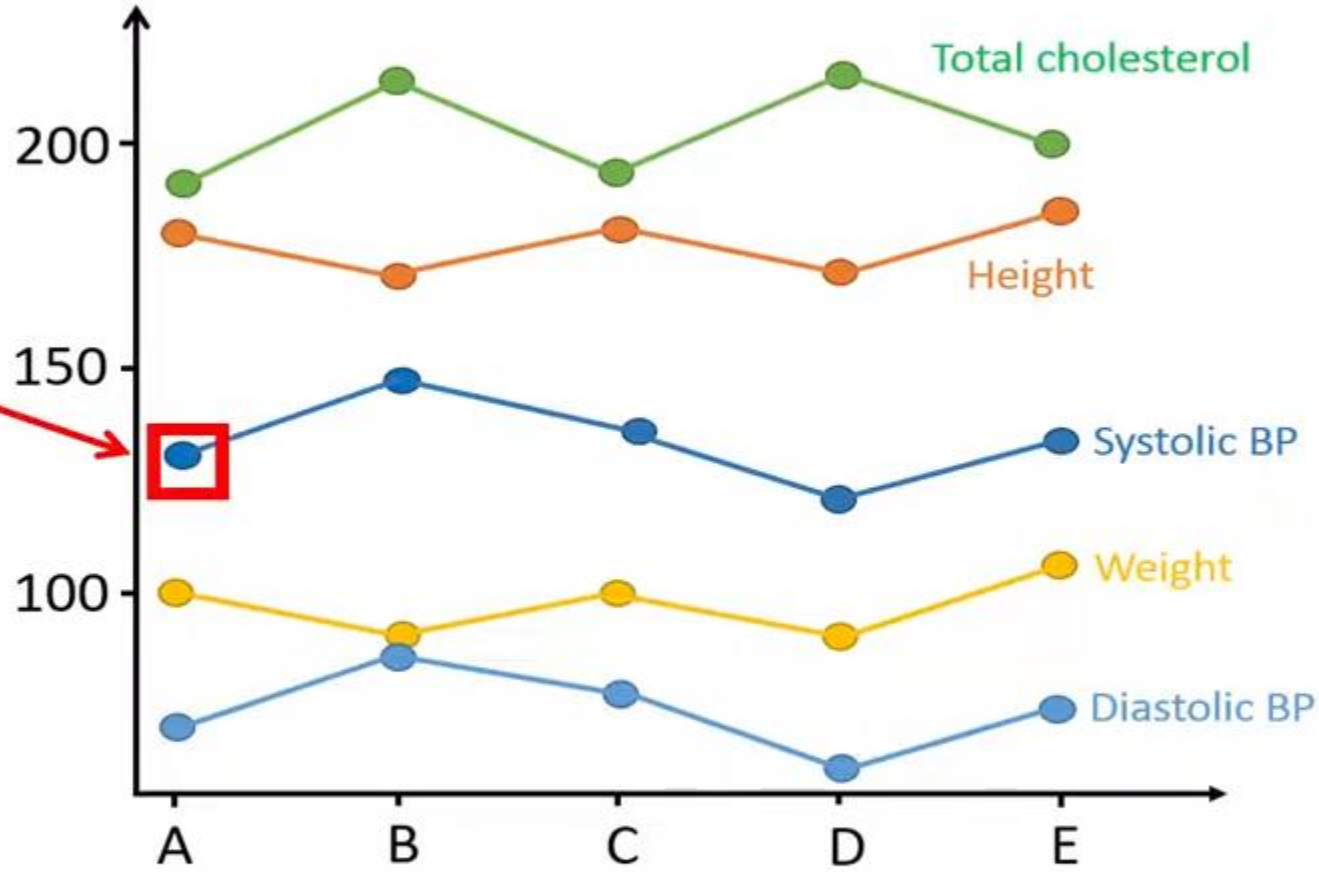
	A	B	C	D	E
Diastolic BP (mmHg)	70	86	78	60	75
Systolic BP (mmHg)	130	147	137	121	134
Weight (kg)	100	90	100	90	106
Height (cm)	180	170	181	171	185
Total cholesterol (mg/dl)	190	215	192	215	200



The following profile plot of the five different variables will help us to understand how clustering works.

Hierarchical clustering

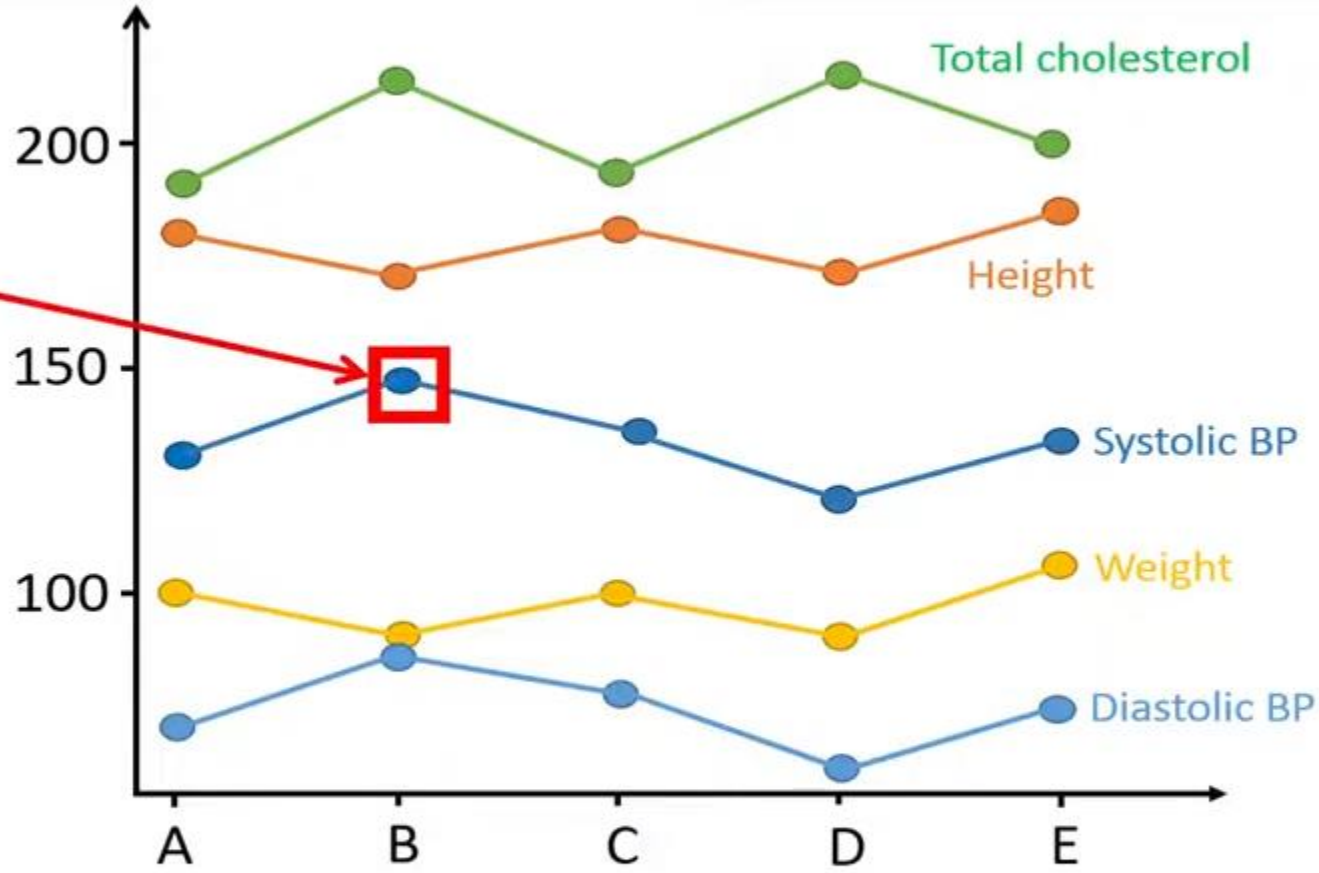
	A	B	C	D	E
Diastolic BP (mmHg)	70	86	78	60	75
Systolic BP (mmHg)	130	147	137	121	134
Weight (kg)	100	90	100	90	106
Height (cm)	180	170	181	171	185
Total cholesterol (mg/dl)	190	215	192	215	200



For example, this point represents the systolic blood pressure of person A,

Hierarchical clustering

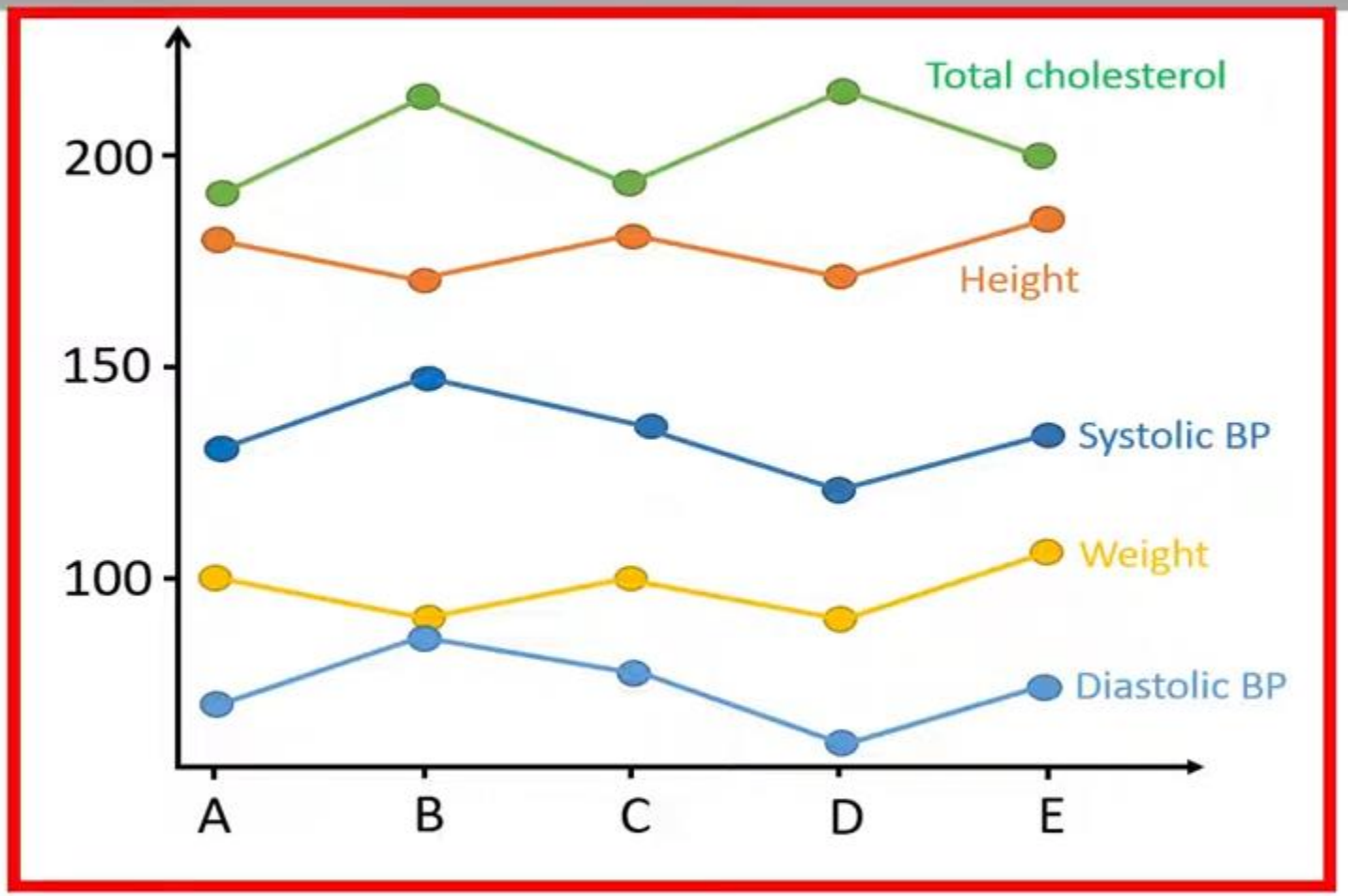
	A	B	C	D	E
Diastolic BP (mmHg)	70	86	78	60	75
Systolic BP (mmHg)	130	147	137	121	134
Weight (kg)	100	90	100	90	106
Height (cm)	180	170	181	171	185
Total cholesterol (mg/dl)	190	215	192	215	200



whereas this point represents the systolic blood pressure of person B and so forth.

Hierarchical clustering

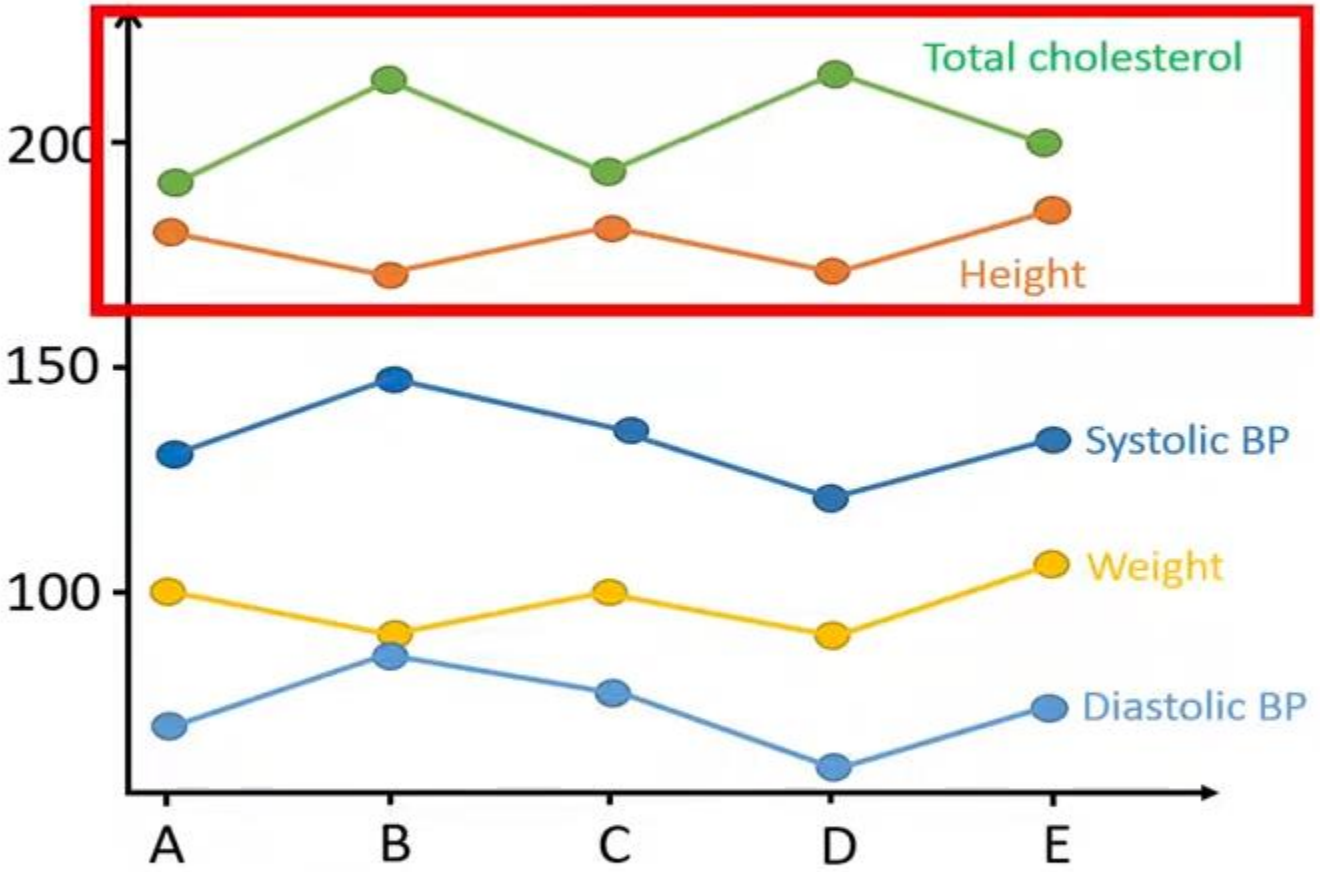
	A	B	C	D	E
Diastolic BP (mmHg)	70	86	78	60	75
Systolic BP (mmHg)	130	147	137	121	134
Weight (kg)	100	90	100	90	106
Height (cm)	180	170	181	171	185
Total cholesterol (mg/dl)	190	215	192	215	200



Let's say that we like to group variables that have similar values.

Hierarchical clustering

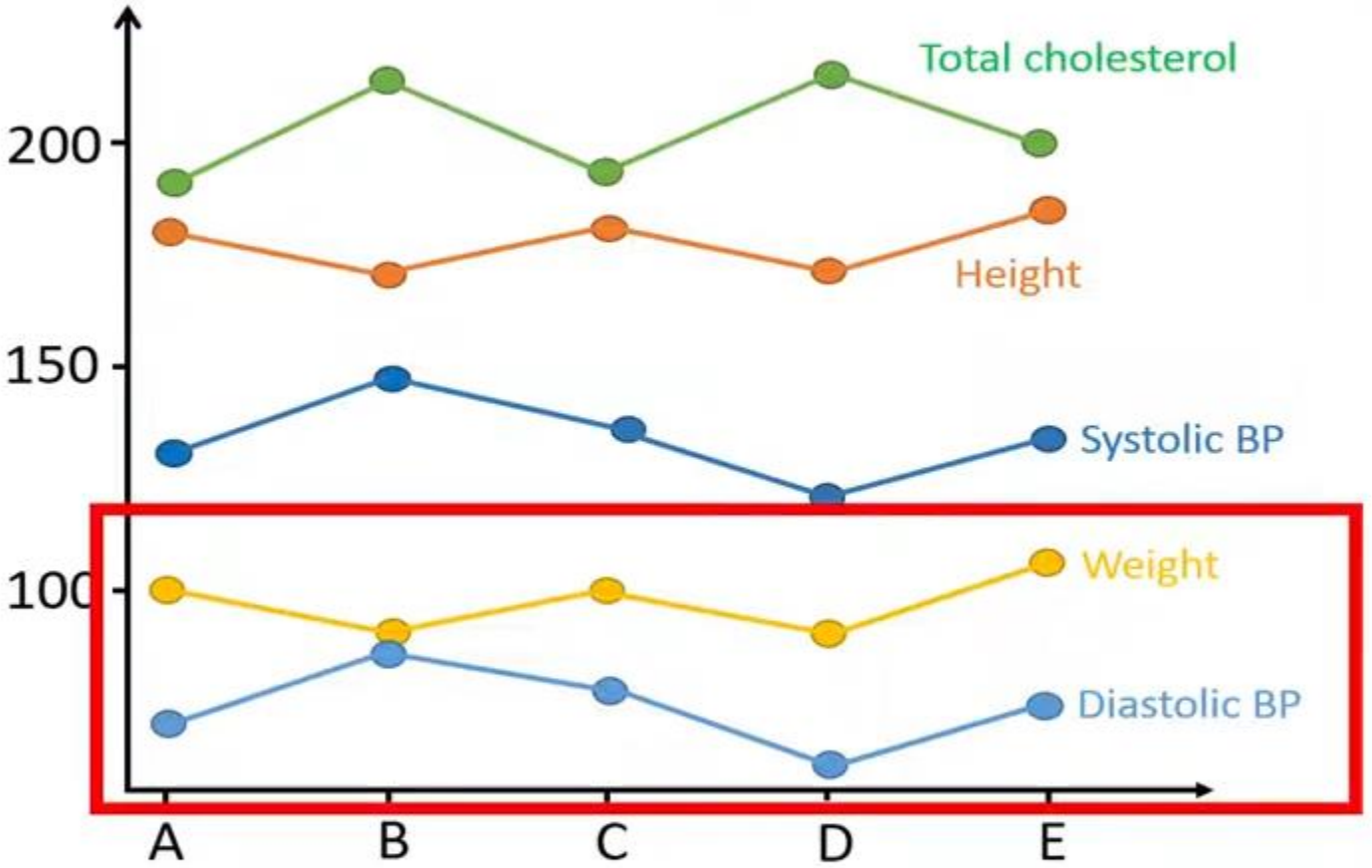
	A	B	C	D	E
Diastolic BP (mmHg)	70	86	78	60	75
Systolic BP (mmHg)	130	147	137	121	134
Weight (kg)	100	90	100	90	106
Height (cm)	180	170	181	171	185
Total cholesterol (mg/dl)	190	215	192	215	200



For example, it seems reasonable to group the total cholesterol level with the body height,

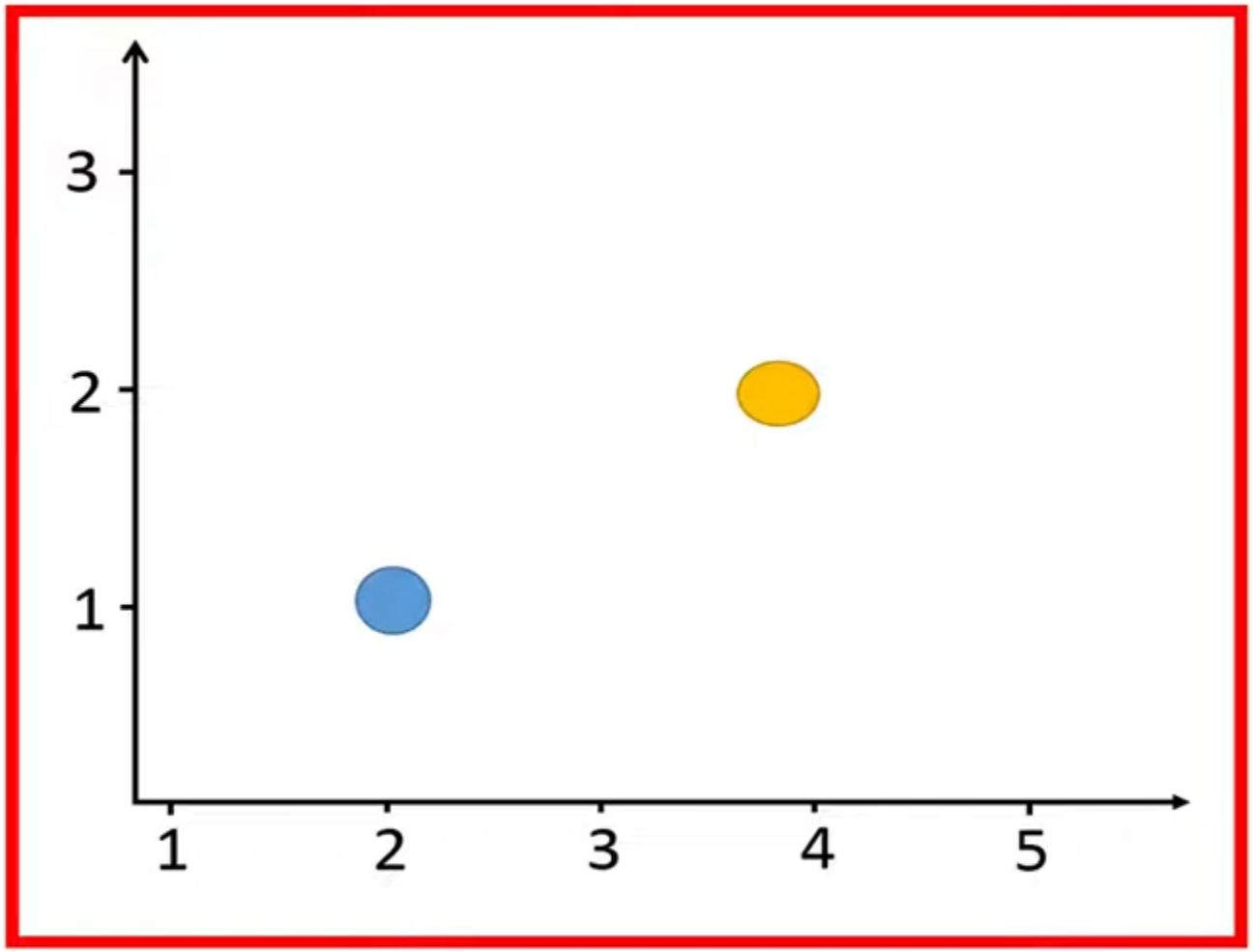
Hierarchical clustering

	A	B	C	D	E
Diastolic BP (mmHg)	70	86	78	60	75
Systolic BP (mmHg)	130	147	137	121	134
Weight (kg)	100	90	100	90	106
Height (cm)	180	170	181	171	185
Total cholesterol (mg/dl)	190	215	192	215	200



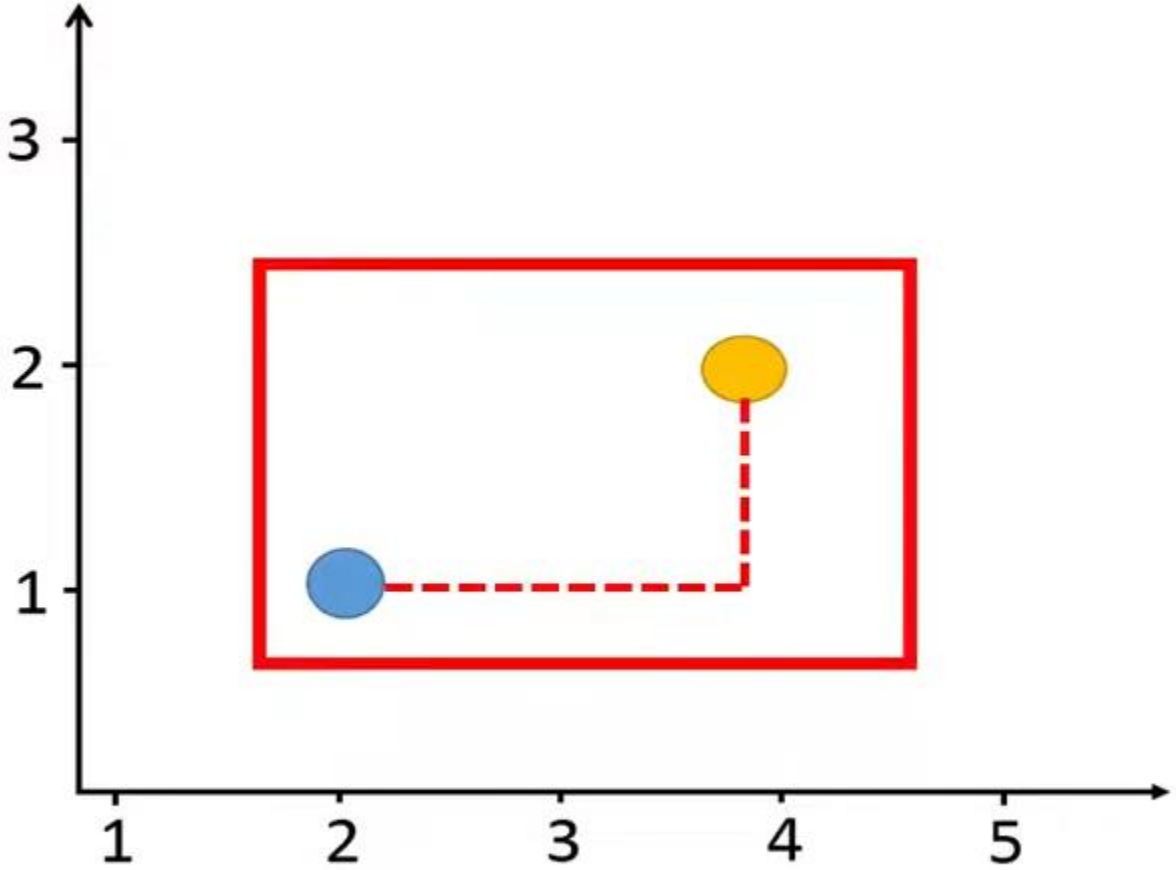
and the variables body weight and diastolic blood pressure because these pairs of variables have similar values.

Manhattan distance



Similarity between two variables, or two objects, is usually measured by calculating the distance between the data points. For example, how would you calculate the distance between these two data points?

Manhattan distance



One way to calculate the distance between these two points is to sum the horizontal and vertical distances between the two points.

Manhattan distance



This can be seen as the walking distance between two points on Manhattan.

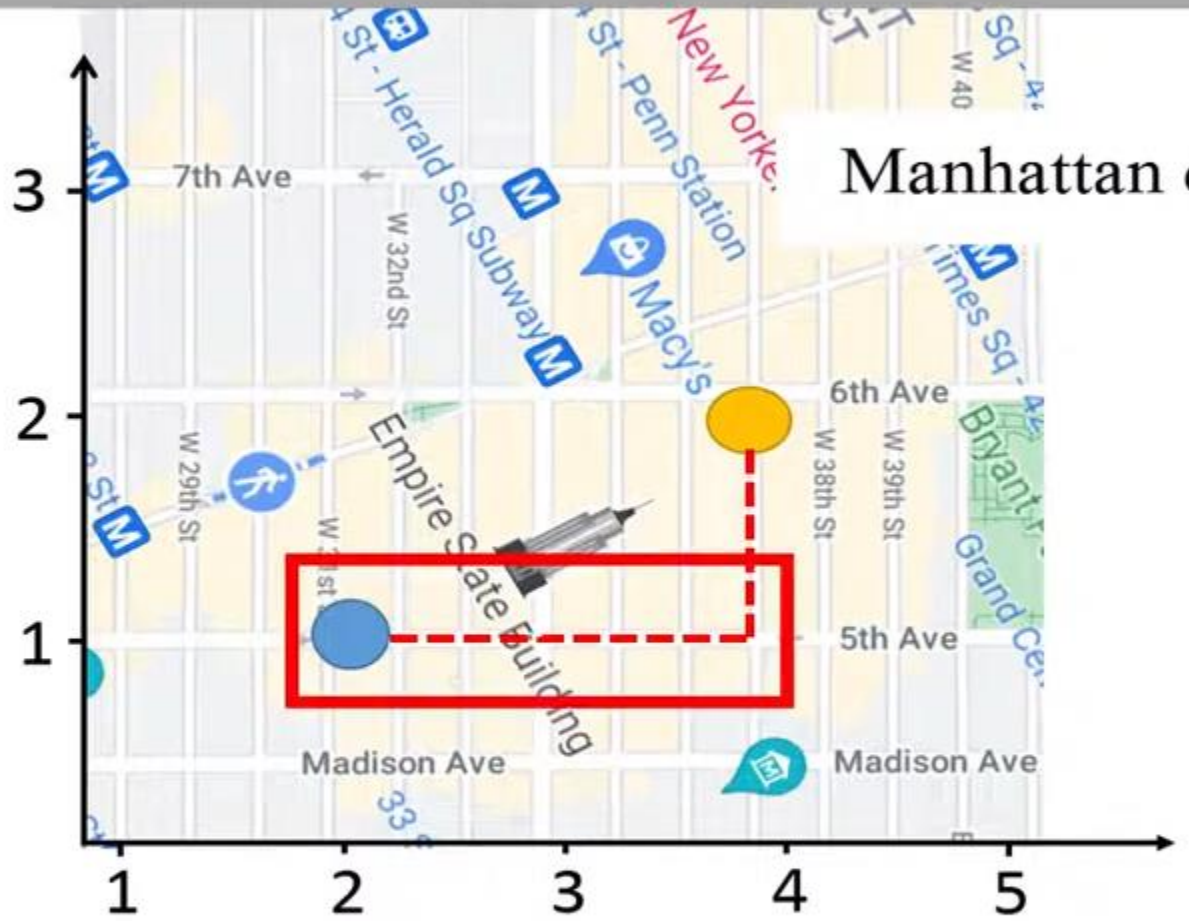
Manhattan distance



$$\text{Manhattan distance} = |2-4| + |1-2| = 3$$

Such type of distance is usually called Manhattan distance.

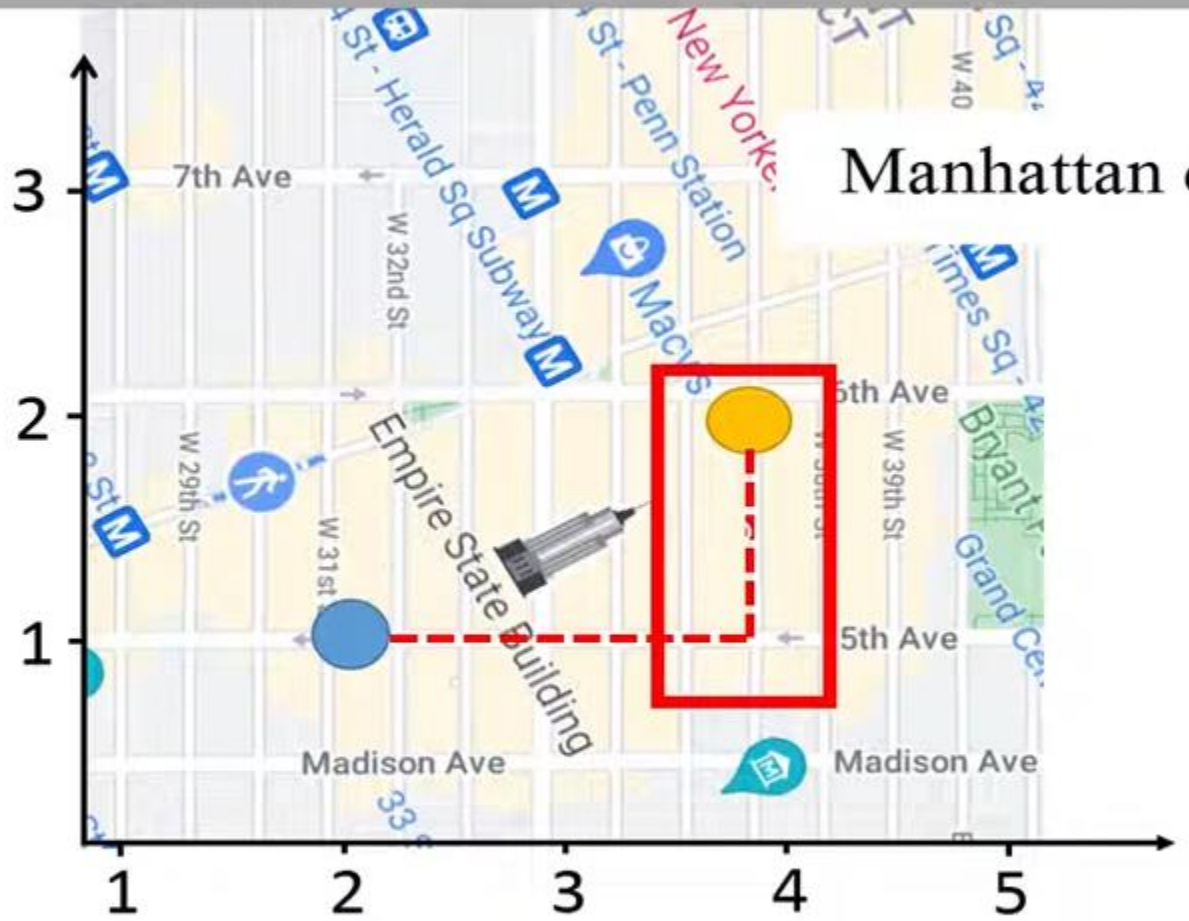
Manhattan distance



$$\text{Manhattan distance} = |2-4| + |1-2| = 3$$

The horizontal distance is here two,

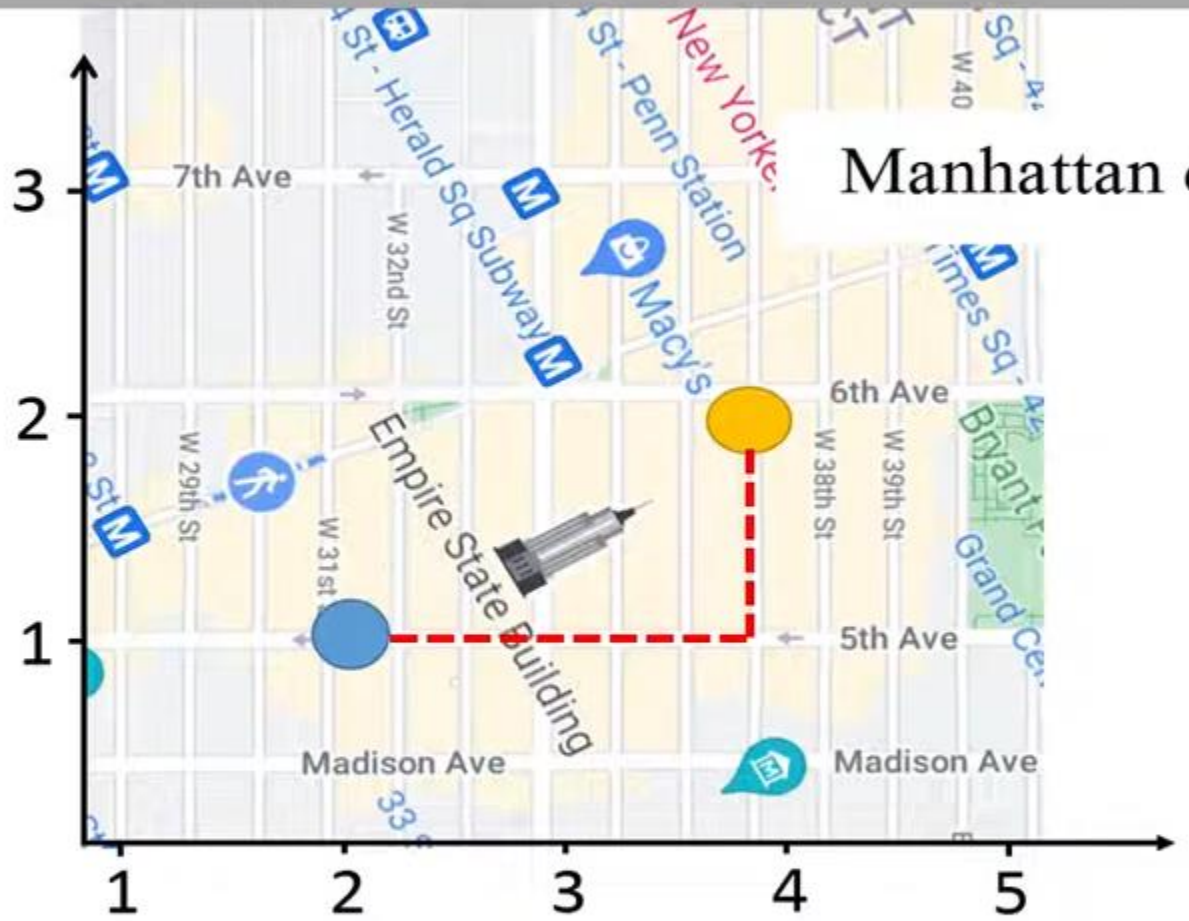
Manhattan distance



$$\text{Manhattan distance} = |2-4| + |1-2| = 3$$

and the vertical distance is one,

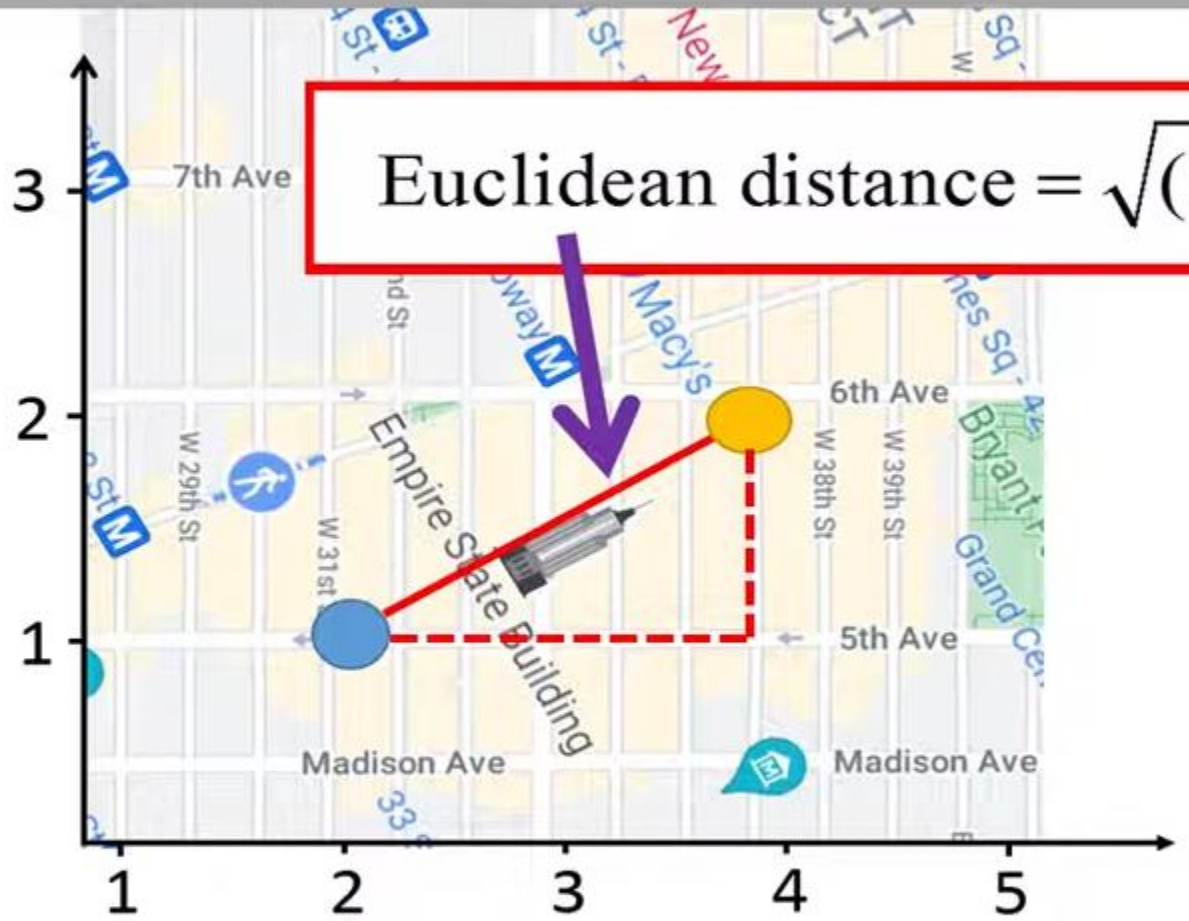
Manhattan distance



$$\text{Manhattan distance} = |2-4| + |1-2| = 3$$

which results in a total distance of three.

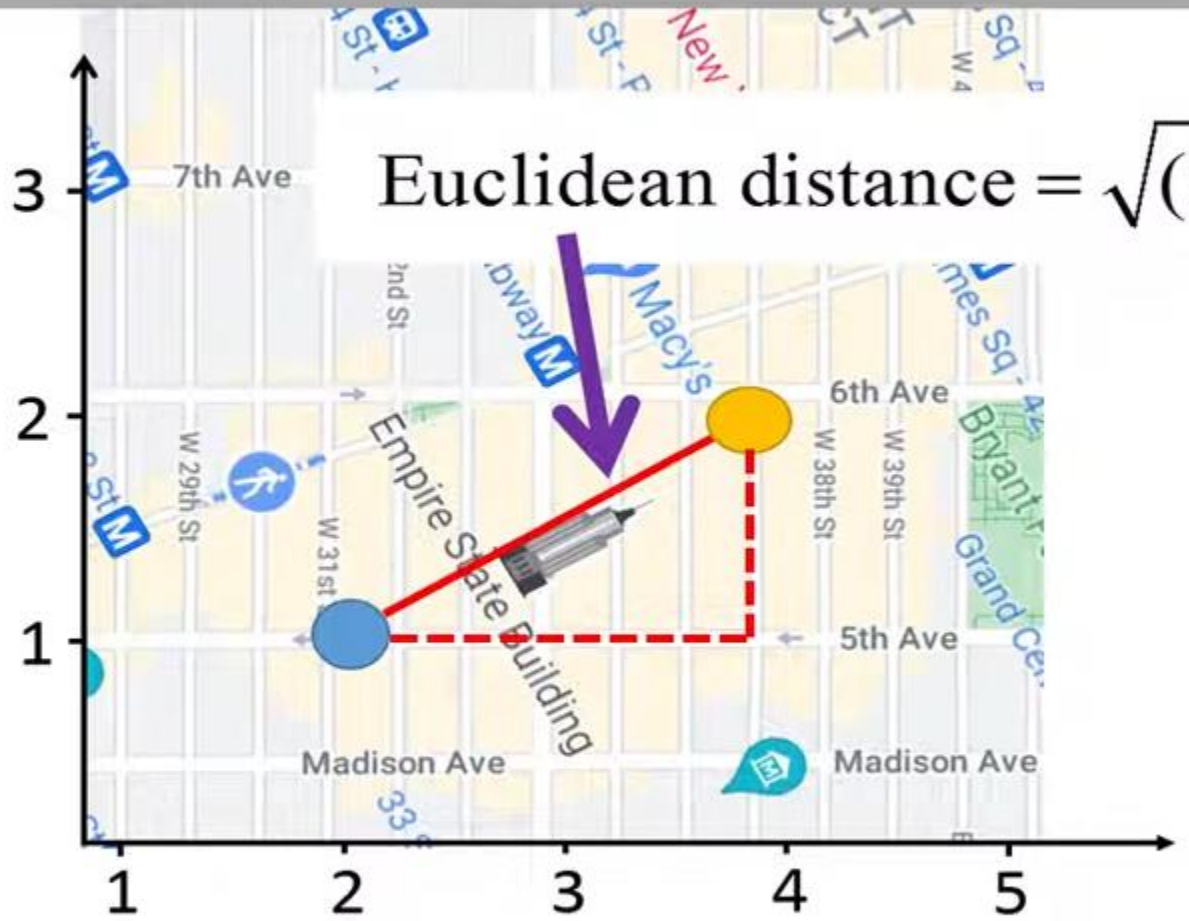
Euclidean distance



$$\text{Euclidean distance} = \sqrt{(2-4)^2 + (1-2)^2} = 2.24$$

If we could take the shortest distance, that would correspond to the Euclidean distance. This is the same calculation we would use to calculate the longest side of a right triangle.

Euclidean distance



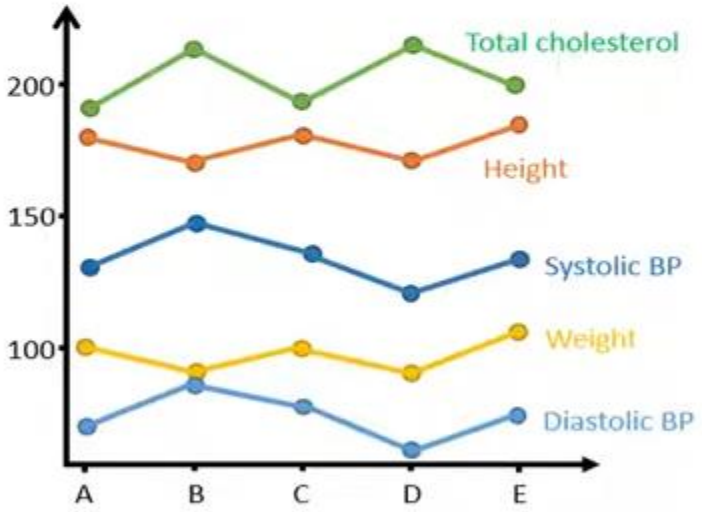
$$\text{Euclidean distance} = \sqrt{(2-4)^2 + (1-2)^2} = 2.24$$

The Euclidean distance in this example is about 2.24.

Calculate the Euclidean distance

	A	B	C	D	E
Diastolic BP (mmHg)	70	86	78	60	75
Systolic BP (mmHg)	130	147	137	121	134
Weight (kg)	100	90	100	90	106
Height (cm)	180	170	181	171	185
Total cholesterol (mg/dl)	190	215	192	215	200

$$d = \sqrt{\left((70-130)^2 + (86-147)^2 + (78-137)^2 + (60-121)^2 + (75-134)^2\right)} = 134.2$$

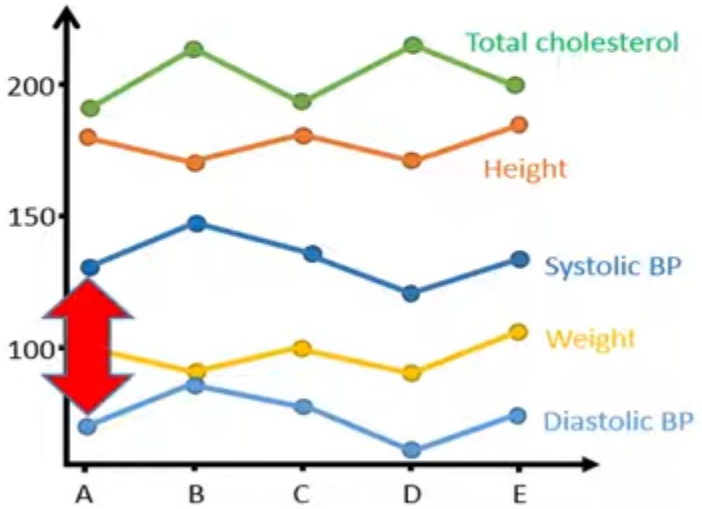


The Euclidean distance is the most common distance measure used in clustering. Let's use the Euclidean distance as a distance measure for how similar the values are between the diastolic and systolic blood pressure.

Calculate the Euclidean distance

	A	B	C	D	E
Diastolic BP (mmHg)	70	86	78	60	75
Systolic BP (mmHg)	130	147	137	121	134
Weight (kg)	100	90	100	90	106
Height (cm)	180	170	181	171	185
Total cholesterol (mg/dl)	190	215	192	215	200

$$d = \sqrt{(70-130)^2 + (86-147)^2 + (78-137)^2 + (60-121)^2 + (75-134)^2} = 134.2$$

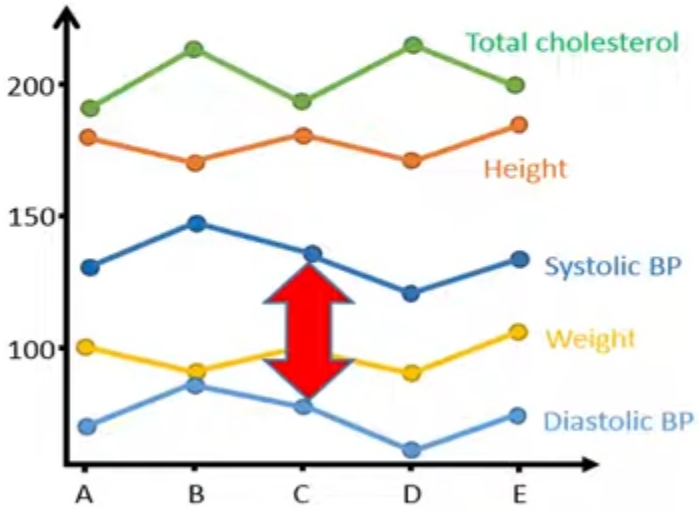


We therefore square the difference between these two variables for the individuals A,

Calculate the Euclidean distance

	A	B	C	D	E
Diastolic BP (mmHg)	70	86	78	60	75
Systolic BP (mmHg)	130	147	137	121	134
Weight (kg)	100	90	100	90	106
Height (cm)	180	170	181	171	185
Total cholesterol (mg/dl)	190	215	192	215	200

$$d = \sqrt{\left((70-130)^2 + (86-147)^2 + (78-137)^2 + (60-121)^2 + (75-134)^2\right)} = 134.2$$

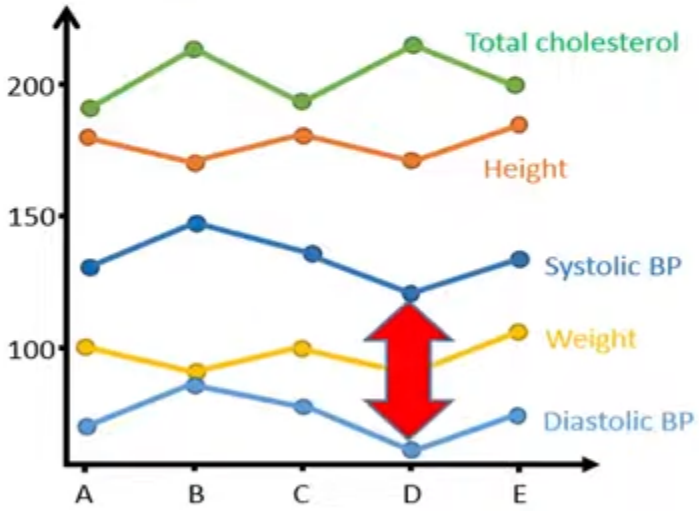


C

Calculate the Euclidean distance

	A	B	C	D	E
Diastolic BP (mmHg)	70	86	78	60	75
Systolic BP (mmHg)	130	147	137	121	134
Weight (kg)	100	90	100	90	106
Height (cm)	180	170	181	171	185
Total cholesterol (mg/dl)	190	215	192	215	200

$$d = \sqrt{\left((70-130)^2 + (86-147)^2 + (78-137)^2 + (60-121)^2 + (75-134)^2\right)} = 134.2$$

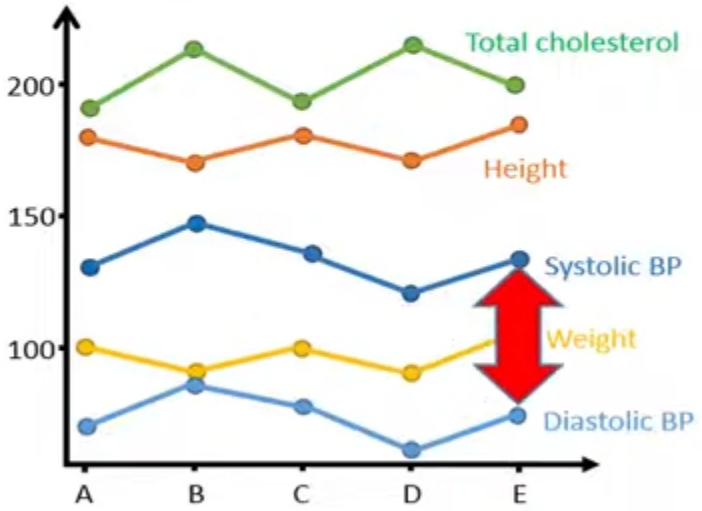


D

Calculate the Euclidean distance

	A	B	C	D	E
Diastolic BP (mmHg)	70	86	78	60	75
Systolic BP (mmHg)	130	147	137	121	134
Weight (kg)	100	90	100	90	106
Height (cm)	180	170	181	171	185
Total cholesterol (mg/dl)	190	215	192	215	200

$$d = \sqrt{(70-130)^2 + (86-147)^2 + (78-137)^2 + (60-121)^2 + (75-134)^2} = 134.2$$

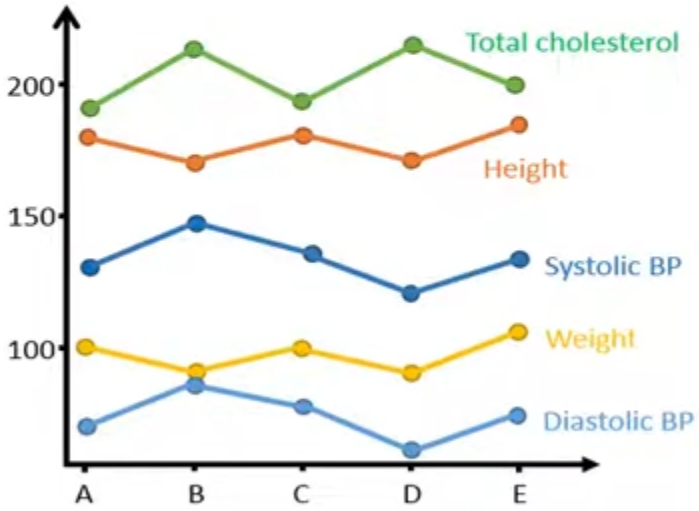


and E.

Calculate the Euclidean distance

	A	B	C	D	E
Diastolic BP (mmHg)	70	86	78	60	75
Systolic BP (mmHg)	130	147	137	121	134
Weight (kg)	100	90	100	90	106
Height (cm)	180	170	181	171	185
Total cholesterol (mg/dl)	190	215	192	215	200

$$d = \sqrt{\left((70-130)^2 + (86-147)^2 + (78-137)^2 + (60-121)^2 + (75-134)^2\right)} = 134.2$$



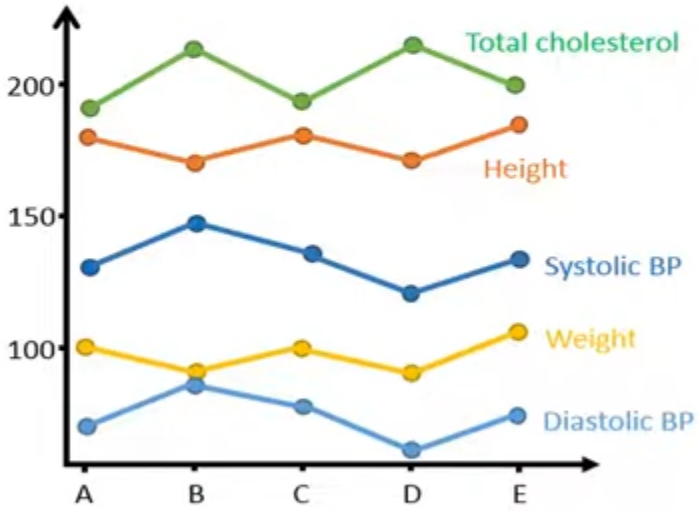
The square root of the sum of the squared differences is equal to 134.2.

Calculate the Euclidean distance

	A	B	C	D	E
Diastolic BP (mmHg)	70	86	78	60	75
Systolic BP (mmHg)	130	147	137	121	134
Weight (kg)	100	90	100	90	106
Height (cm)	180	170	181	171	185
Total cholesterol (mg/dl)	190	215	192	215	200

$$d = \sqrt{\left((70-130)^2 + (86-147)^2 + (78-137)^2 + (60-121)^2 + (75-134)^2\right)} = 134.2$$

$$d = \sqrt{\left((70-100)^2 + (86-90)^2 + (78-100)^2 + (60-90)^2 + (75-106)^2\right)} = 57.1$$



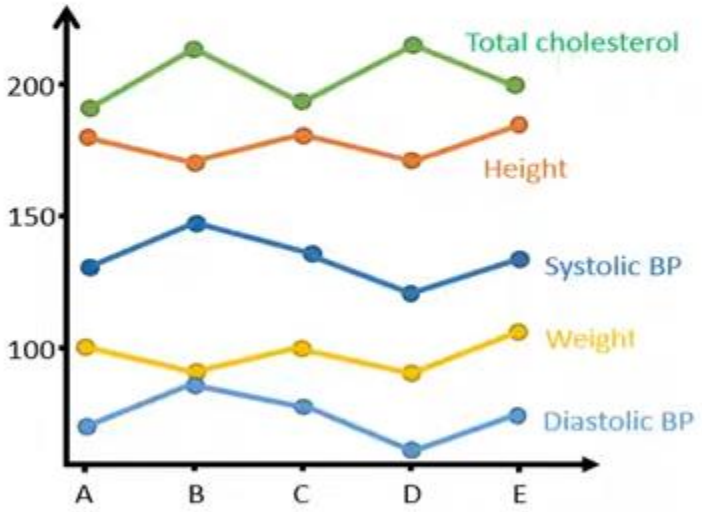
Then we calculate the Euclidean distance between the two variables diastolic blood pressure and body weight.

Calculate the Euclidean distance

	A	B	C	D	E
Diastolic BP (mmHg)	70	86	78	60	75
Systolic BP (mmHg)	130	147	137	121	134
Weight (kg)	100	90	100	90	106
Height (cm)	180	170	181	171	185
Total cholesterol (mg/dl)	190	215	192	215	200

$$d = \sqrt{\left((70-130)^2 + (86-147)^2 + (78-137)^2 + (60-121)^2 + (75-134)^2\right)} = 134.2$$

$$d = \sqrt{\left((70-100)^2 + (86-90)^2 + (78-100)^2 + (60-90)^2 + (75-106)^2\right)} = 57.1$$



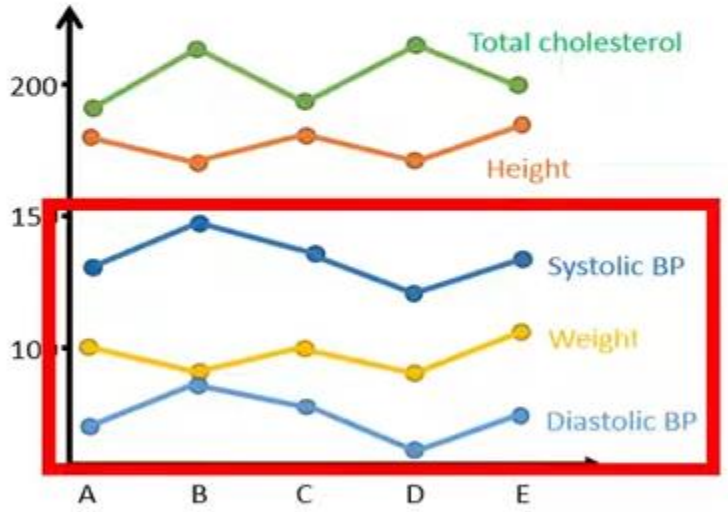
We see that the variable diastolic blood pressure has a shorter distance to the variable body weight compared to the systolic blood pressure.

Calculate the Euclidean distance

	A	B	C	D	E
Diastolic BP (mmHg)	70	86	78	60	75
Systolic BP (mmHg)	130	147	137	121	134
Weight (kg)	100	90	100	90	106
Height (cm)	180	170	181	171	185
Total cholesterol (mg/dl)	190	215	192	215	200

$$d = \sqrt{\left((70-130)^2 + (86-147)^2 + (78-137)^2 + (60-121)^2 + (75-134)^2\right)} = 134.2$$

$$d = \sqrt{\left((70-100)^2 + (86-90)^2 + (78-100)^2 + (60-90)^2 + (75-106)^2\right)} = 57.1$$



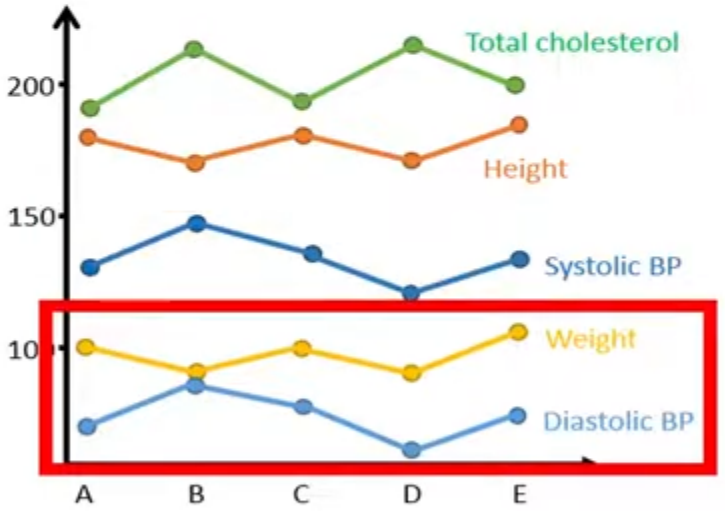
This seems reasonable if we study the profile plot,

Calculate the Euclidean distance

	A	B	C	D	E
Diastolic BP (mmHg)	70	86	78	60	75
Systolic BP (mmHg)	130	147	137	121	134
Weight (kg)	100	90	100	90	106
Height (cm)	180	170	181	171	185
Total cholesterol (mg/dl)	190	215	192	215	200

$$d = \sqrt{\left((70-130)^2 + (86-147)^2 + (78-137)^2 + (60-121)^2 + (75-134)^2\right)} = 134.2$$

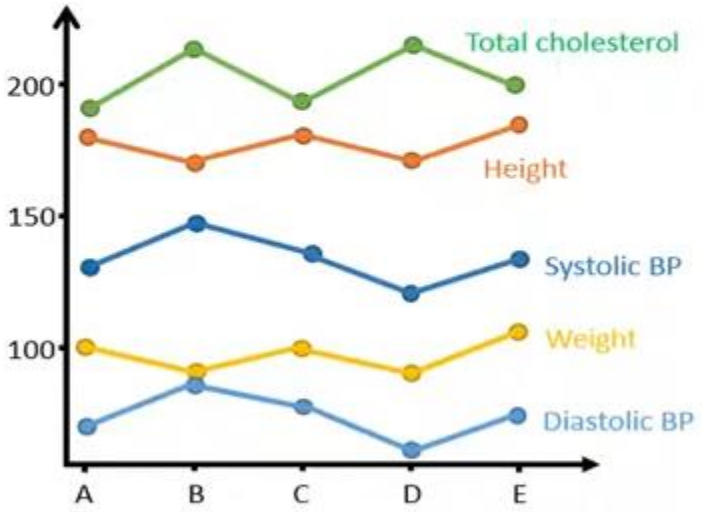
$$d = \sqrt{\left((70-100)^2 + (86-90)^2 + (78-100)^2 + (60-90)^2 + (75-106)^2\right)} = 57.1$$



because the values of the diastolic blood pressure are closer to the values of the body weight.

Distance matrix

	A	B	C	D	E
Diastolic BP (mmHg)	70	86	78	60	75
Systolic BP (mmHg)	130	147	137	121	134
Weight (kg)	100	90	100	90	106
Height (cm)	180	170	181	171	185
Total cholesterol (mg/dl)	190	215	192	215	200



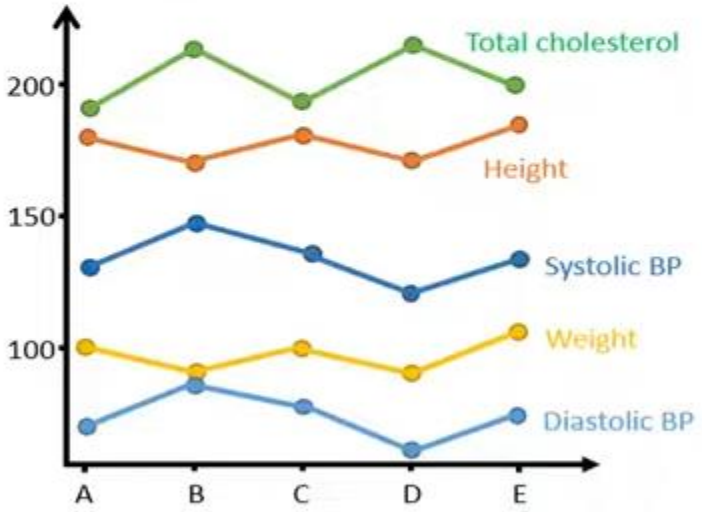
Distance matrix:

	Diastolic BP	Systolic BP	Weight	Height
Systolic BP	134.2			
Weight	57.1	85.2		
Height	232.8	100.3	179.3	
Total cholesterol	289.3	156.3	238.0	66.4

If we calculate the Euclidean distance between all pairs of variables, we can create the so-called distance matrix.

Distance matrix

	A	B	C	D	E
Diastolic BP (mmHg)	70	86	78	60	75
Systolic BP (mmHg)	130	147	137	121	134
Weight (kg)	100	90	100	90	106
Height (cm)	180	170	181	171	185
Total cholesterol (mg/dl)	190	215	192	215	200



Distance matrix:

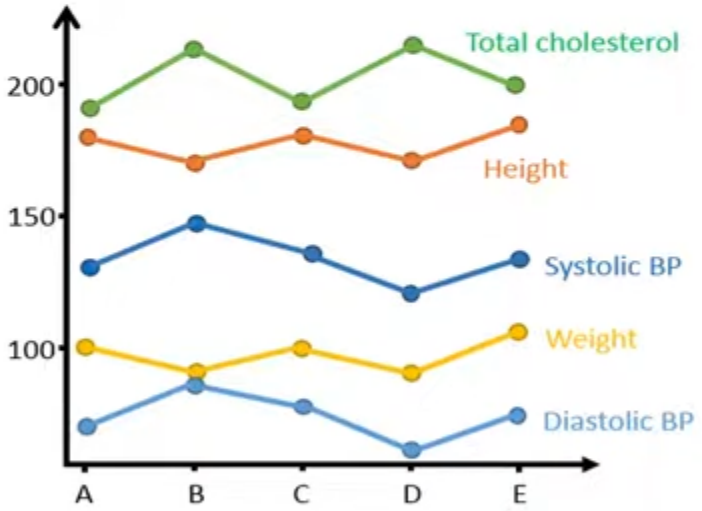
	Diastolic BP	Systolic BP	Weight	Height
Systolic BP	134.2			
Weight	57.1	85.2		
Height	232.8	100.3	179.3	
Total cholesterol	289.3	156.3	238.0	66.4

For example, the Euclidean distance between the systolic blood pressure and the total cholesterol level is 156.3.

Distance matrix

Distance matrix:

	Diastolic BP	Systolic BP	Weight	Height
Systolic BP	134.2			
Weight	57.1	85.2		
Height	232.8	100.3	179.3	
Total cholesterol	289.3	156.3	238.0	66.4

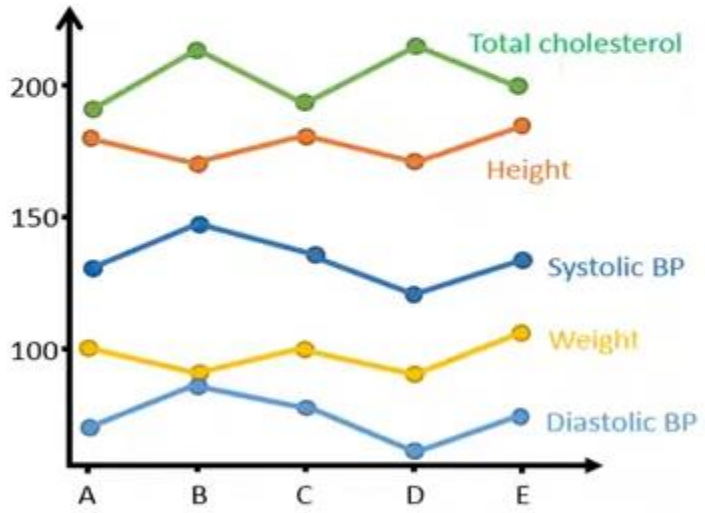


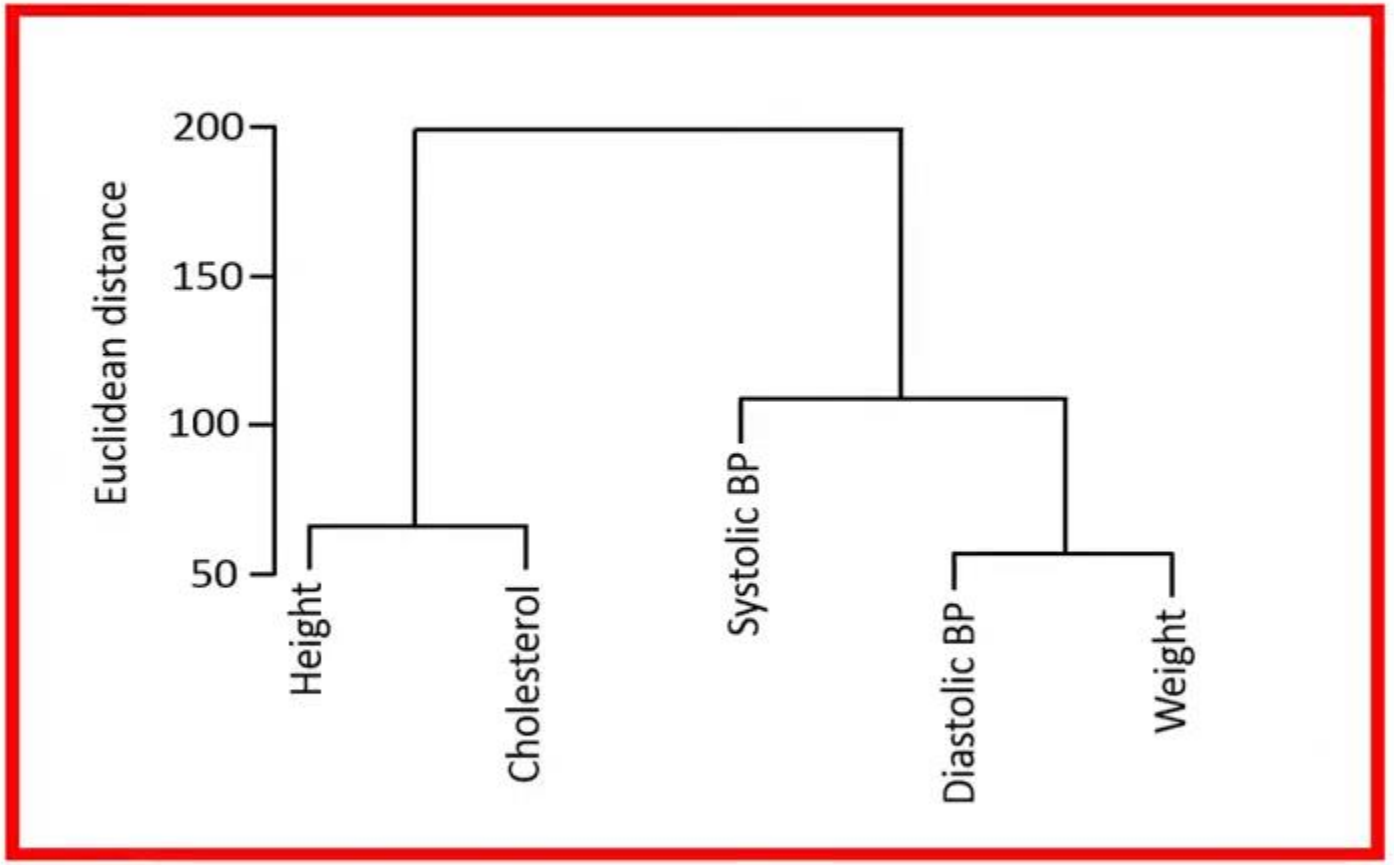
Let's move the distance matrix up here.

Create dendrogram

Distance matrix:

	Diastolic BP	Systolic BP	Weight	Height
Systolic BP	134.2			
Weight	57.1	85.2		
Height	232.8	100.3	179.3	
Total cholesterol	289.3	156.3	238.0	66.4



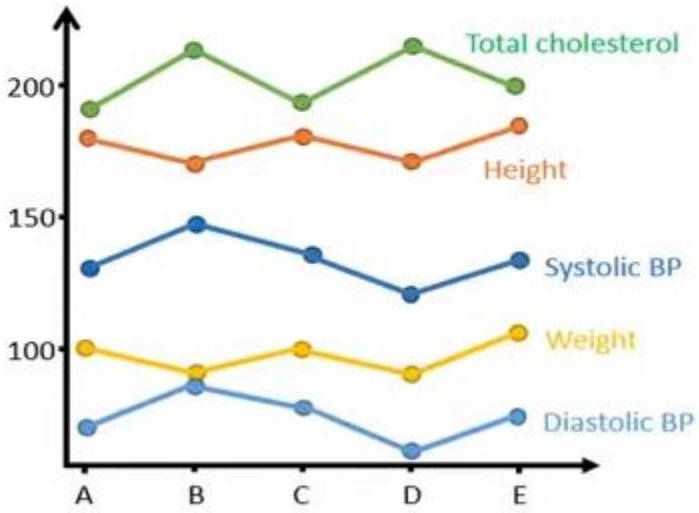


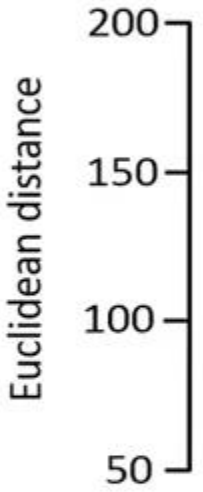
We will now draw the following dendrogram.

Create dendrogram

Distance matrix:

	Diastolic BP	Syst		
Systolic BP	134.2			
Weight	57.1			
Height	232.8			
Total cholesterol	289.3	156.3	238.0	66.4



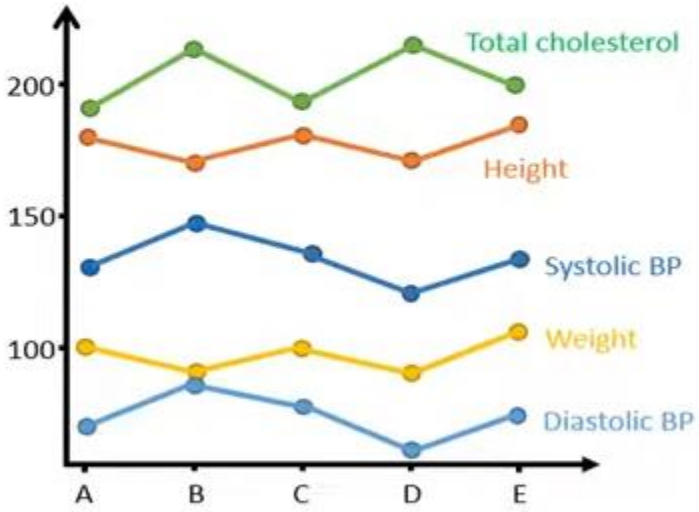


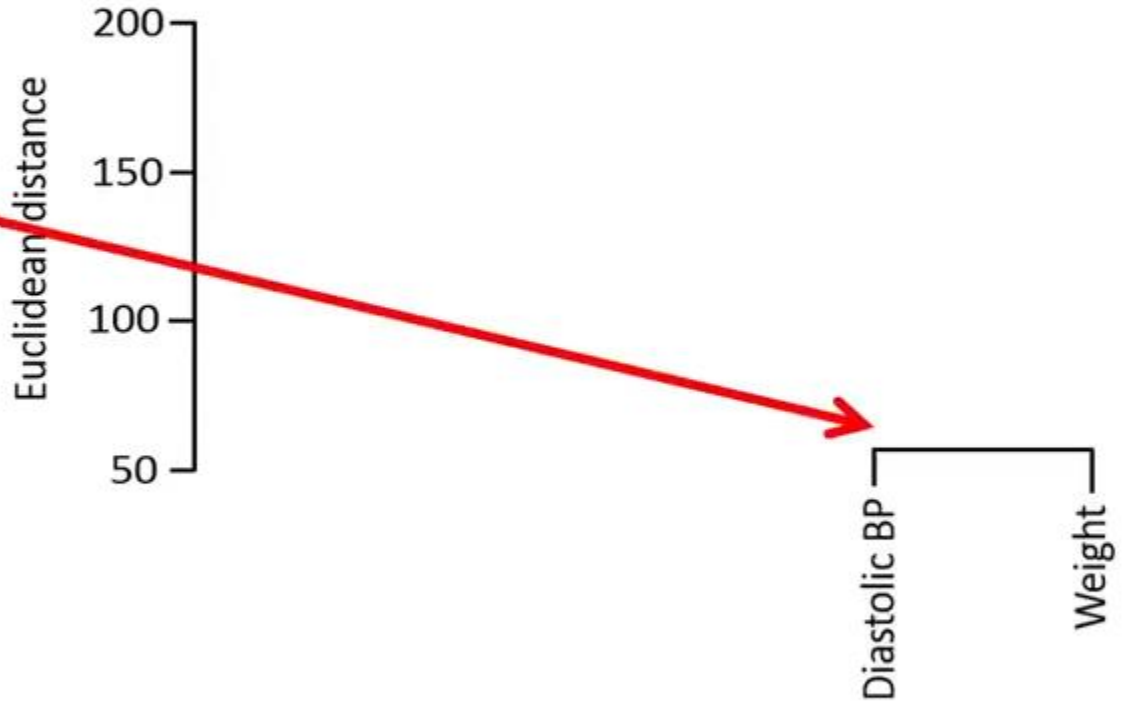
We first start to identify the shortest distance in the matrix, which is the distance between the body weight and the diastolic blood pressure.

Create dendrogram

Distance matrix:

	Diastolic BP	Systolic BP	Weight	Height
Systolic BP	134.2			
Weight	57.1	85.2		
Height	232.8	[illegible]	179.3	
Total cholesterol	289.3	156.3	238.6	66.4



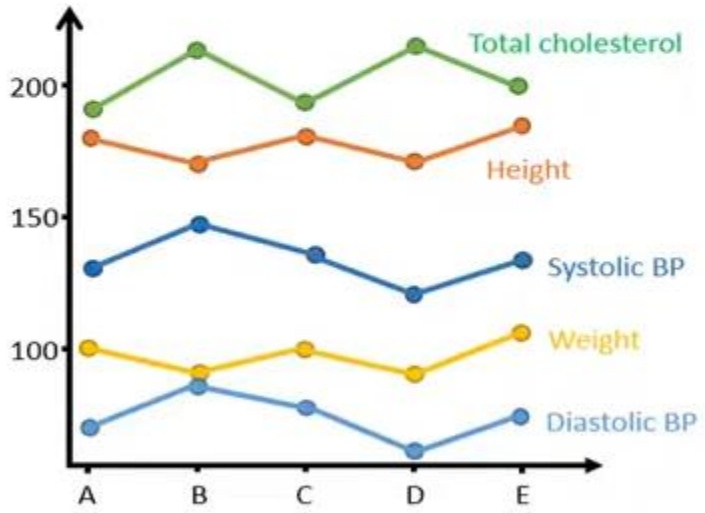


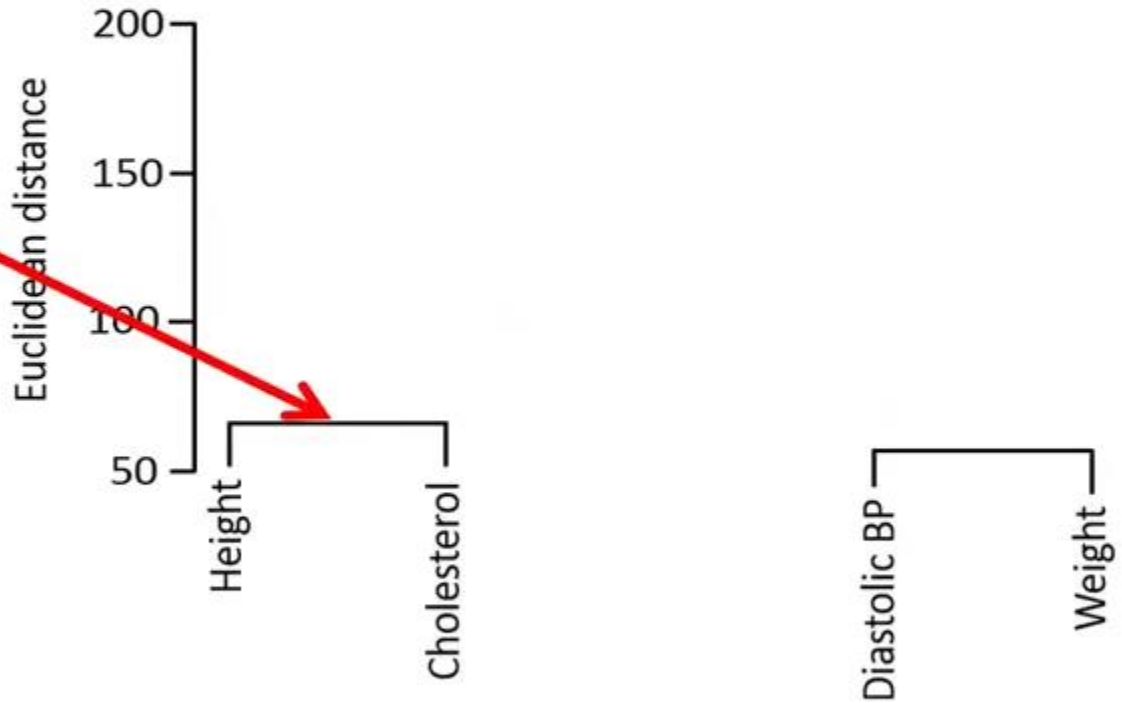
We therefore draw a horizontal line at the y-coordinate 57.1 in the plot.

Create dendrogram

Distance matrix:

	Diastolic BP	Systolic BP	Weight	Height
Systolic BP	134.2			
Weight	57.1	85.2		
Height	232.8	100.3	179.3	
Total cholesterol	289.3	156.3	238.0	66.4



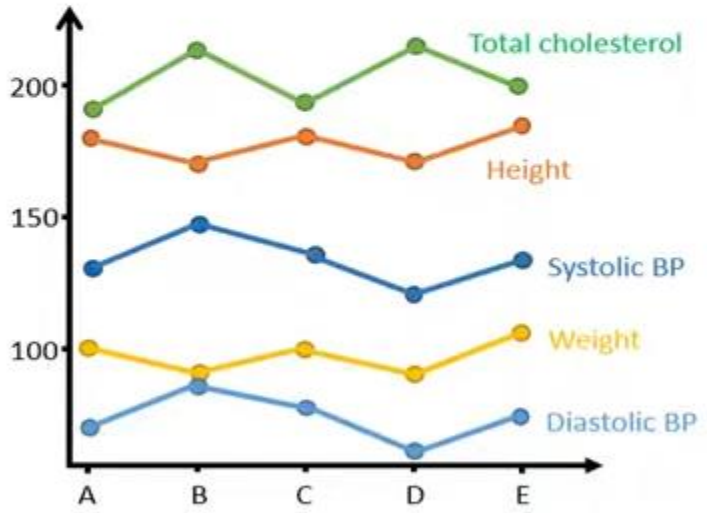


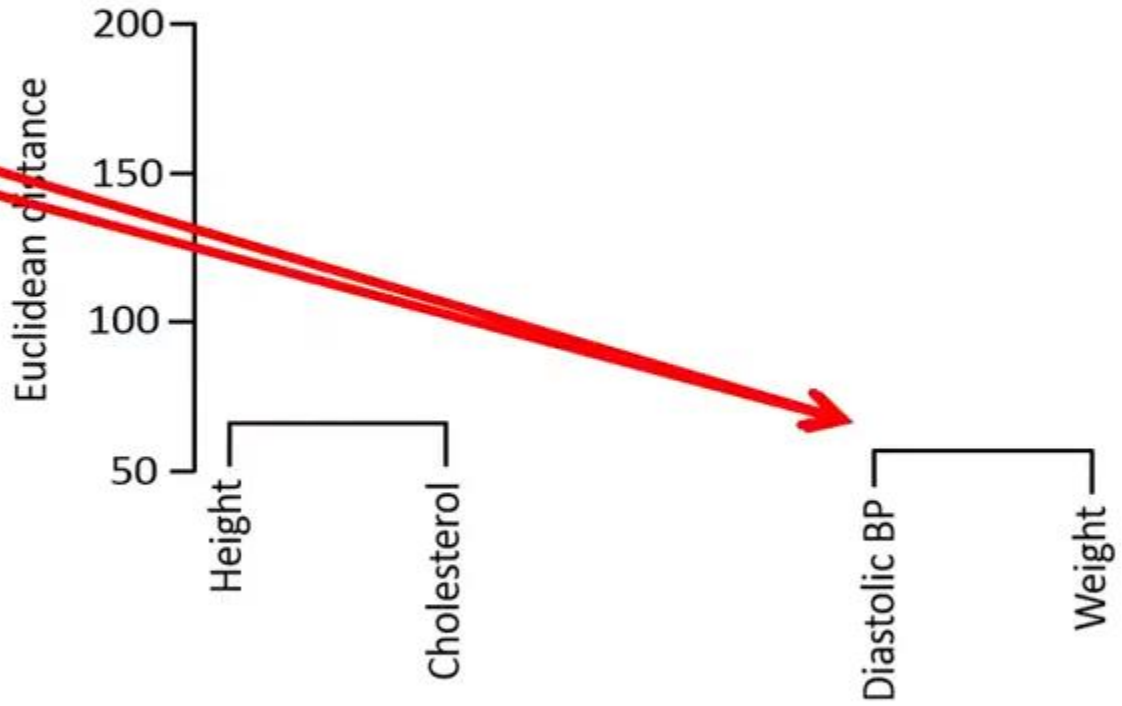
The next shortest distance is between the body height and the cholesterol level.

Create dendrogram

Distance matrix:

	Diastolic BP	Systolic BP	Weight	Height
Systolic BP	134.2			
Weight	57.1	85.2		
Height	232.8	100.3	179.3	
Total cholesterol	289.3	156.3	238.0	66.4



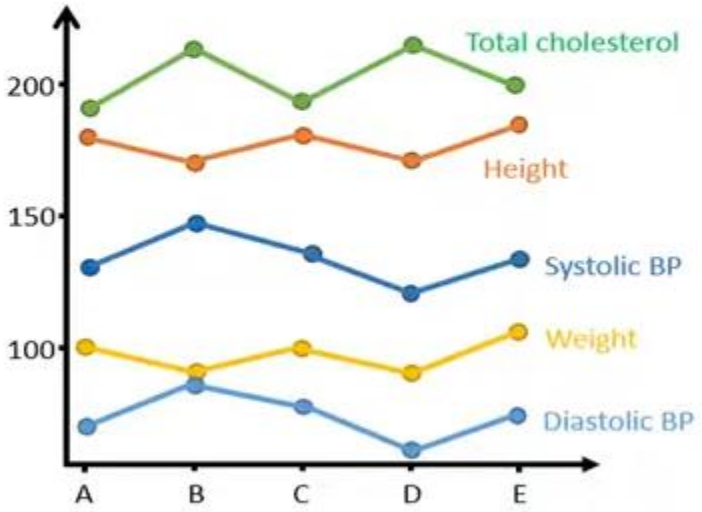


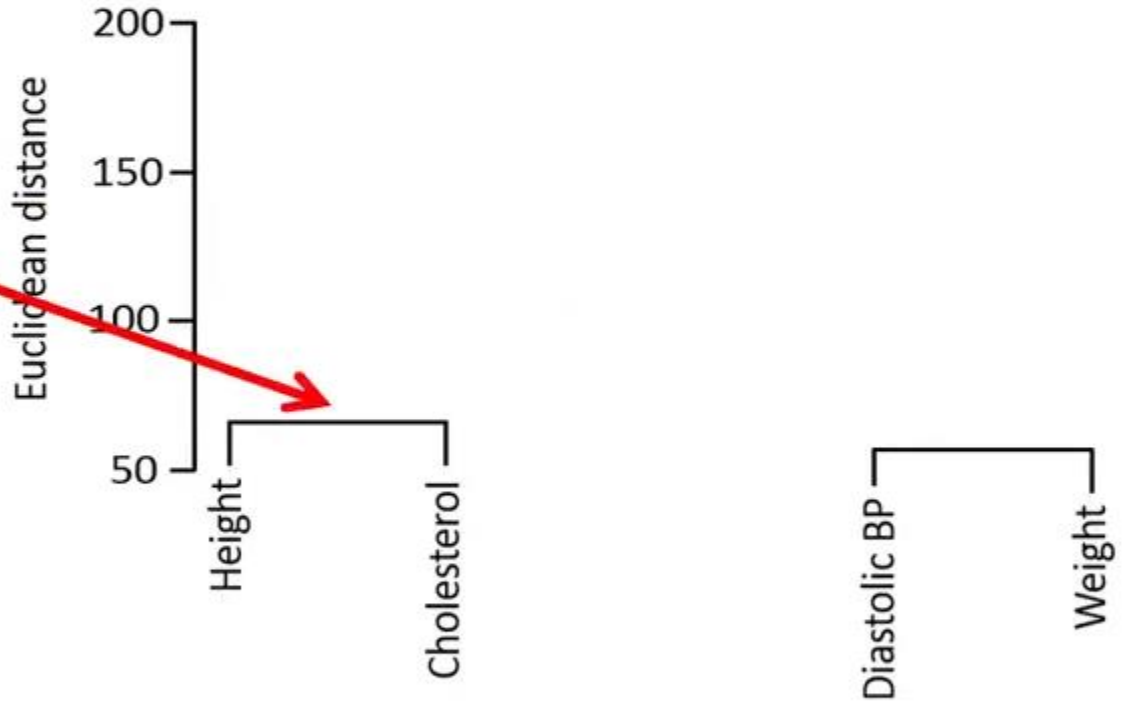
Next, we need to figure out if the systolic blood pressure is closer to this cluster,

Create dendrogram

Distance matrix:

	Diastolic BP	Systolic BP	Weight	Height
Systolic BP	134.2			
Weight	57.1	85.2		
Height	232.8	100.3	179.3	
Total cholesterol	289.3	156.3	238.0	66.4



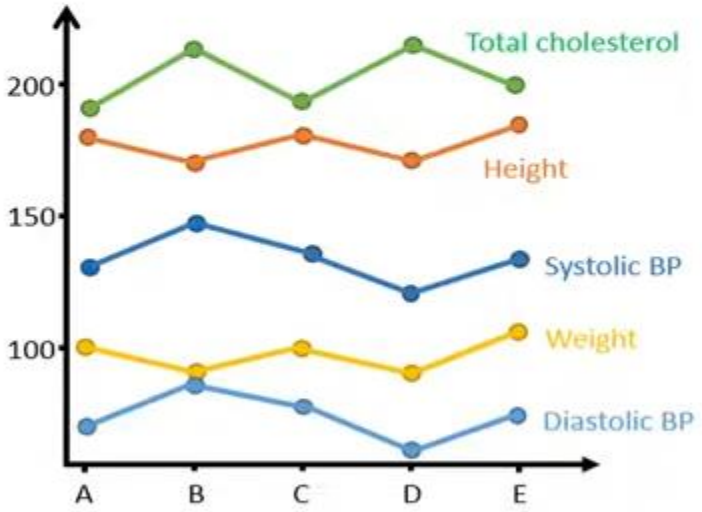


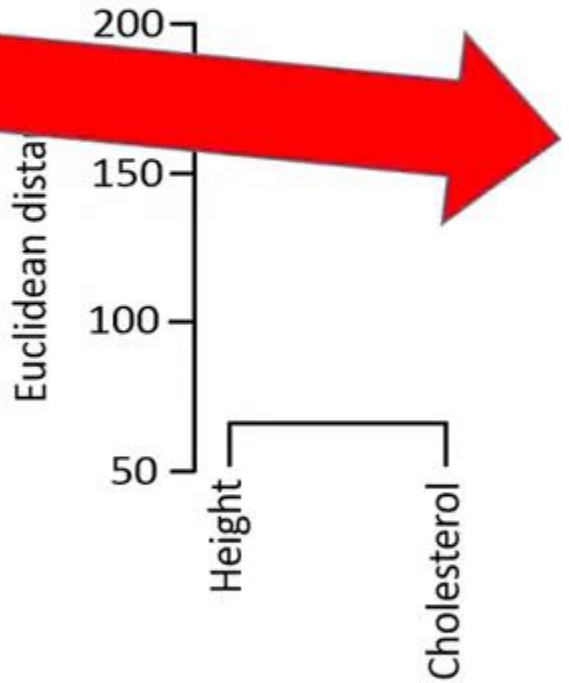
or this cluster.

Create dendrogram

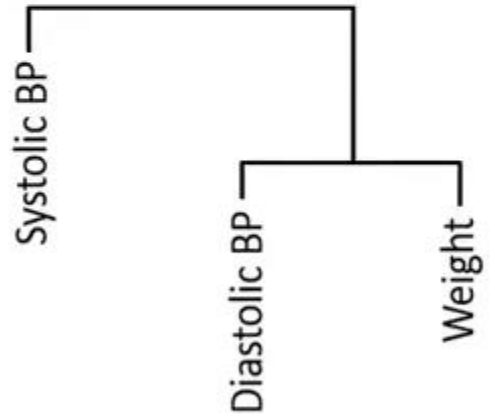
Distance matrix:

	Diastolic BP	Systolic BP	Weight	Height
Systolic BP	134.2			
Weight	57.1	85.2		
Height	232.8	100.3		
Total cholesterol	289.3	156.3	238.0	66.4





$$d = \frac{134.2 + 85.2}{2} = 109.7$$

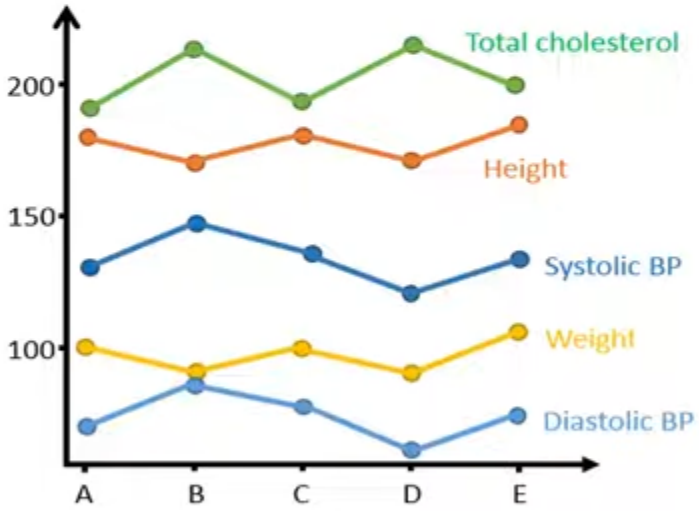


The mean distance between the systolic blood pressure and the two variables in this cluster is 109.7,

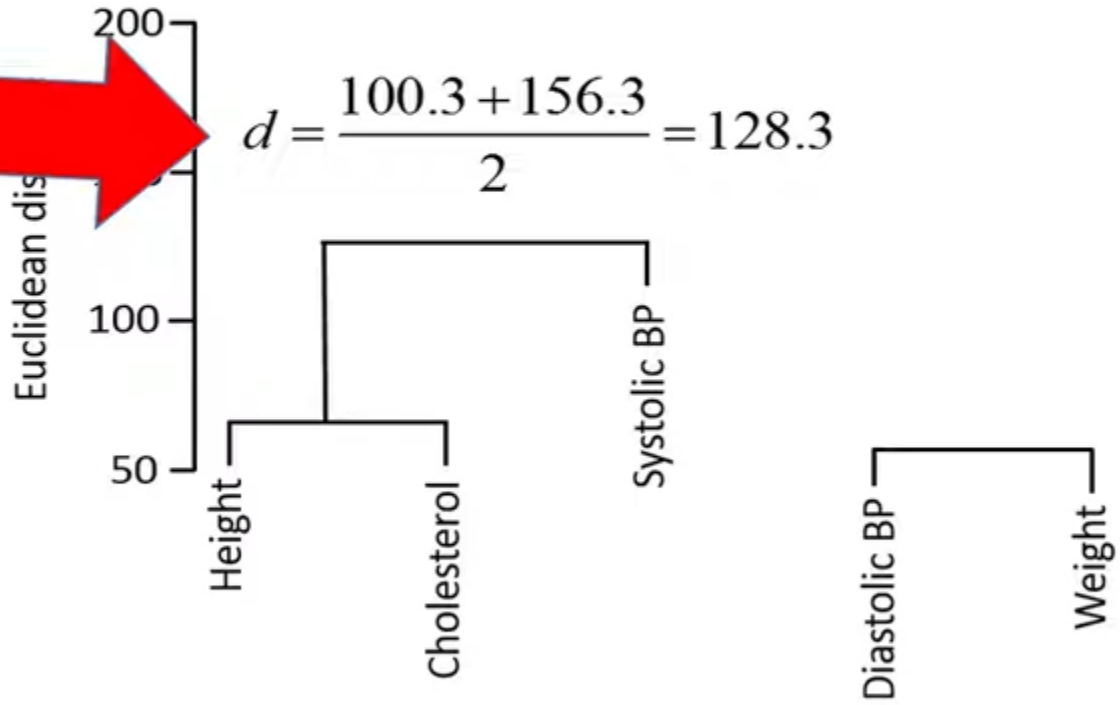
Create dendrogram

Distance matrix:

	Diastolic BP	Systolic BP	Weight	Height
Systolic BP	134.2			
Weight	57.1	85.2		
Height	232.8	100.3		
Total cholesterol	289.3	156.3		



$$d = \frac{100.3 + 156.3}{2} = 128.3$$

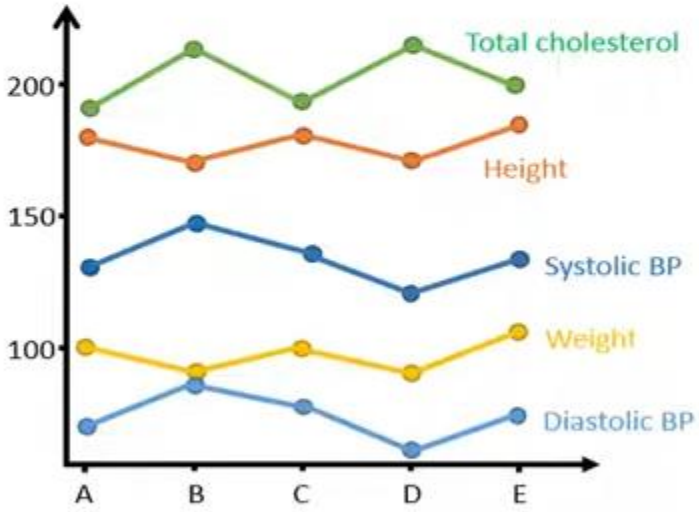


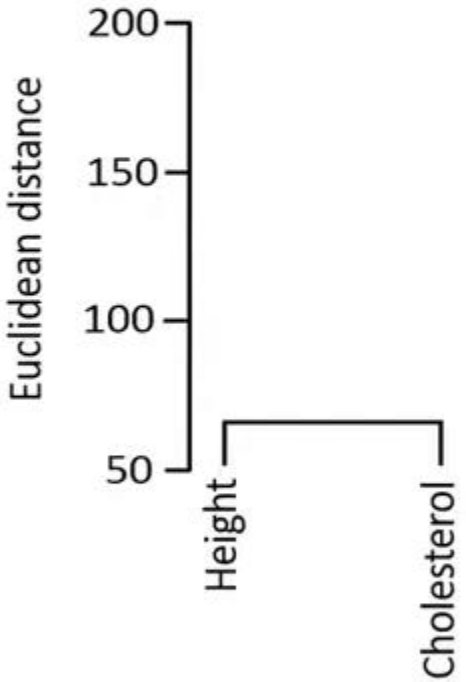
whereas the mean distance to the two variables body height and cholesterol is 128.3.

Create dendrogram

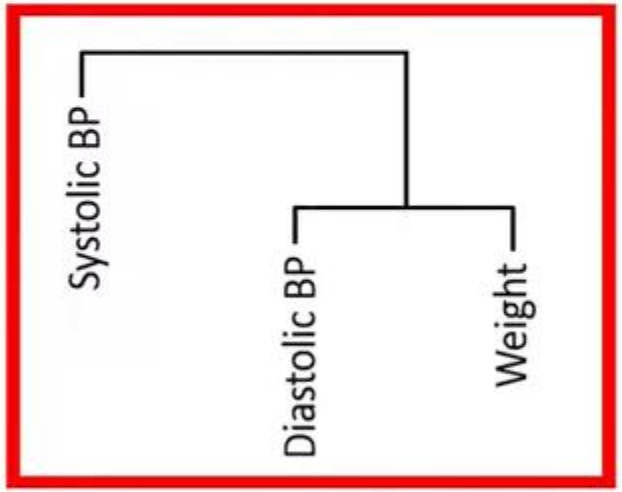
Distance matrix:

	Diastolic BP	Systolic BP	Weight	Height
Systolic BP	134.2			
Weight	57.1	85.2		
Height	232.8	100.3	179.3	
Total cholesterol	289.3	156.3	238.0	66.4





$$d = \frac{134.2 + 85.2}{2} = 109.7$$

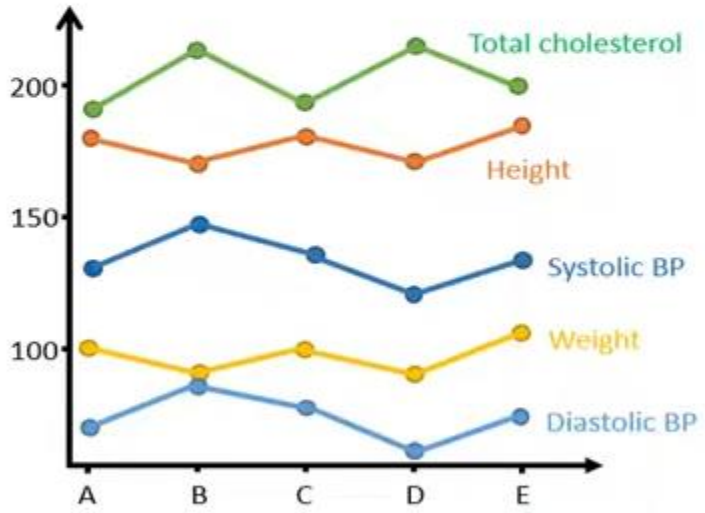


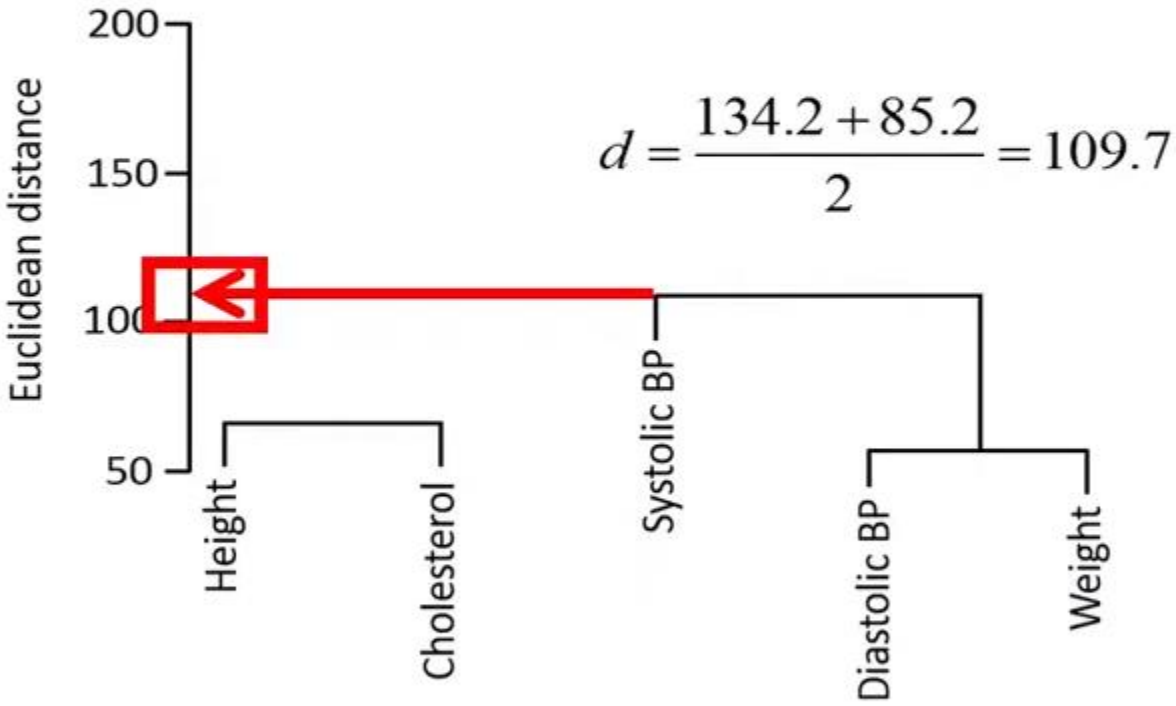
We therefore cluster the systolic blood pressure with the body weight and the diastolic blood pressure because the mean distance is shorter to this cluster.

Create dendrogram

Distance matrix:

	Diastolic BP	Systolic BP	Weight	Height
Systolic BP	134.2			
Weight	57.1	85.2		
Height	232.8	100.3	179.3	
Total cholesterol	289.3	156.3	238.0	66.4





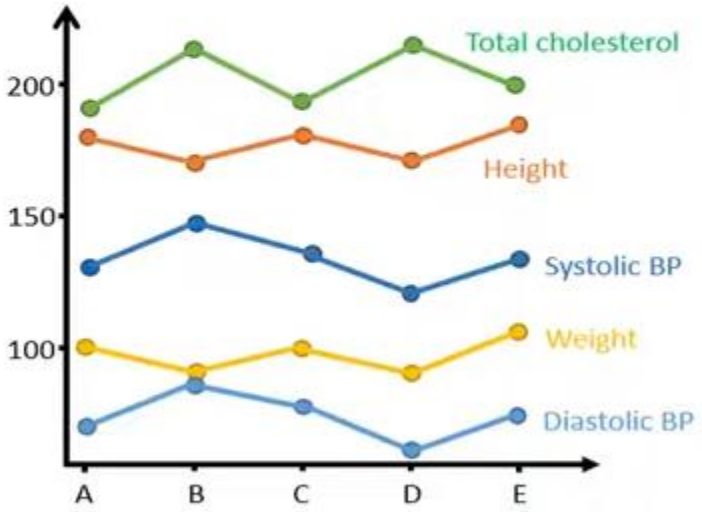
$$d = \frac{134.2 + 85.2}{2} = 109.7$$

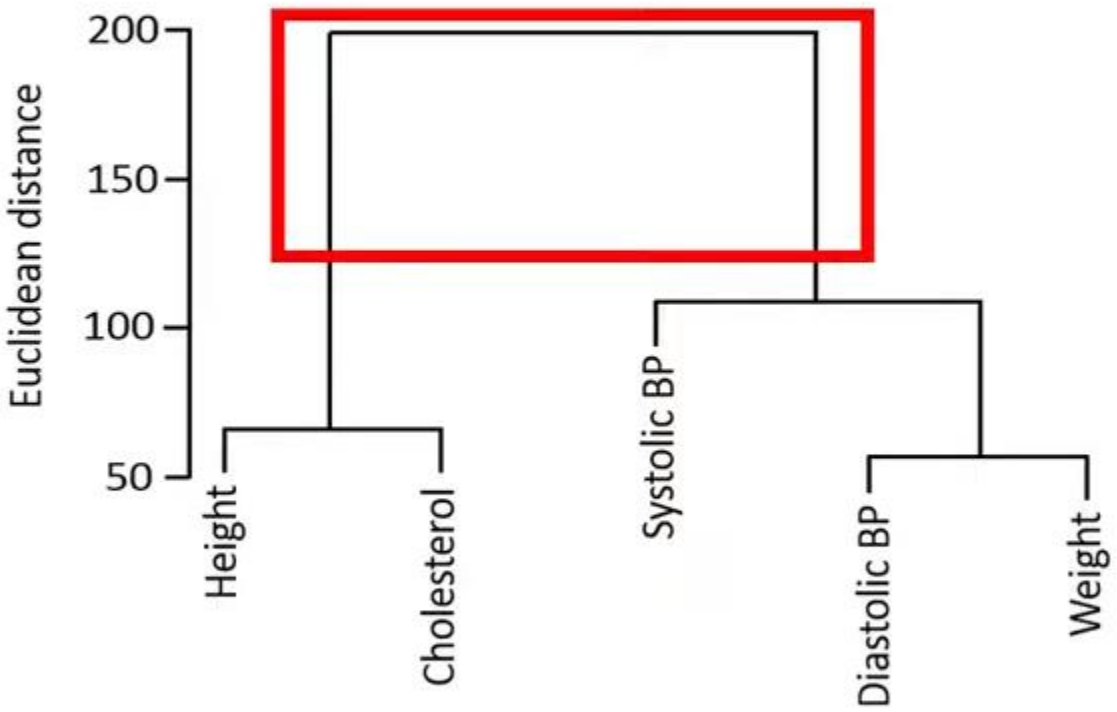
Note that the y-coordinate of this horizontal line should therefore be equal to 109.7.

Create dendrogram

Distance matrix:

	Diastolic BP	Systolic BP	Weight	Height
Systolic BP	134.2			
Weight	57.1	85.2		
Height	232.8	100.3	179.3	
Total cholesterol	289.3	156.3	238.0	66.4





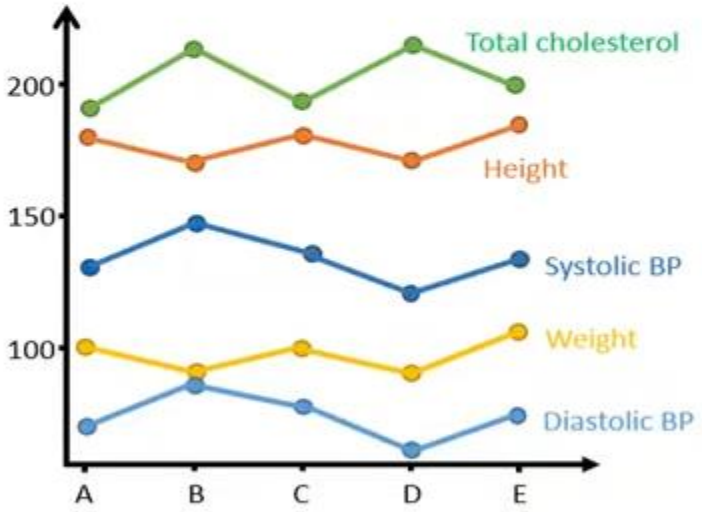
Finally, we connect the two clusters. The y-coordinate of this horizontal line is the mean of,

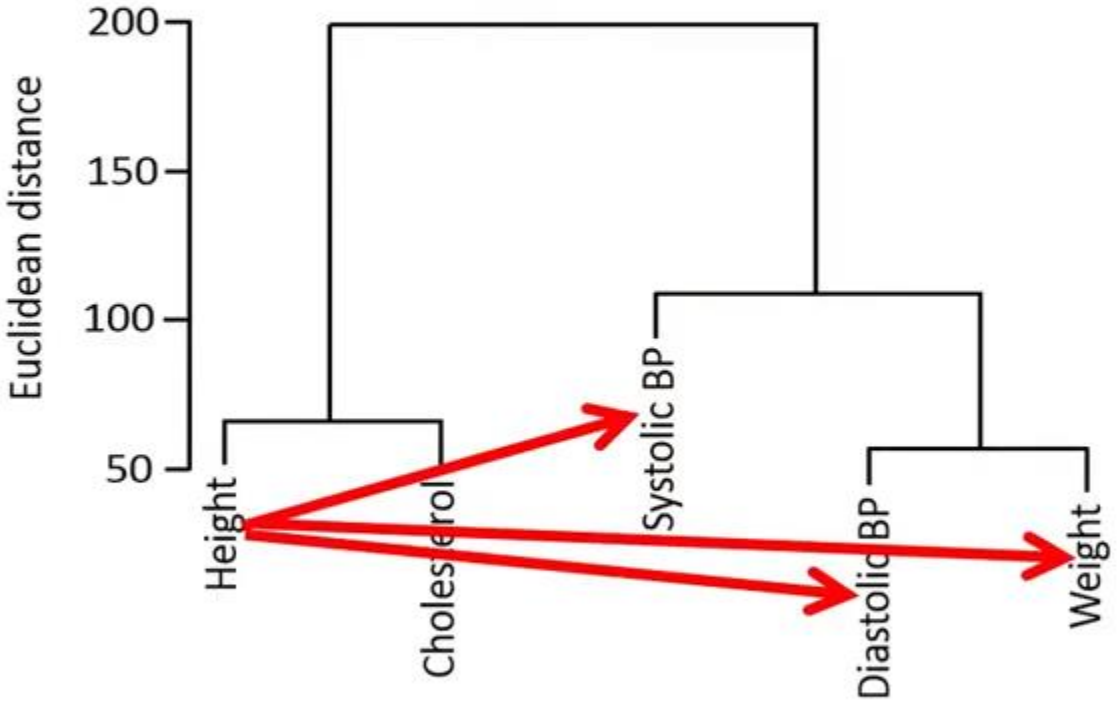
Create dendrogram

Distance matrix:

	Diastolic BP	Systolic BP	Weight	Height
Systolic BP	134.2			
Weight	57.1	85.2		
Height	232.8	100.3	179.3	
Total cholesterol	289.3	156.3	238.0	66.4

$$d = \frac{232.8 + 100.3 + 179.3 + 289.3 + 156.3 + 238.0}{6} = 199.3$$





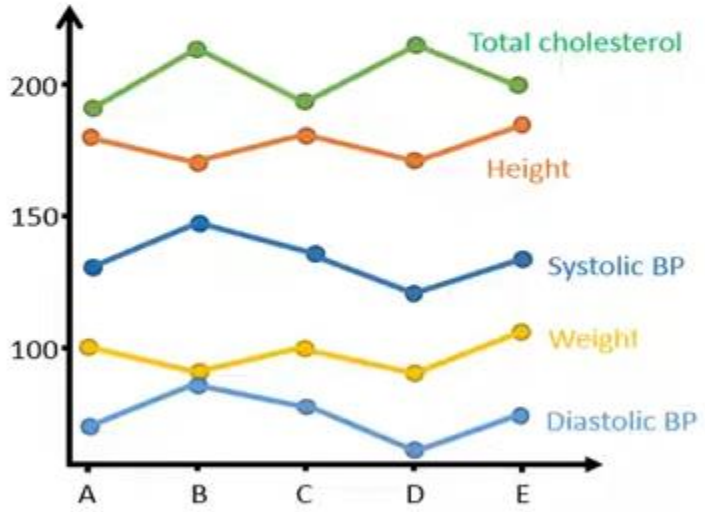
the distance between the body height and the three variables in the other cluster,

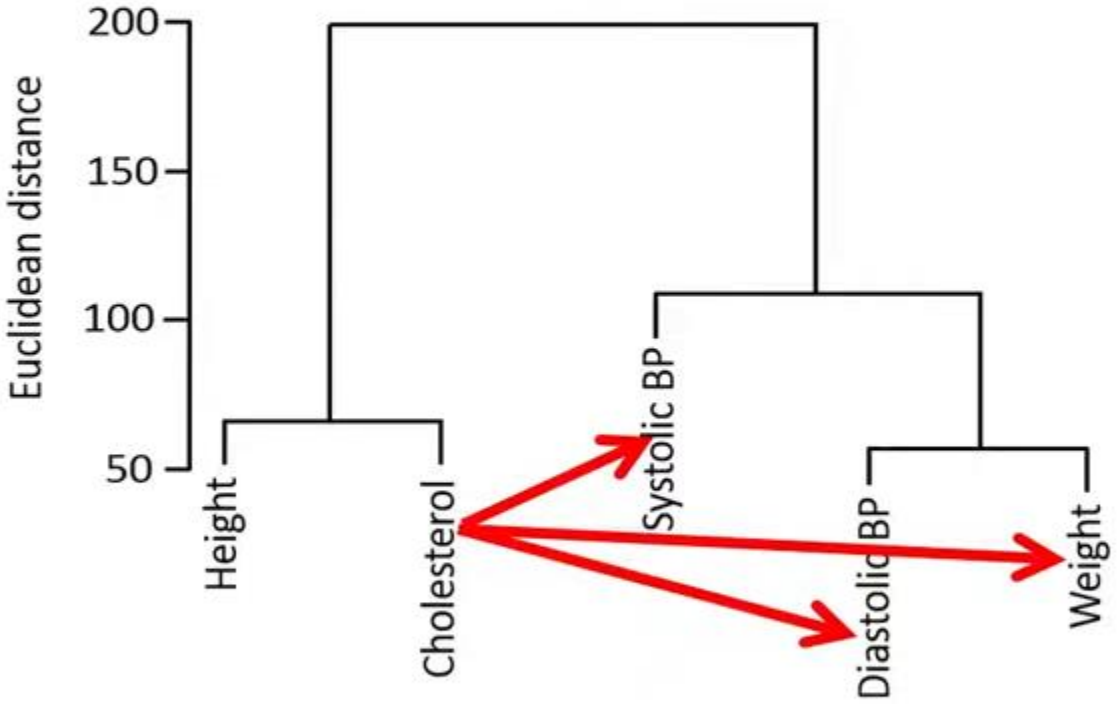
Create dendrogram

Distance matrix:

	Diastolic BP	Systolic BP	Weight	Height
Systolic BP	134.2			
Weight	57.1	85.2		
Height	232.8	100.3	179.3	
Total cholesterol	289.3	156.3	238.0	66.4

$$d = \frac{232.8 + 100.3 + 179.3 + 289.3 + 156.3 + 238.0}{6} = 199.3$$



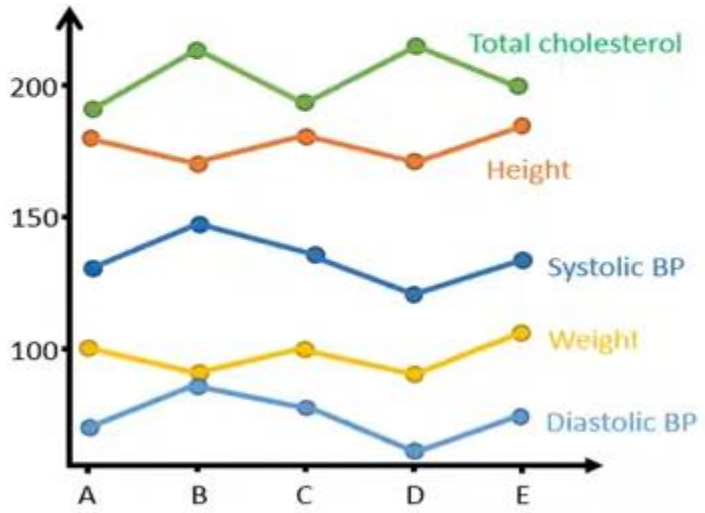


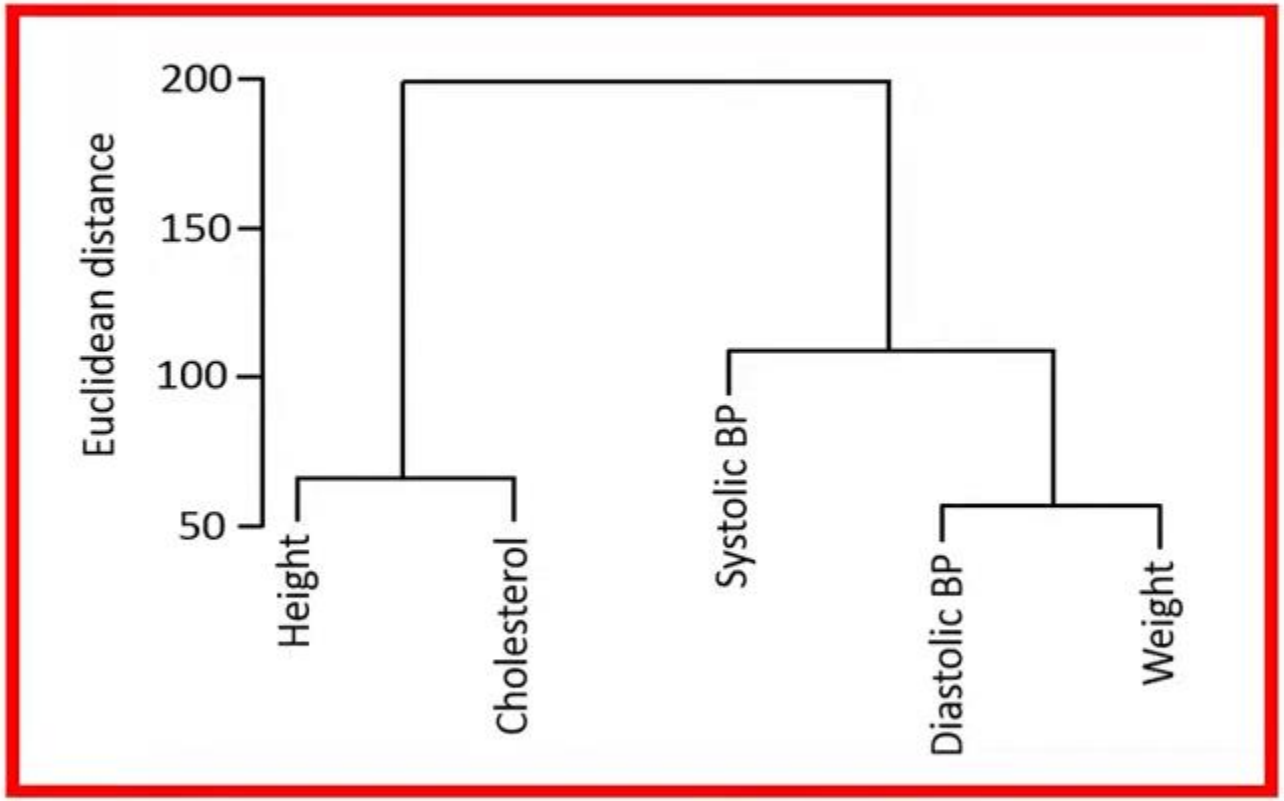
and the distance between the cholesterol variable and the three variables in the other cluster.

Interpret dendrogram

Distance matrix:

	Diastolic BP	Systolic BP	Weight	Height
Systolic BP	134.2			
Weight	57.1	85.2		
Height	232.8	100.3	179.3	
Total cholesterol	289.3	156.3	238.0	66.4



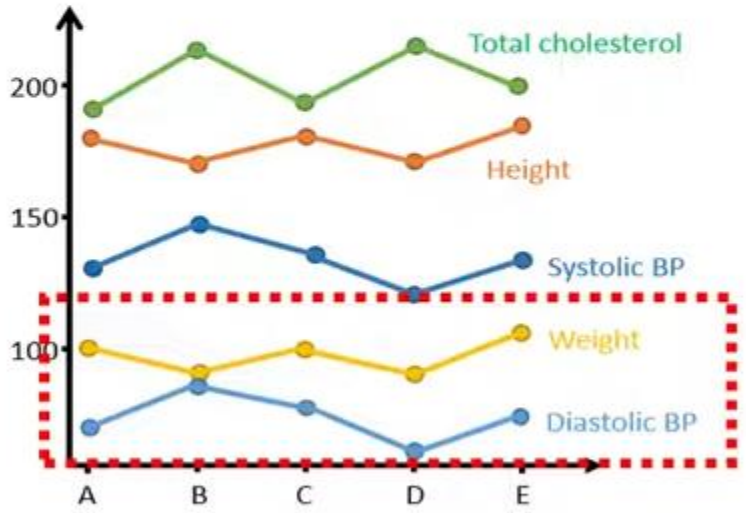


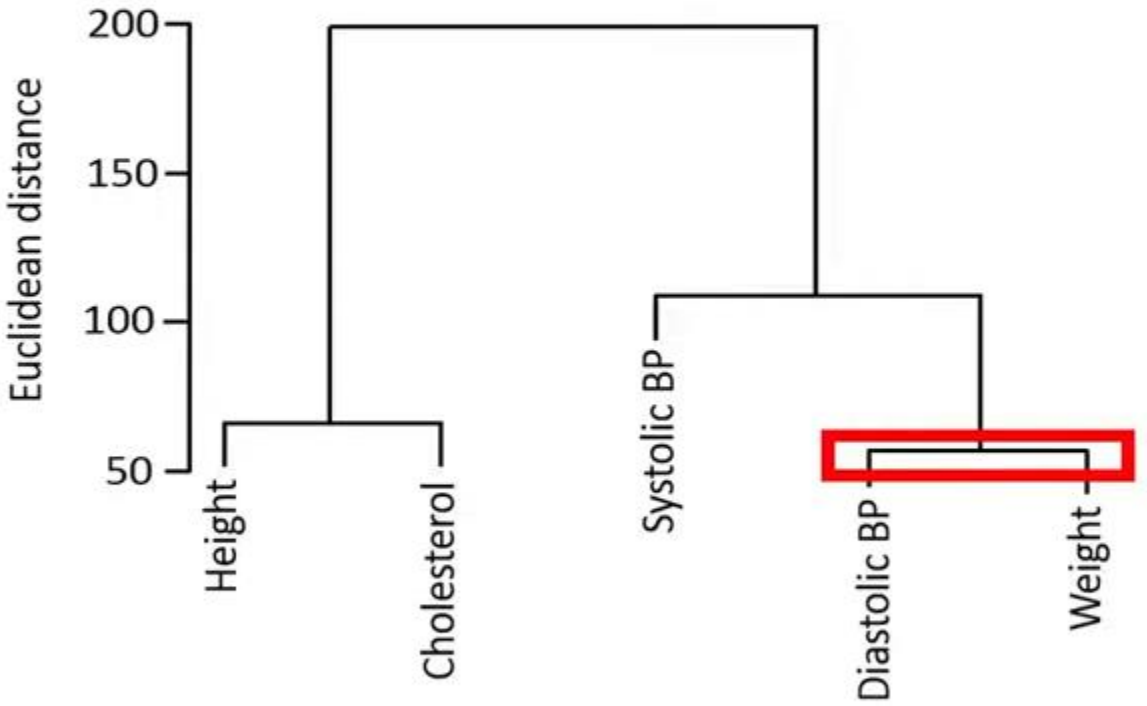
So how do we interpret this dendrogram?

Interpret dendrogram

Distance matrix:

	Diastolic BP	Systolic BP	Weight	Height
Systolic BP	134.2			
Weight	57.1	85.2		
Height	232.8	100.3	179.3	
Total cholesterol	289.3	156.3	238.0	66.4



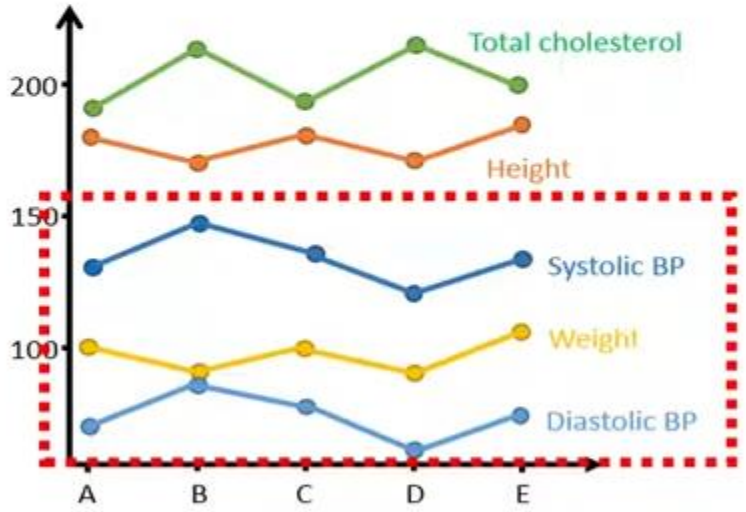


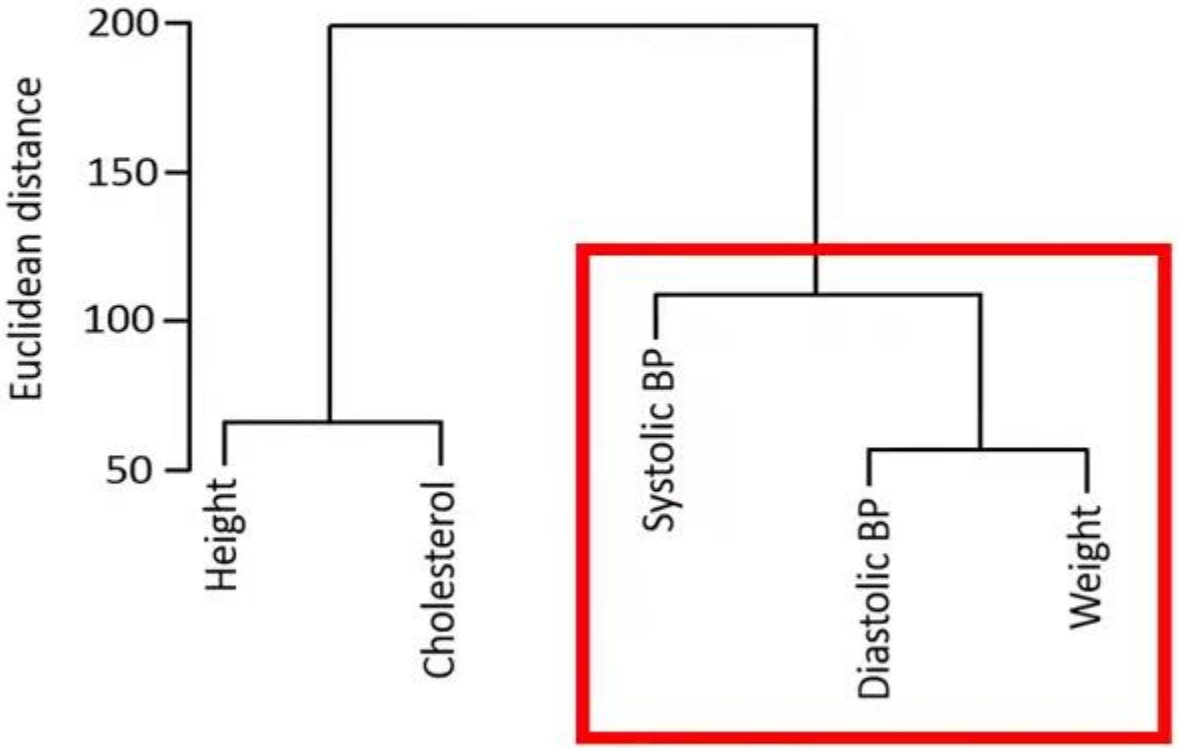
Since the branch that connects these two variables are at the bottom of the dendrogram, we know that these two variables are the most similar out of all possible pairs.

Interpret dendrogram

Distance matrix:

	Diastolic BP	Systolic BP	Weight	Height
Systolic BP	134.2			
Weight	57.1	85.2		
Height	232.8	100.3	179.3	
Total cholesterol	289.3	156.3	238.0	66.4



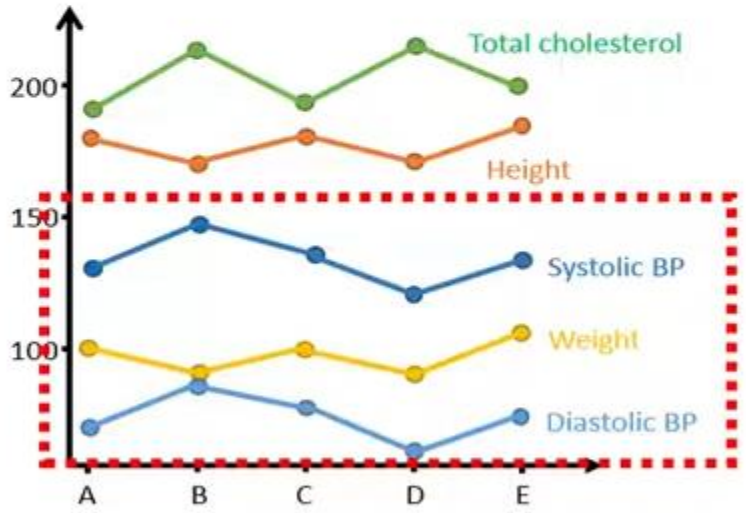


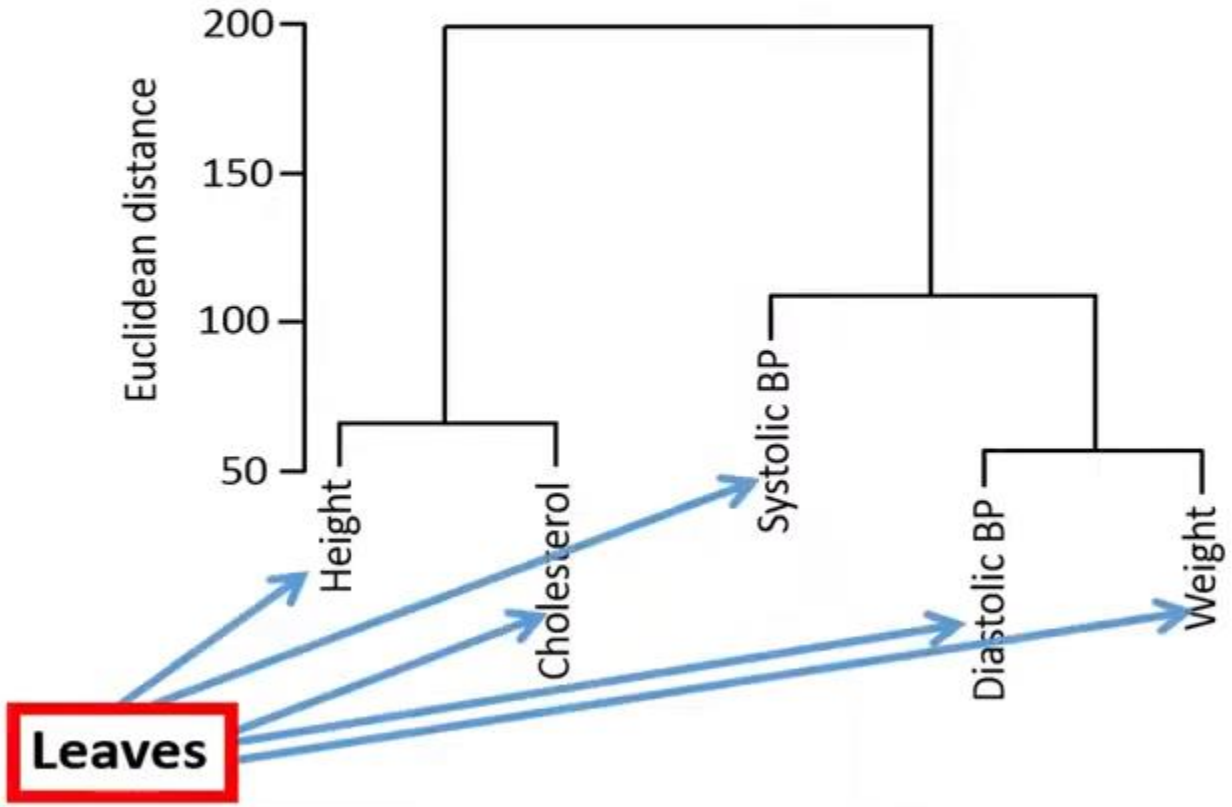
We also see that the systolic blood pressure is more similar to this cluster than the clustering involving the variables height and cholesterol level.

Interpret dendrogram

Distance matrix:

	Diastolic BP	Systolic BP	Weight	Height
Systolic BP	134.2			
Weight	57.1	85.2		
Height	232.8	100.3	179.3	
Total cholesterol	289.3	156.3	238.0	66.4



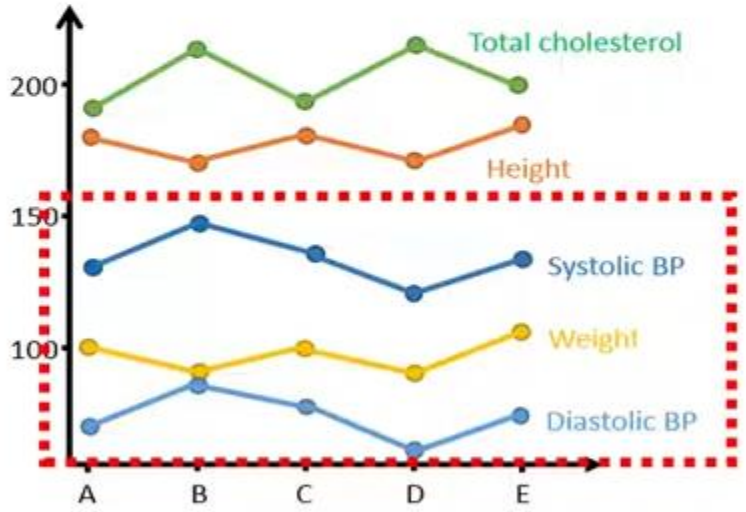


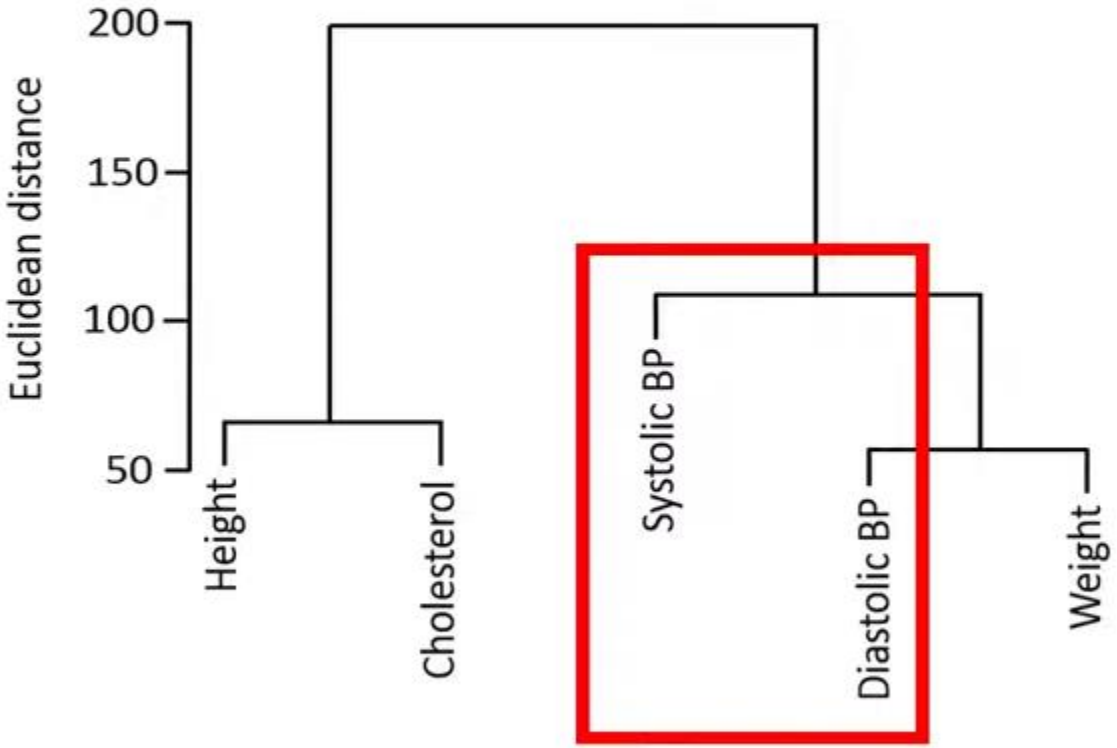
However, the ordering of the leaves in this dendrogram is irrelevant.

Interpret dendrogram

Distance matrix:

	Diastolic BP	Systolic BP	Weight	Height
Systolic BP	134.2			
Weight	57.1	85.2		
Height	232.8	100.3	179.3	
Total cholesterol	289.3	156.3	238.0	66.4



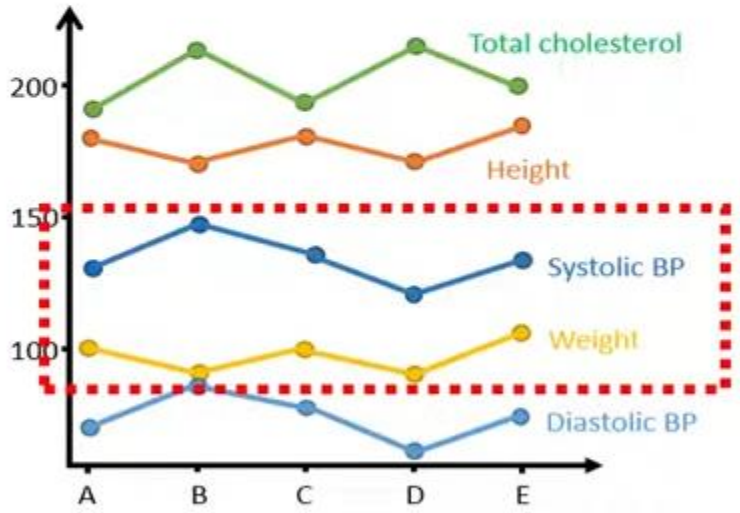


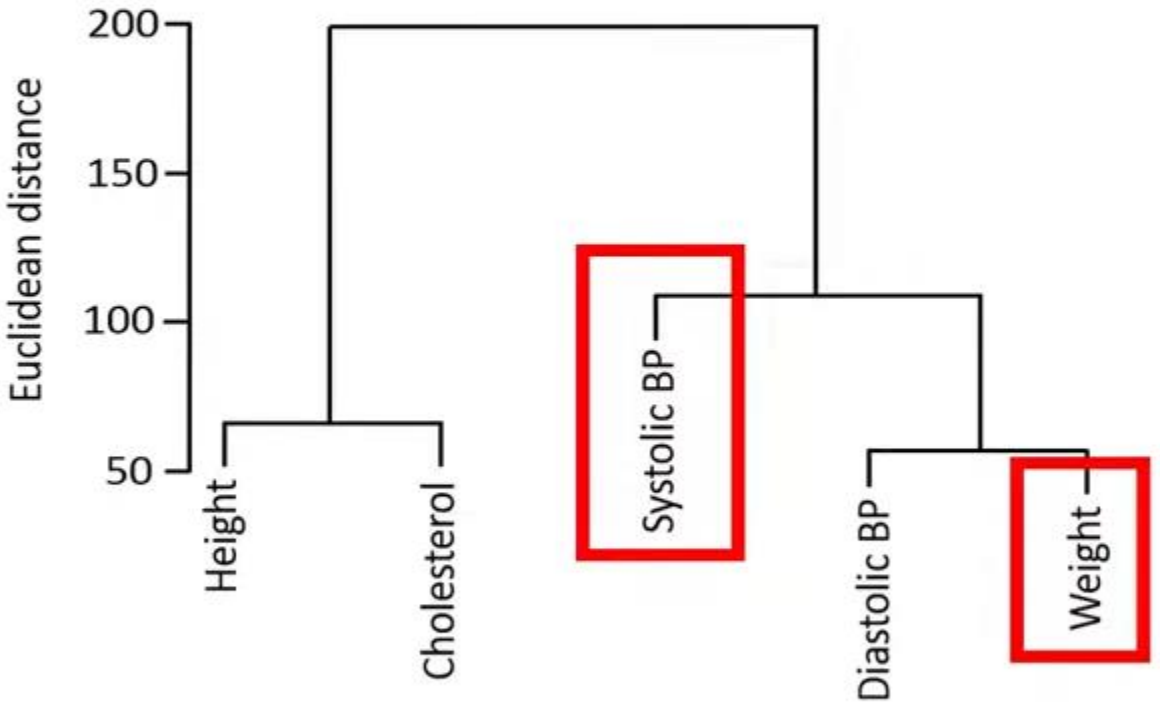
Although the systolic blood pressure is ordered so that it is next to the diastolic blood pressure, it does not mean that it is more similar to this variable.

Interpret dendrogram

Distance matrix:

	Diastolic BP	Systolic BP	Weight	Height
Systolic BP	134.2			
Weight	57.1	85.2		
Height	232.8	100.3	179.3	
Total cholesterol	289.3	156.3	238.0	66.4



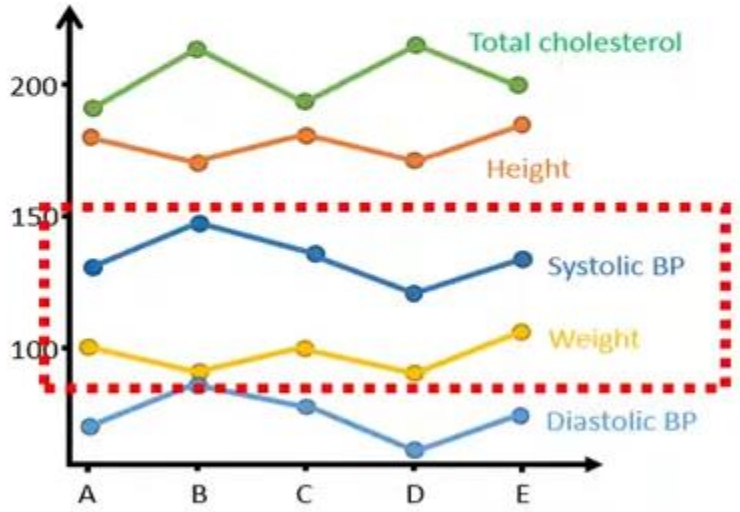


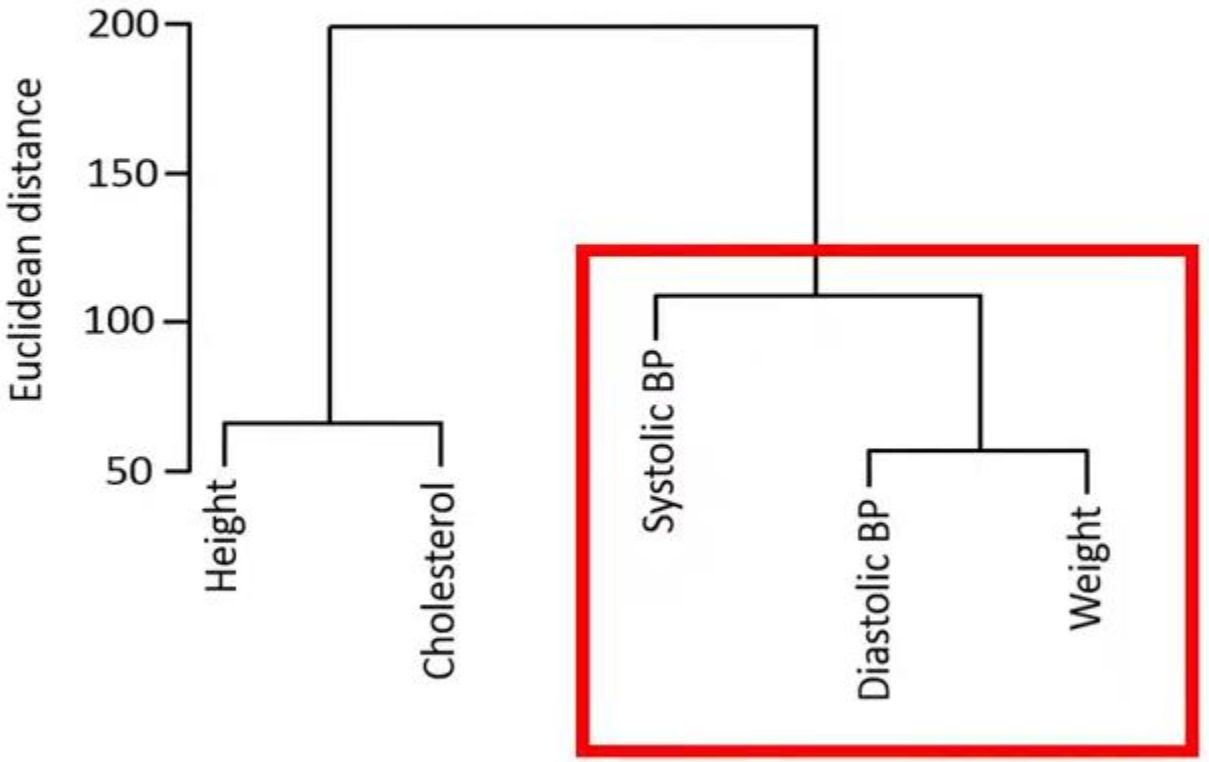
The systolic blood pressure is in fact closer to the body weight if we study the distance matrix.

Interpret dendrogram

Distance matrix:

	Diastolic BP	Systolic BP	Weight	Height
Systolic BP	134.2			
Weight	57.1	85.2		
Height	232.8	100.3	179.3	
Total cholesterol	289.3	156.3	238.0	66.4



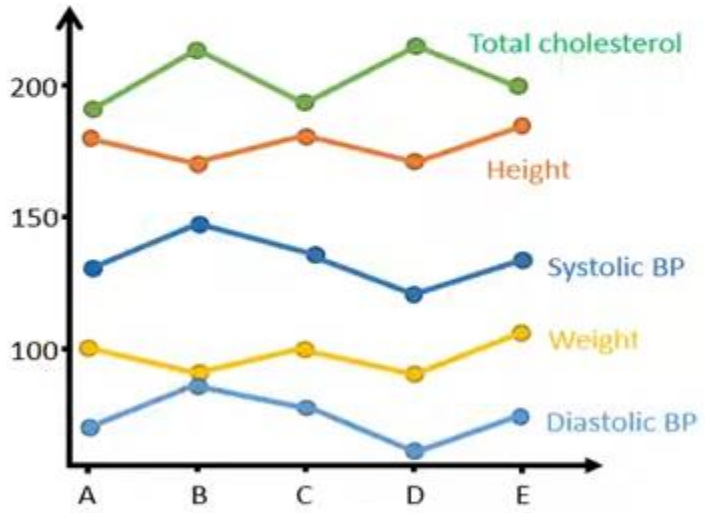


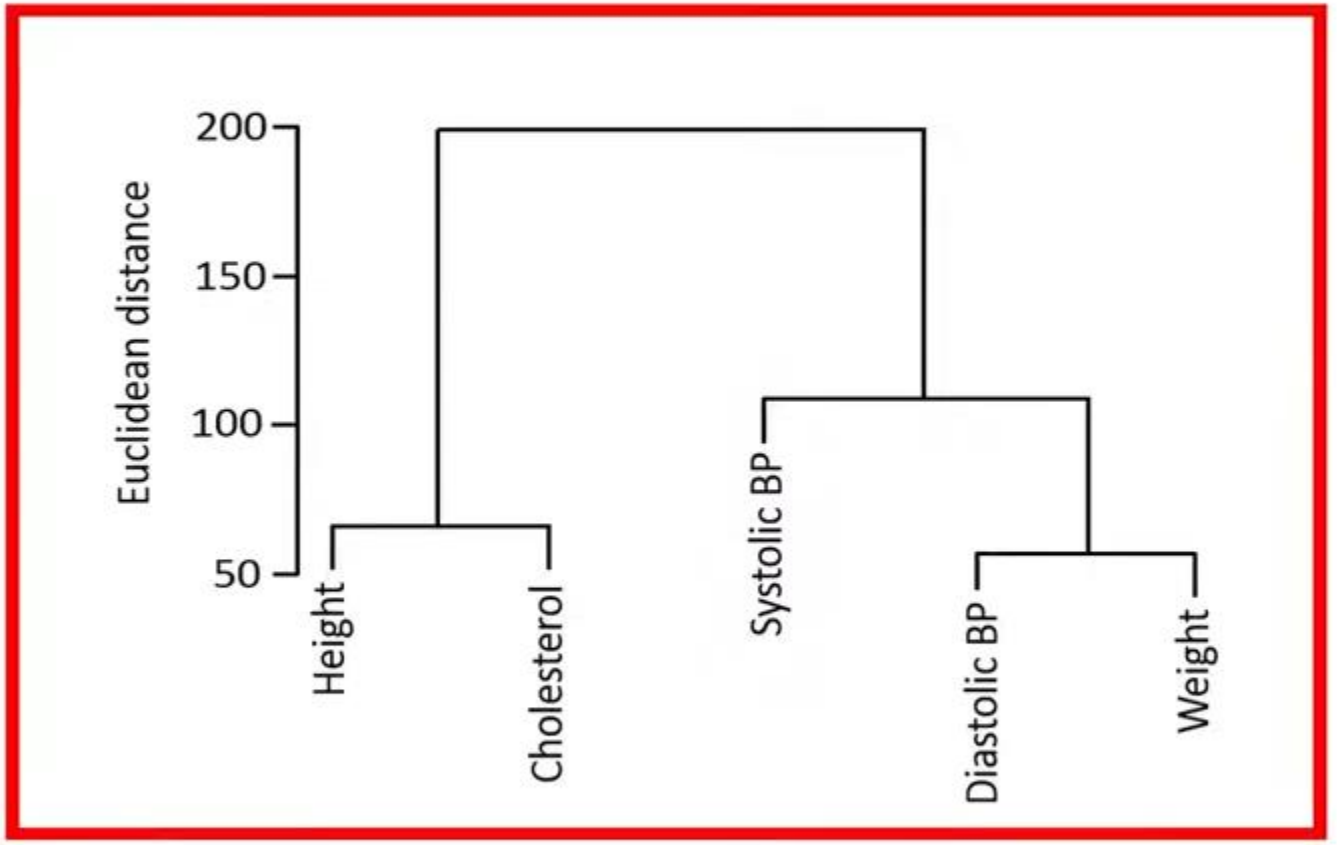
Most software tools order the leaves to make a nice dendrogram. The ordering of the leaves is therefore irrelevant unless your software orders the leaves in a certain way.

Identify clusters

Distance matrix:

	Diastolic BP	Systolic BP	Weight	Height
Systolic BP	134.2			
Weight	57.1	85.2		
Height	232.8	100.3	179.3	
Total cholesterol	289.3	156.3	238.0	66.4



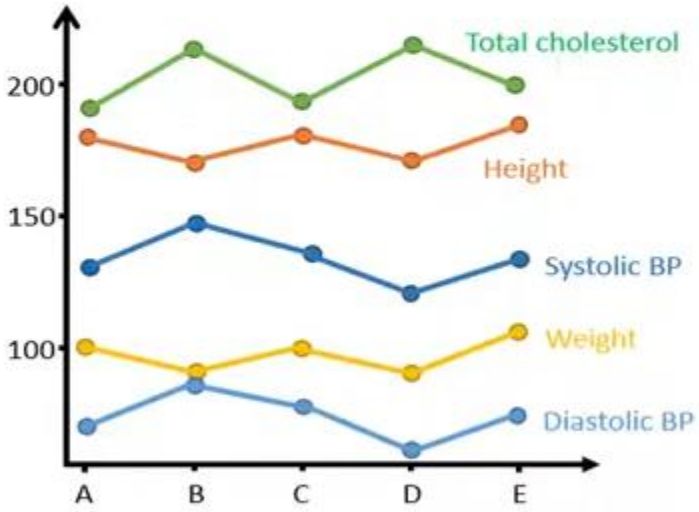


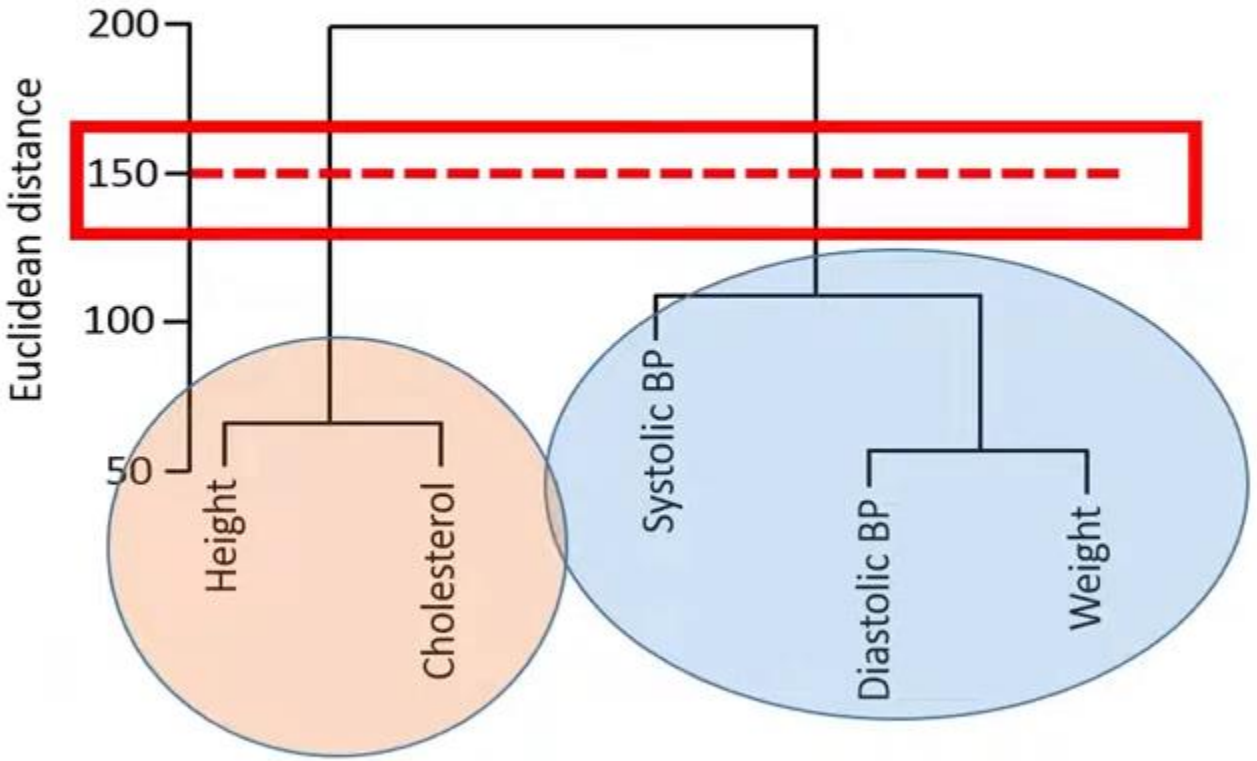
Once we have generated the dendrogram we can identify interesting clusters.

Identify clusters

Distance matrix:

	Diastolic BP	Systolic BP	Weight	Height
Systolic BP	134.2			
Weight	57.1	85.2		
Height	232.8	100.3	179.3	
Total cholesterol	289.3	156.3	238.0	66.4



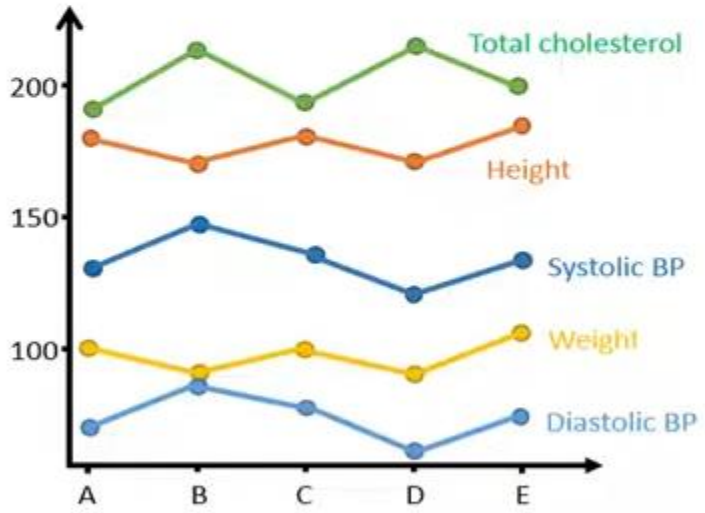


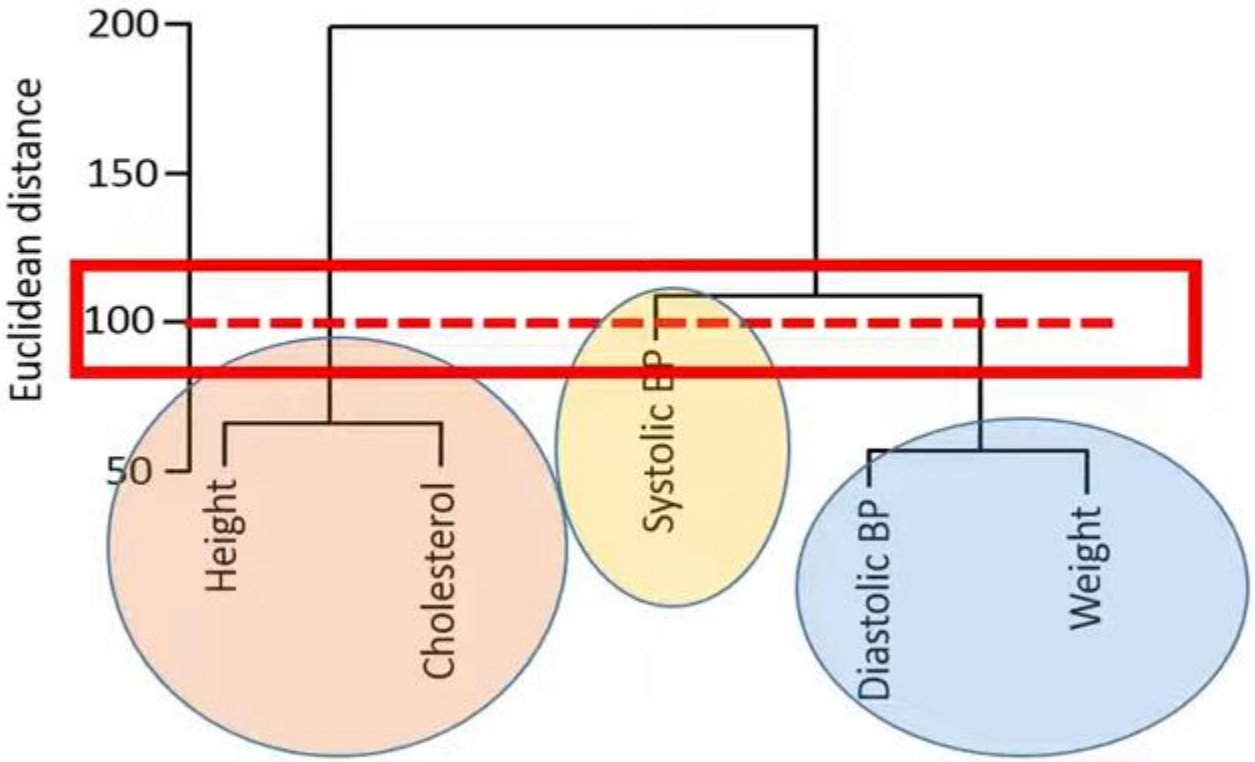
If we set an arbitrary cutoff line at, for example, 150, we see that we have grouped the five variables into two clusters.

Identify clusters

Distance matrix:

	Diastolic BP	Systolic BP	Weight	Height
Systolic BP	134.2			
Weight	57.1	85.2		
Height	232.8	100.3	179.3	
Total cholesterol	289.3	156.3	238.0	66.4



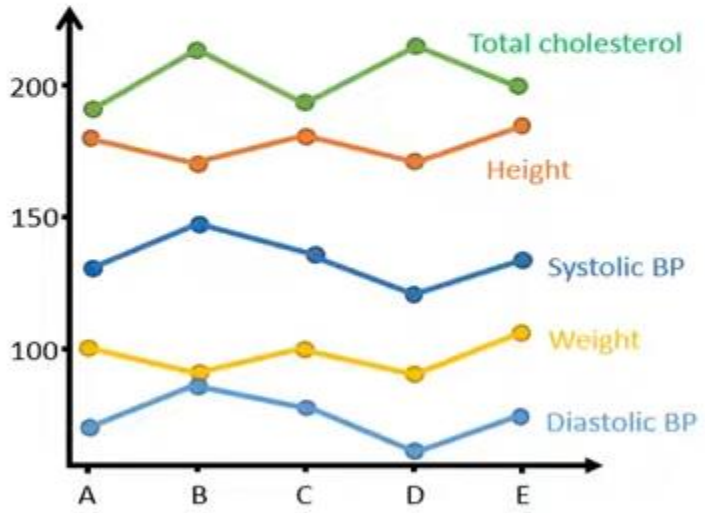


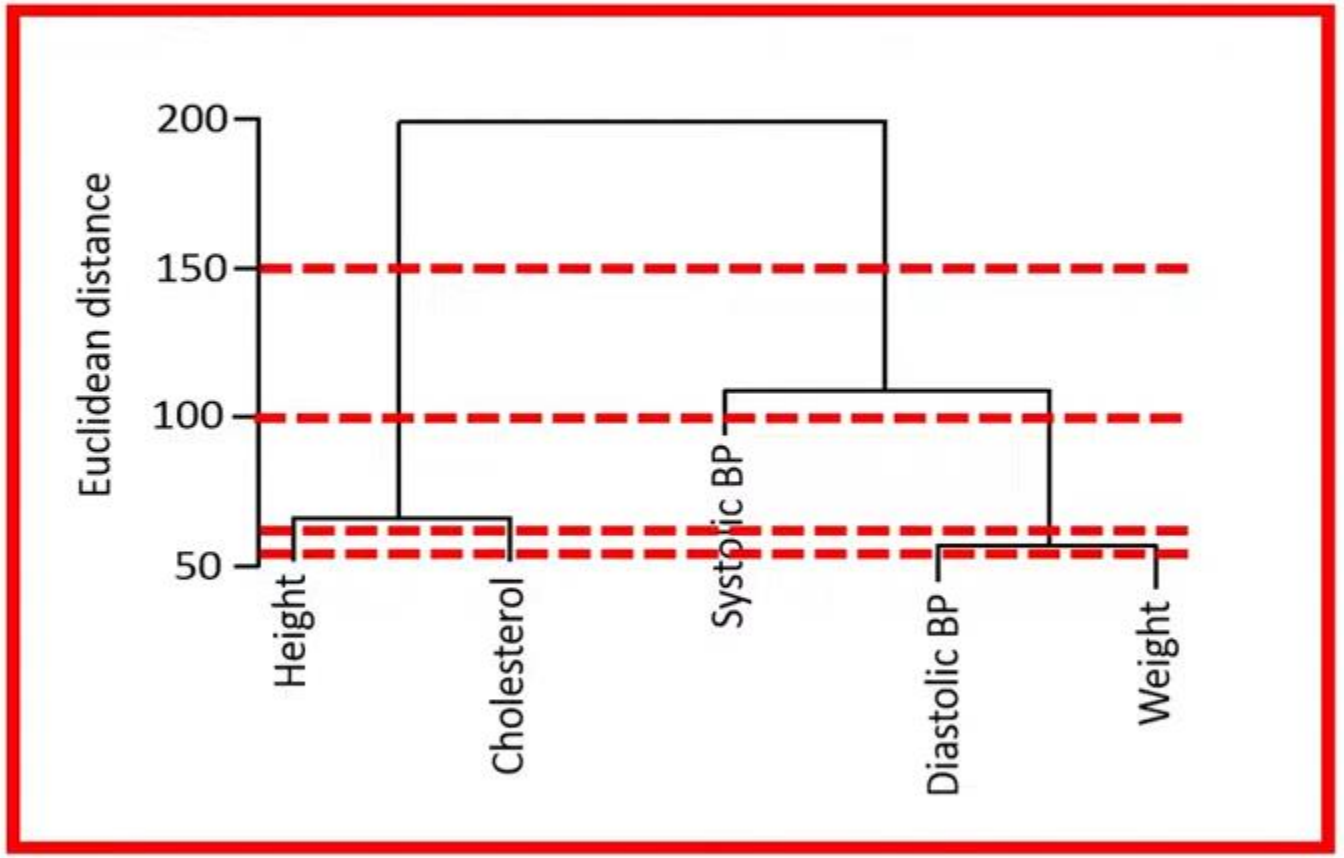
And if we set the cutoff at about 100, we will group the variables into three distinct clusters.

Identify clusters

Distance matrix:

	Diastolic BP	Systolic BP	Weight	Height
Systolic BP	134.2			
Weight	57.1	85.2		
Height	232.8	100.3	179.3	
Total cholesterol	289.3	156.3	238.0	66.4



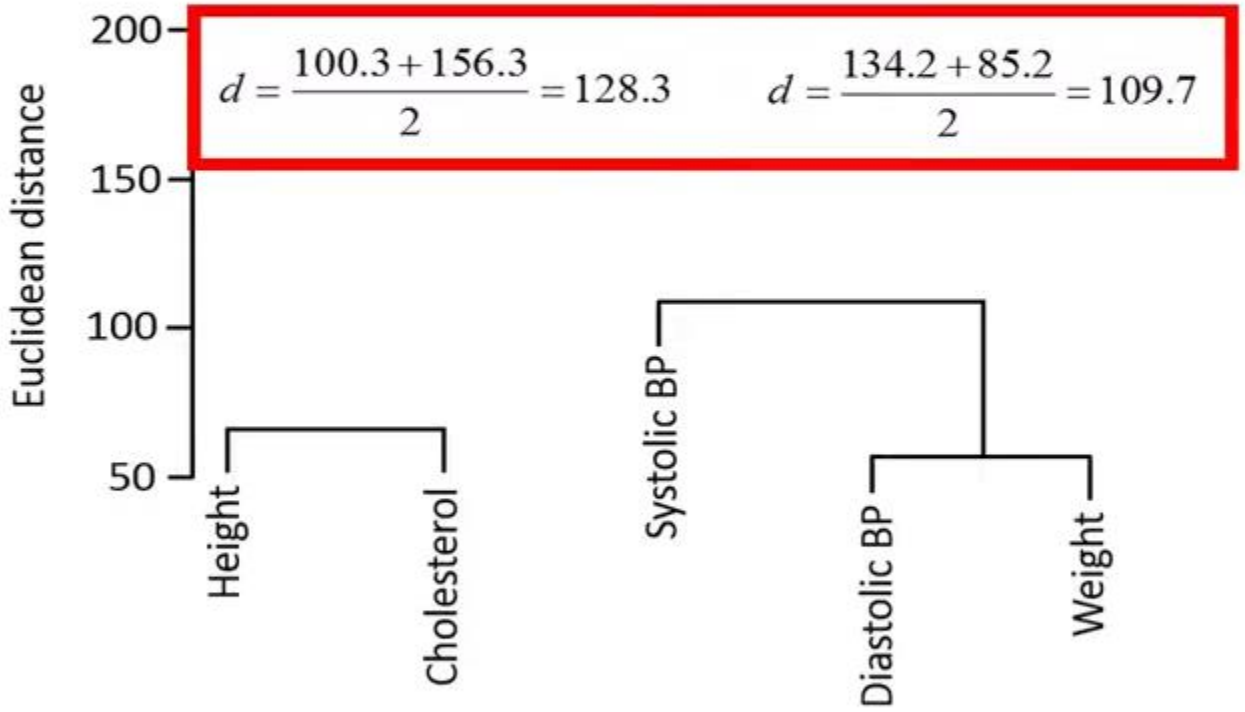


The cutoff value that we select will determine how many clusters we will generate. The cutoff value is determined by the user and depends on how many clusters the user wants to identify.

Linkage function – mean (UPGMA)

Distance matrix:

	Diastolic BP	Systolic BP	Weight	Height
Systolic BP	134.2			
Weight	57.1	85.2		
Height	232.8	100.3	179.3	
Total cholesterol	289.3	156.3	238.0	66.4

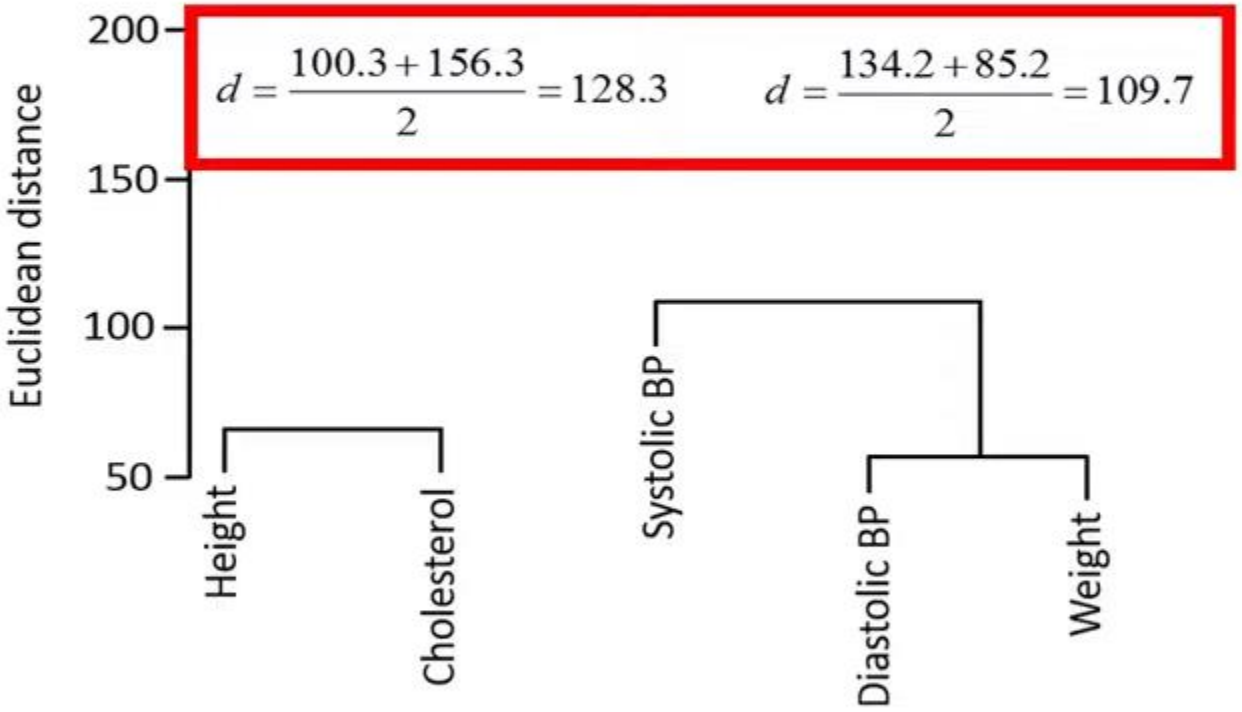


Remember that we previously determined that the systolic blood pressure should form a cluster with the two variables diastolic blood pressure and weight because the mean distance to that cluster was shorter compared to the other cluster.

Linkage function – median

Distance matrix:

	Diastolic BP	Systolic BP	Weight	Height
Systolic BP	134.2			
Weight	57.1	85.2		
Height	232.8	100.3	179.3	
Total cholesterol	289.3	156.3	238.0	66.4

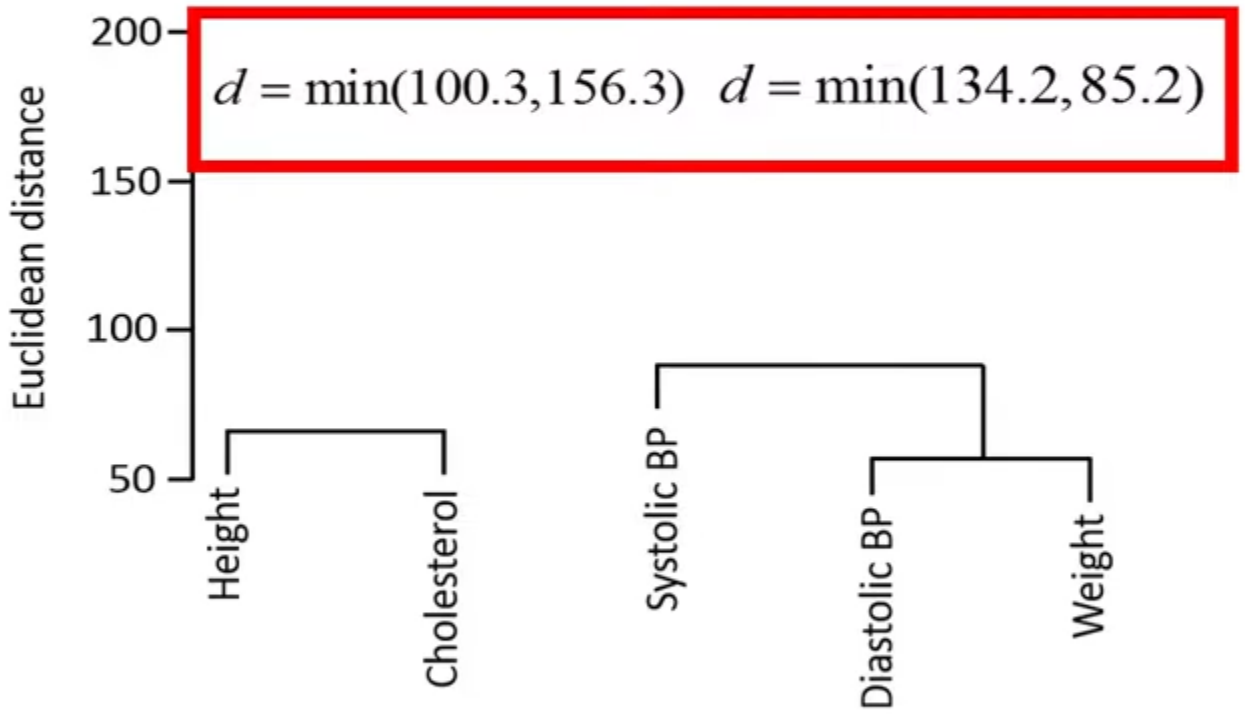


However, we could instead have used the median value, which in this case would give the same result because we only have two numbers. The measure that we use is called linkage function.

Single linkage function

Distance matrix:

	Diastolic BP	Systolic BP	Weight	Height
Systolic BP	134.2			
Weight	57.1	85.2		
Height	232.8	100.3	179.3	
Total cholesterol	289.3	156.3	238.0	66.4



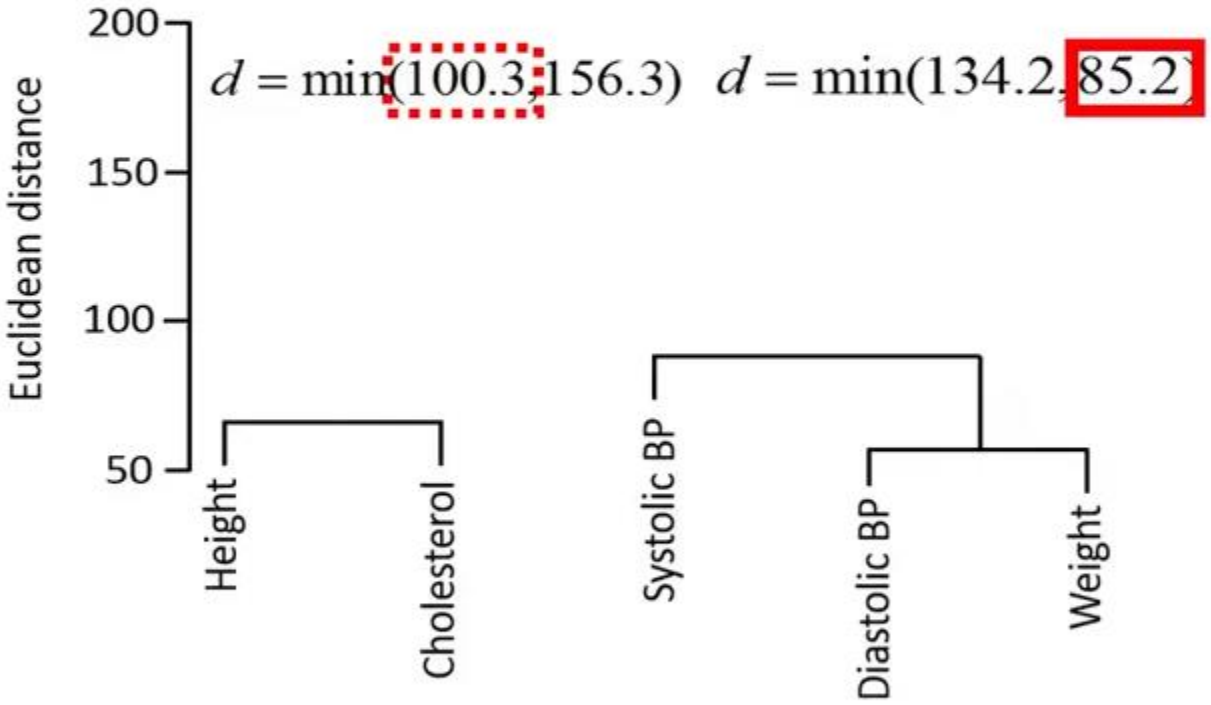
Another way is to use the so-called single-linkage function, where we check the minimum distance to the clusters.

Single linkage function

Distance matrix:

	Diastolic BP	Systolic BP	Weight	Height
Systolic BP	134.2			
Weight	57.1	85.2		
Height	232.8	100.3	179.3	
Total cholesterol	289.3	156.3	238.0	66.4

$d = \min(100.3, 156.3)$ $d = \min(134.2, 85.2)$

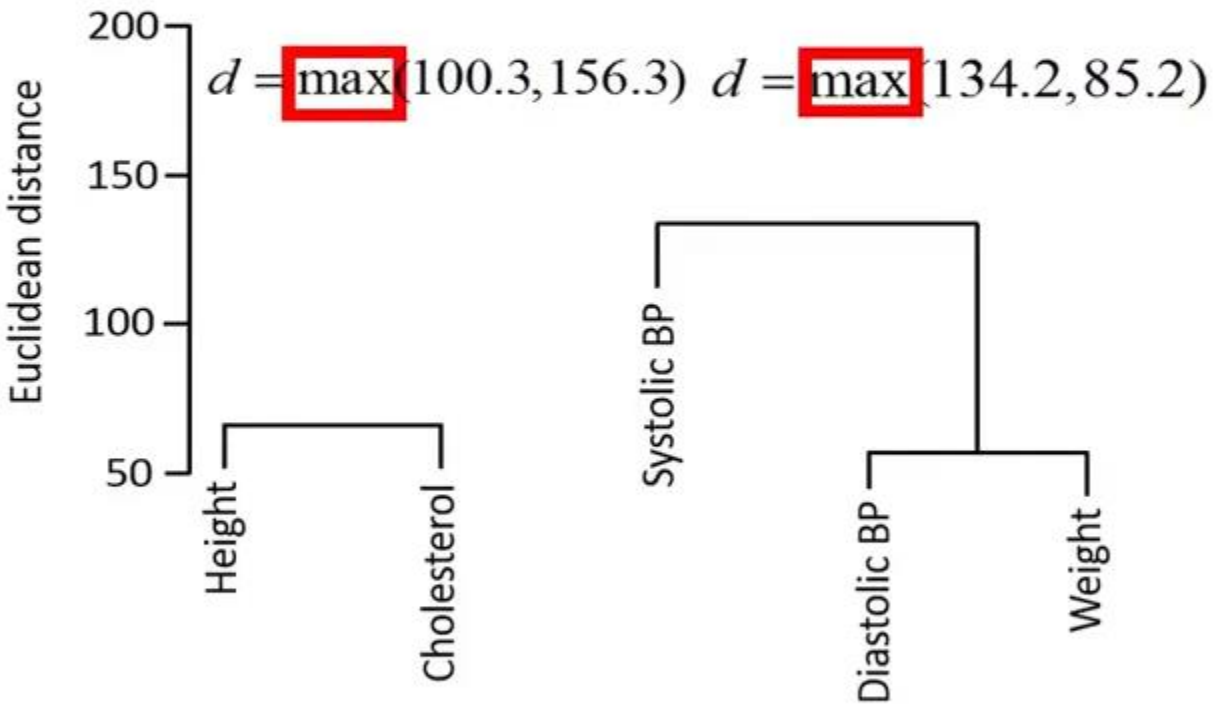


Since the shortest minimum distance between the systolic blood pressure and the variables in the two clusters is to the body weight, the variable systolic blood pressure would still group with the second cluster.

Complete linkage function

Distance matrix:

	Diastolic BP	Systolic BP	Weight	Height
Systolic BP	134.2			
Weight	57.1	85.2		
Height	232.8	100.3	179.3	
Total cholesterol	289.3	156.3	238.0	66.4

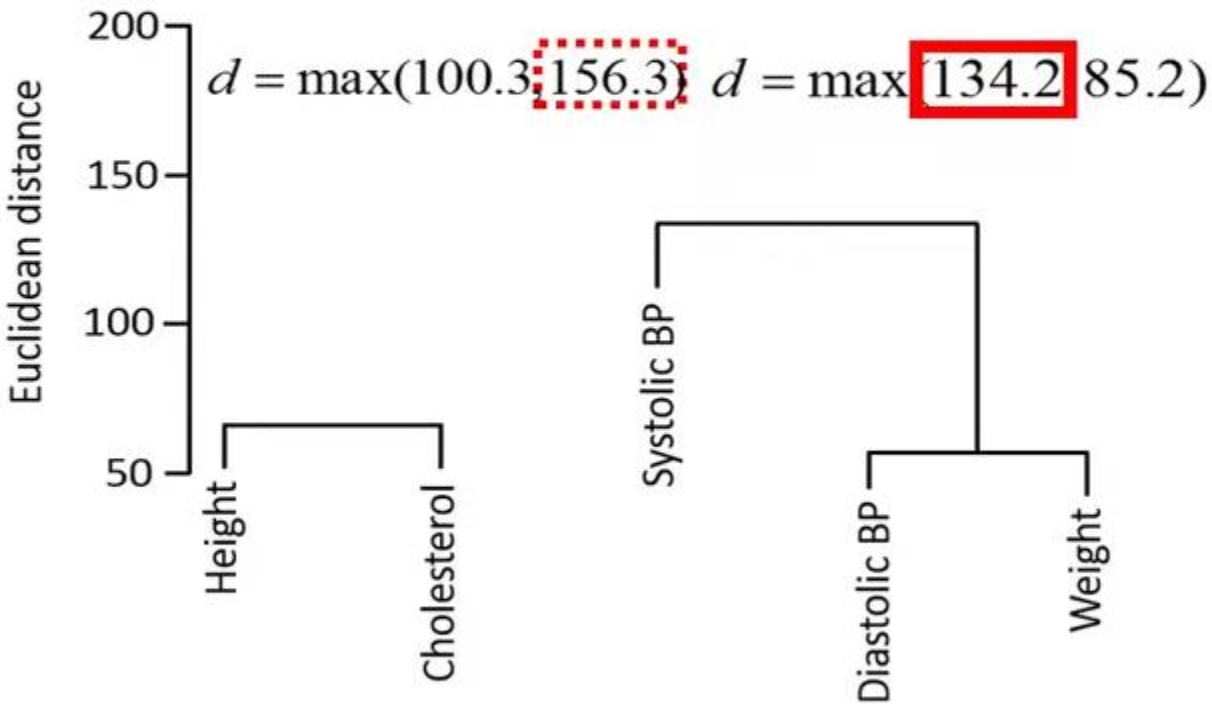


Another option is to use the so-called complete-linkage function, where we check the maximum distance to the clusters.

Complete linkage function

Distance matrix:

	Diastolic BP	Systolic BP	Weight	Height
Systolic BP	134.2			
Weight	57.1	85.2		
Height	232.8	100.3	179.3	
Total cholesterol	289.3	156.3	238.0	66.4

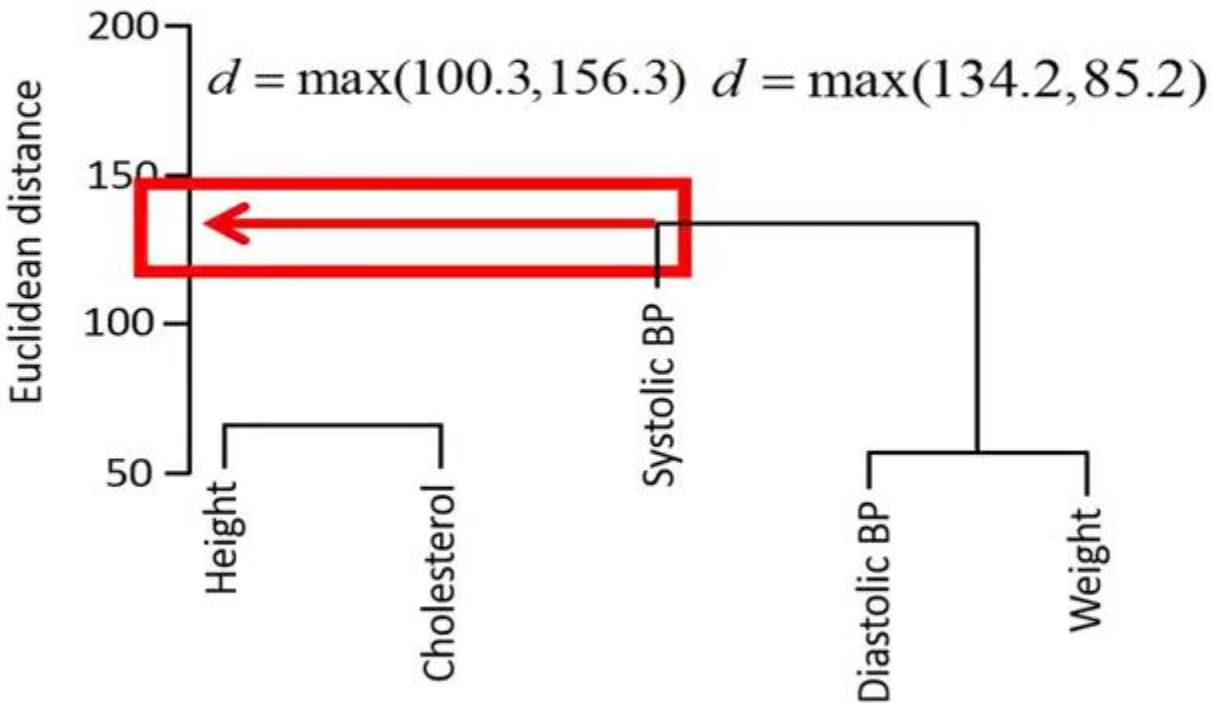


Since the shortest maximum distance between the systolic blood pressure and the variables in the two clusters is to the diastolic blood pressure, the systolic blood pressure would again group with the second cluster.

Complete linkage function

Distance matrix:

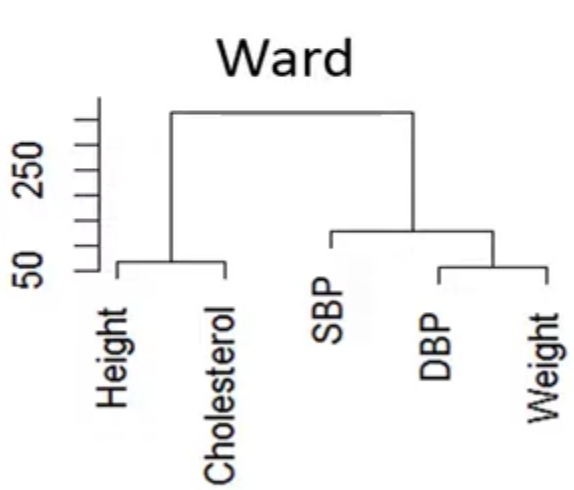
	Diastolic BP	Systolic BP	Weight	Height
Systolic BP	134.2			
Weight	57.1	85.2		
Height	232.8	100.3	179.3	
Total cholesterol	289.3	156.3	238.0	66.4

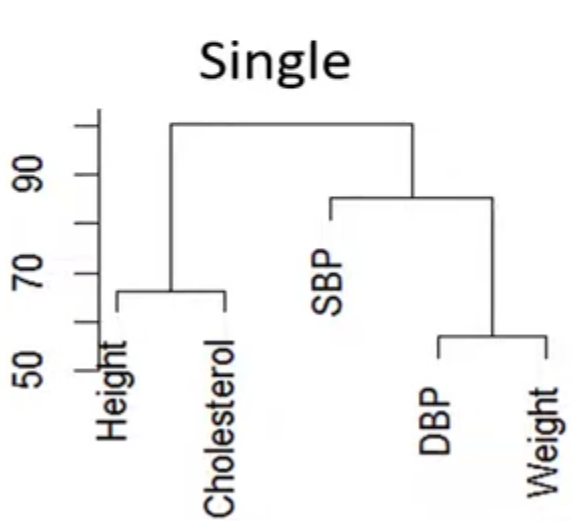


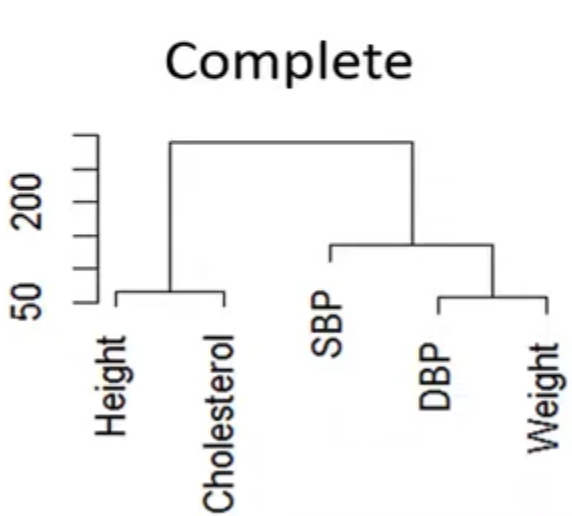
Note that the height of this part of the dendrogram now represents the maximum distance, which is 134.2.

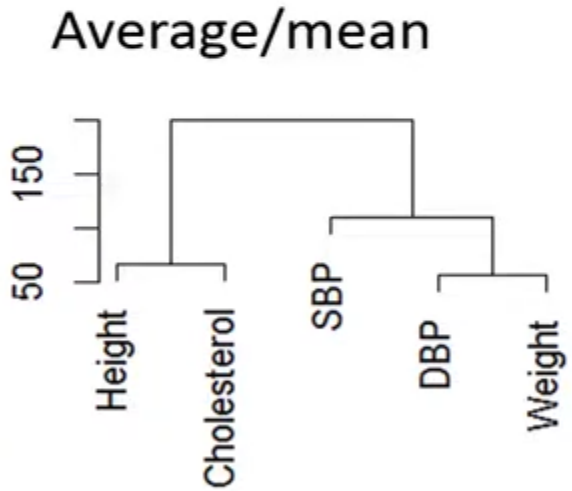
Select a linkage function

There are a number of different linkage functions to select between. You can try many different functions and select the one that generates a dendrogram that makes sense according to the aim of your study.



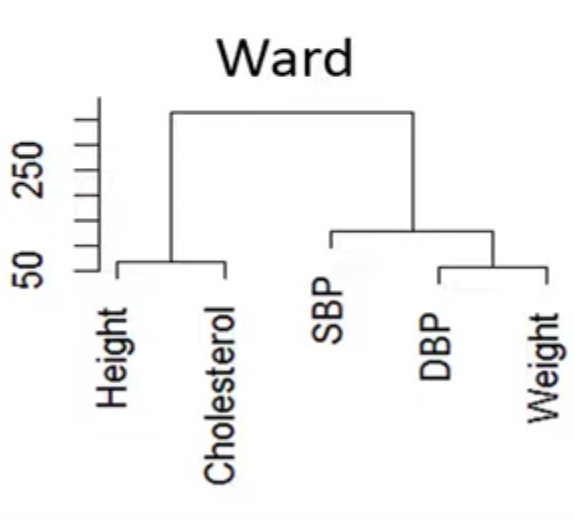


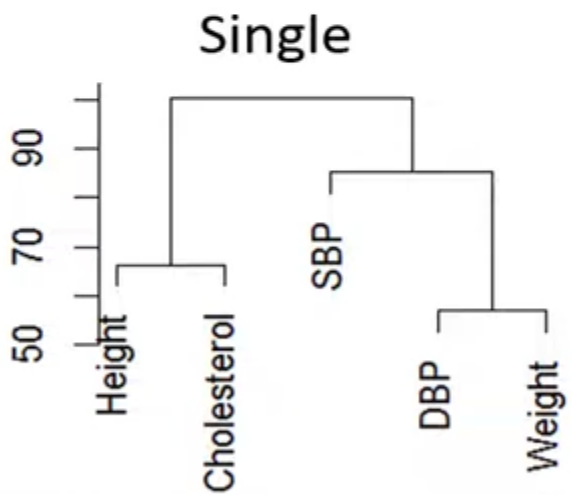


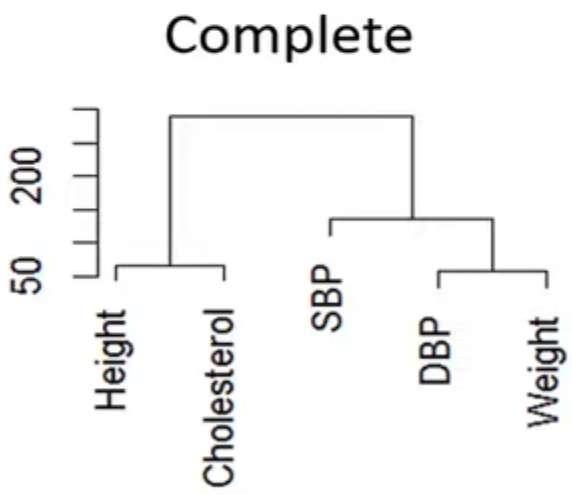


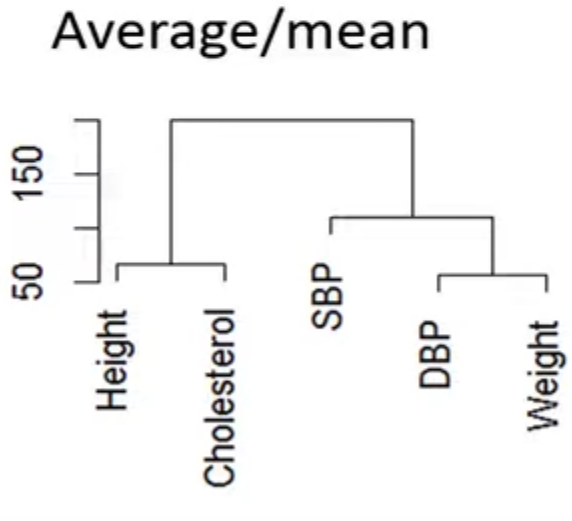
Select a linkage function

Another way is to use some sort of validation technique, where the linkage function that, for example, generates the most robust cluster is selected.







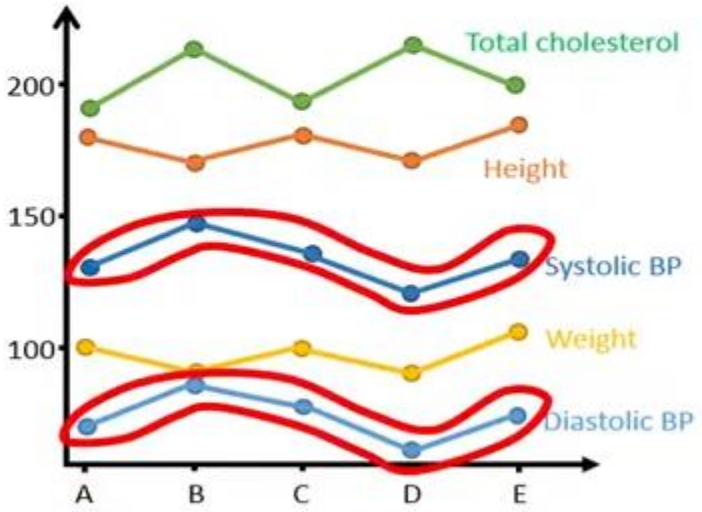


Distance metrics

In addition to different linkage functions, it is important to select an appropriate distance metric that reflects how we define similarity.

Distance metrics

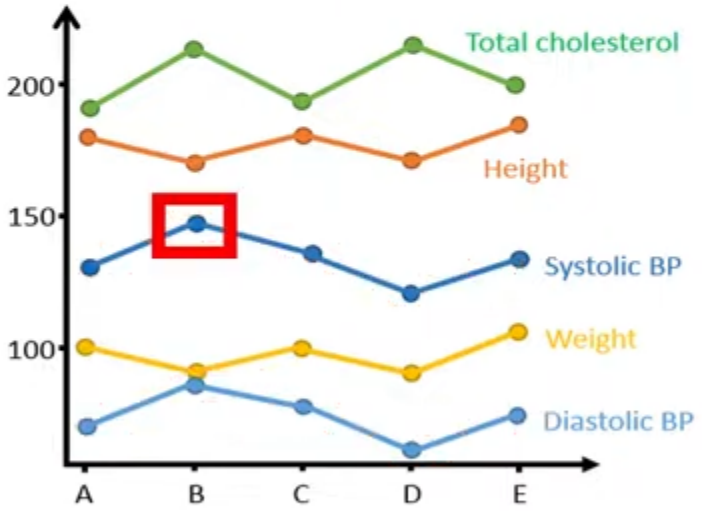
	A	B	C	D	E
Diastolic BP (mmHg)	70	86	78	60	75
Systolic BP (mmHg)	130	147	137	121	134
Weight (kg)	100	90	100	90	106
Height (cm)	180	170	181	171	185
Total cholesterol (mg/dl)	190	215	192	215	200



However, it might make more sense to group the diastolic blood pressure with the systolic blood pressure because these two variables have similar patterns across the individuals.

Distance metrics

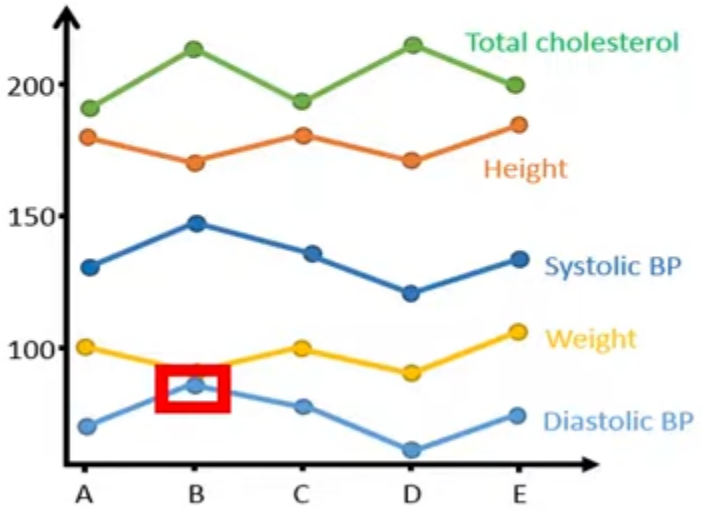
	A	B	C	D	E
Diastolic BP (mmHg)	70	86	78	60	75
Systolic BP (mmHg)	130	147	137	121	134
Weight (kg)	100	90	100	90	106
Height (cm)	180	170	181	171	185
Total cholesterol (mg/dl)	190	215	192	215	200



If one individual has a high systolic blood pressure,

Distance metrics

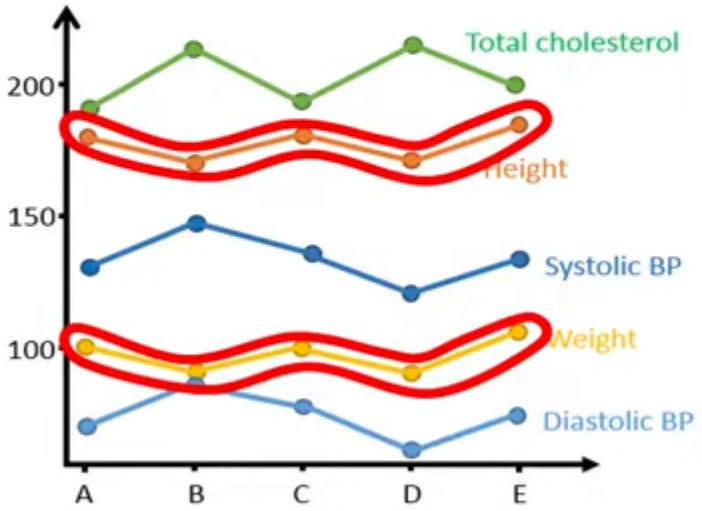
	A	B	C	D	E
Diastolic BP (mmHg)	70	86	78	60	75
Systolic BP (mmHg)	130	147	137	121	134
Weight (kg)	100	90	100	90	106
Height (cm)	180	170	181	171	185
Total cholesterol (mg/dl)	190	215	192	215	200



the same individual is likely to also have a high diastolic blood pressure.

Distance metrics

	A	B	C	D	E
Diastolic BP (mmHg)	70	86	78	60	75
Systolic BP (mmHg)	130	147	137	121	134
Weight (kg)	100	90	100	90	106
Height (cm)	180	170	181	171	185
Total cholesterol (mg/dl)	190	215	192	215	200

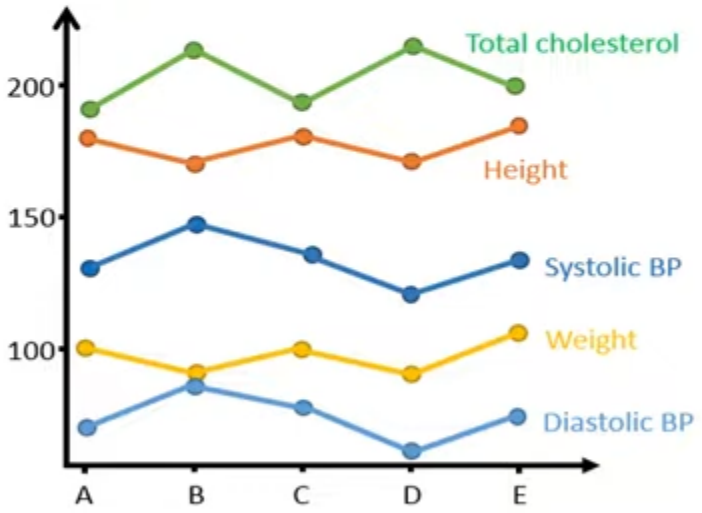


The same is also true for the body weight and body height.

Standardize the variables

	A	B	C	D	E
Diastolic BP (mmHg)	70	86	78	60	75
Systolic BP (mmHg)	130	147	137	121	134
Weight (kg)	100	90	100	90	106
Height (cm)	180	170	181	171	185
Total cholesterol (mg/dl)	190	215	192	215	200

$$Z = \frac{x_i - \bar{x}}{SD}$$

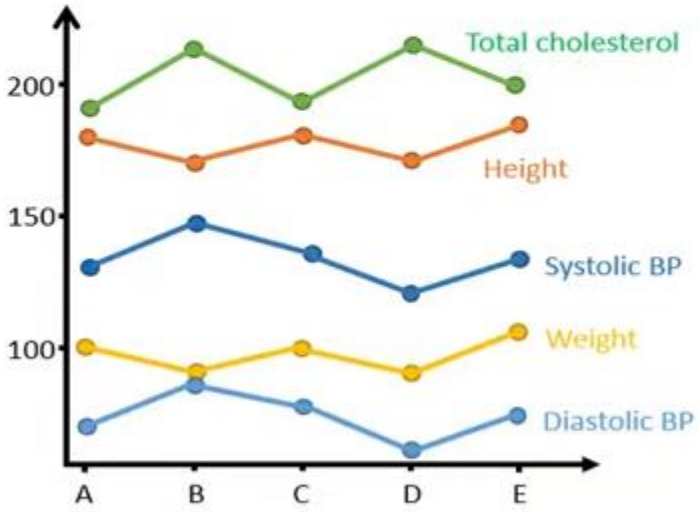


One way to group the variables based on similarity in their pattern, or profile, is to first standardize the variables so that all variables have a mean of zero and a standard deviation of one.

Standardize the variables

	A	B	C	D	E
Diastolic BP (mmHg)	70	86	78	60	75
Systolic BP (mmHg)	130	147	137	121	134
Weight (kg)	100	90	100	90	106
Height (cm)	180	170	181	171	185
Total cholesterol (mg/dl)	190	215	192	215	200

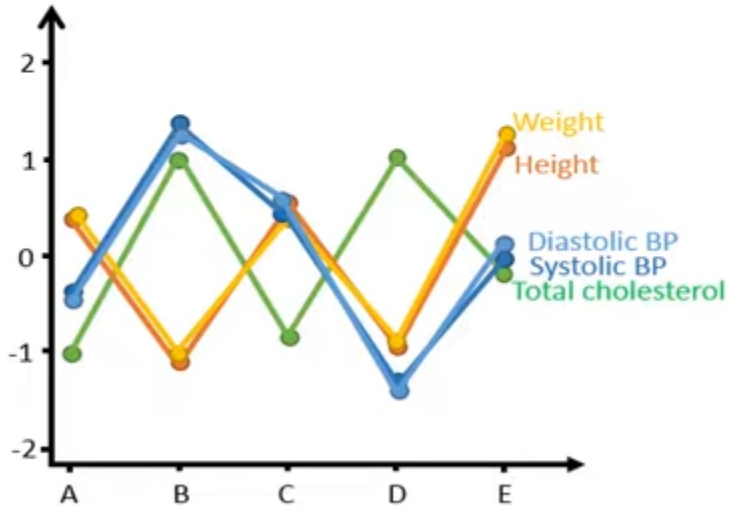
$$Z = \frac{x_i - \bar{x}}{SD}$$



For example, if we would standardize the variable diastolic blood pressure, we should subtract the mean diastolic blood pressure from these values and divide by the standard deviation.

Standardize the variables

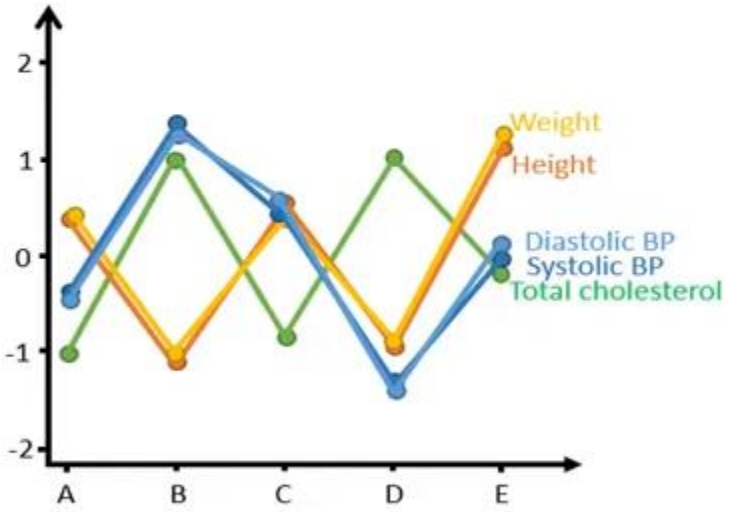
	A	B	C	D	E
Diastolic BP (mmHg)	-0.39	1.26	0.44	-1.43	0.12
Systolic BP (mmHg)	-0.40	1.39	0.34	-1.34	0.02
Weight (kg)	0.40	-1.03	0.40	-1.03	1.25
Height (cm)	0.40	-1.12	0.55	-0.97	1.15
Total cholesterol (mg/dl)	-1.03	1.04	-0.86	1.04	-0.20



Standardizing this variable results in values that have a mean of zero and a standard deviation of one.

Standardize the variables

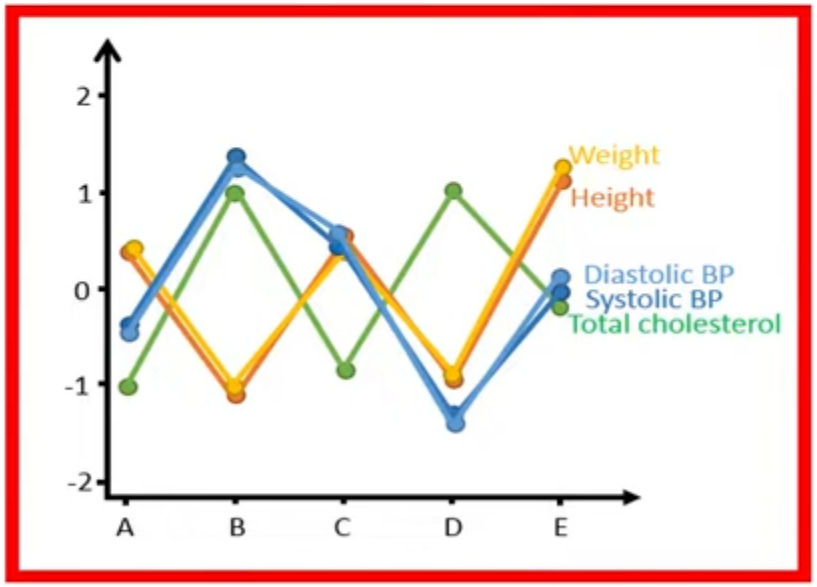
	A	B	C	D	E
Diastolic BP (mmHg)	-0.39	1.26	0.44	-1.43	0.12
Systolic BP (mmHg)	-0.40	1.39	0.34	-1.34	0.02
Weight (kg)	0.40	-1.03	0.40	-1.03	1.25
Height (cm)	0.40	-1.12	0.55	-0.97	1.15
Total cholesterol (mg/dl)	-1.03	1.04	-0.86	1.04	-0.20



For example, this value tells us that person B has a diastolic blood pressure that is 1.26 standard deviations greater than the mean diastolic blood pressure of the five individuals.

Standardize the variables

	A	B	C	D	E
Diastolic BP (mmHg)	-0.39	1.26	0.44	-1.43	0.12
Systolic BP (mmHg)	-0.40	1.39	0.34	-1.34	0.02
Weight (kg)	0.40	-1.03	0.40	-1.03	1.25
Height (cm)	0.40	-1.12	0.55	-0.97	1.15
Total cholesterol (mg/dl)	-1.03	1.04	-0.86	1.04	-0.20

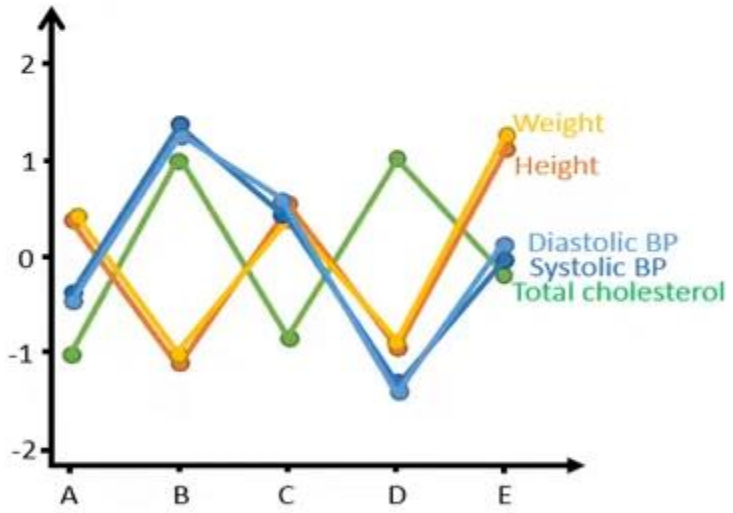


When we standardize the data, variables that have a similar pattern will have a short distance to each other.

Standardize the variables

	A	B	C	D	E
Diastolic BP (mmHg)	-0.39	1.26	0.44	-1.43	0.12
Systolic BP (mmHg)	-0.40	1.39	0.34	-1.34	0.02
Weight (kg)	0.40	-1.03	0.40	-1.03	1.25
Height (cm)	0.40	-1.12	0.55	-0.97	1.15
Total cholesterol (mg/dl)	-1.03	1.04	-0.86	1.04	-0.20

$$d = \sqrt{((-0.39-(-0.4))^2 + (1.26-1.39)^2 + (0.44-0.34)^2 + (-1.43-(-1.34))^2 + (0.12-0.02)^2)} = 0.21$$

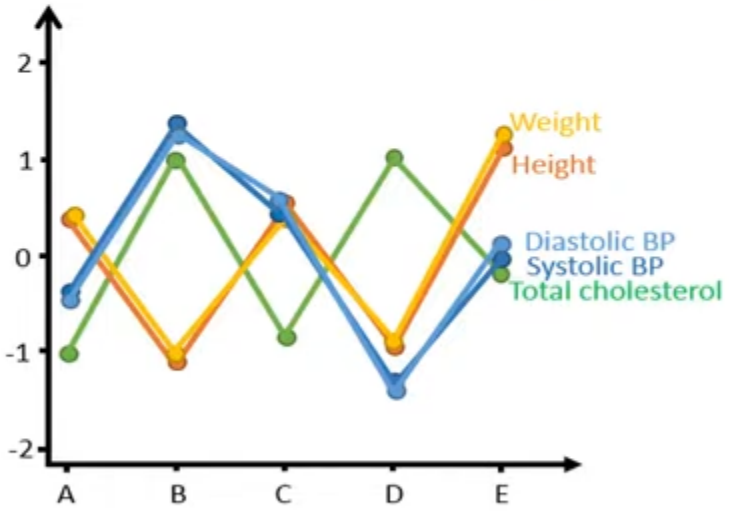


Let's calculate the Euclidean distance based on the standardized variables. For example, the Euclidean distance between the two blood pressure variables is now 0.21.

Standardize the variables

Distance matrix:

	Diastolic BP	Systolic BP	Weight	Height
Systolic BP	0.21			
Weight	2.70	2.84		
Height	2.76	2.90	0.21	
Total cholesterol	2.89	2.77	3.78	3.82

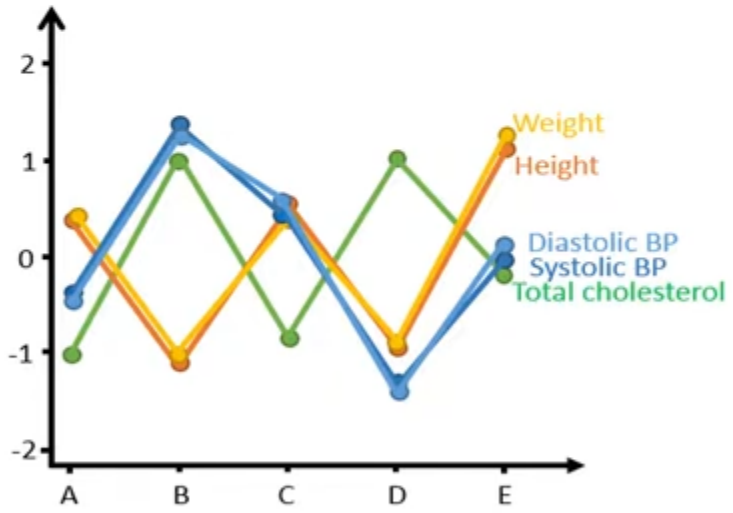


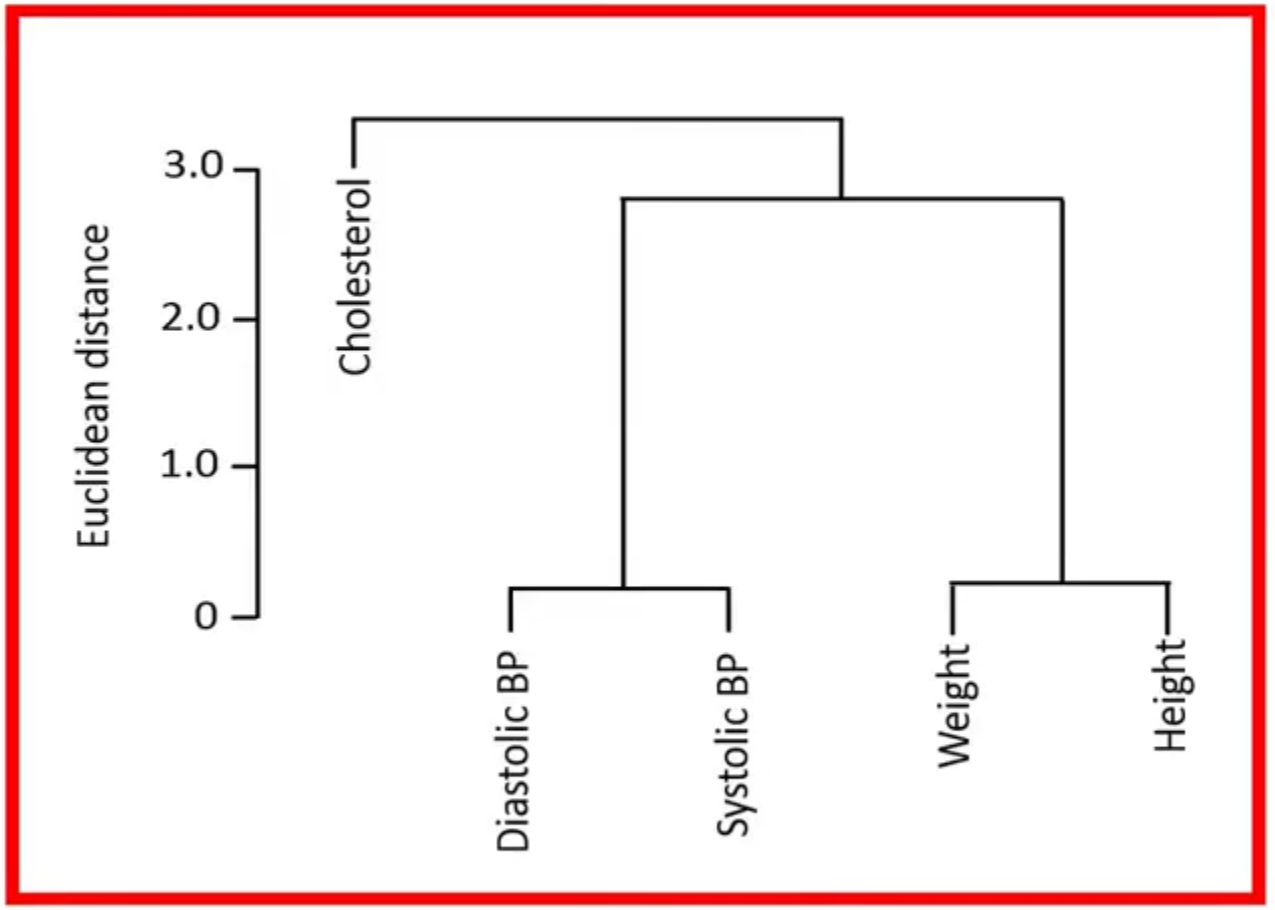
If we calculate the Euclidean distance between all pairs of variables, we will get the following distance matrix.

Standardize the variables

Distance matrix:

	Diastolic BP	Systolic BP	Weight	Height
Systolic BP	0.21			
Weight	2.70	2.84		
Height	2.76	2.90	0.21	
Total cholesterol	2.89	2.77	3.78	3.82



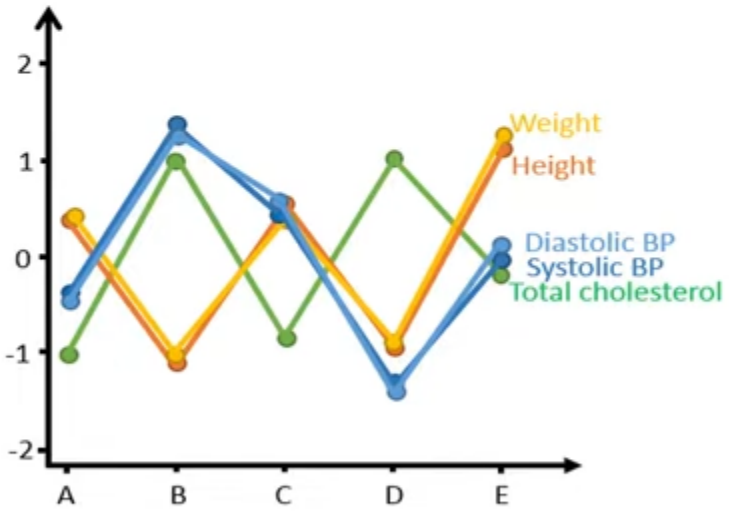


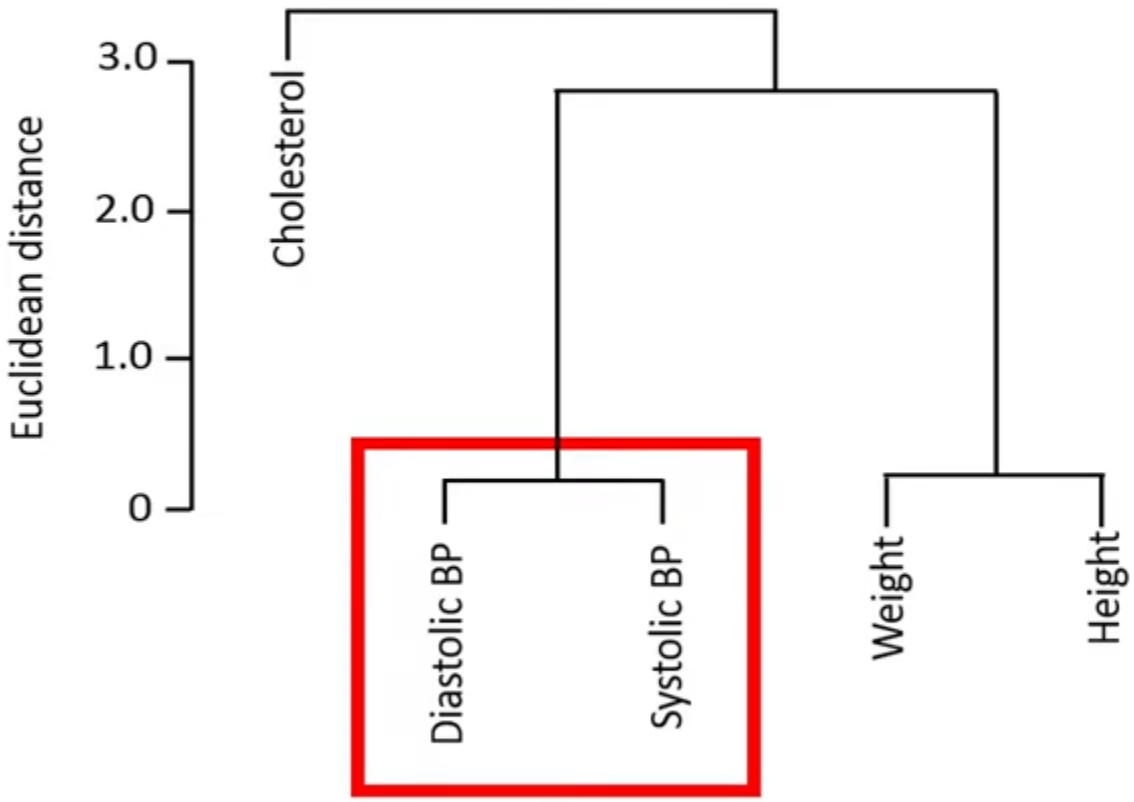
We can then generate the following dendrogram with the same method as we have seen previously.

Standardize the variables

Distance matrix:

	Diastolic BP	Systolic BP	Weight	Height
Systolic BP	0.21			
Weight	2.70	2.84		
Height	2.76	2.90	0.21	
Total cholesterol	2.89	2.77	3.78	3.82



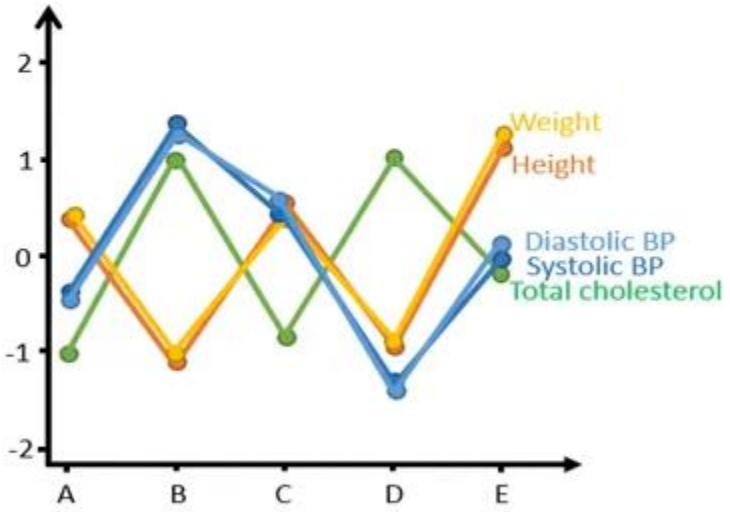


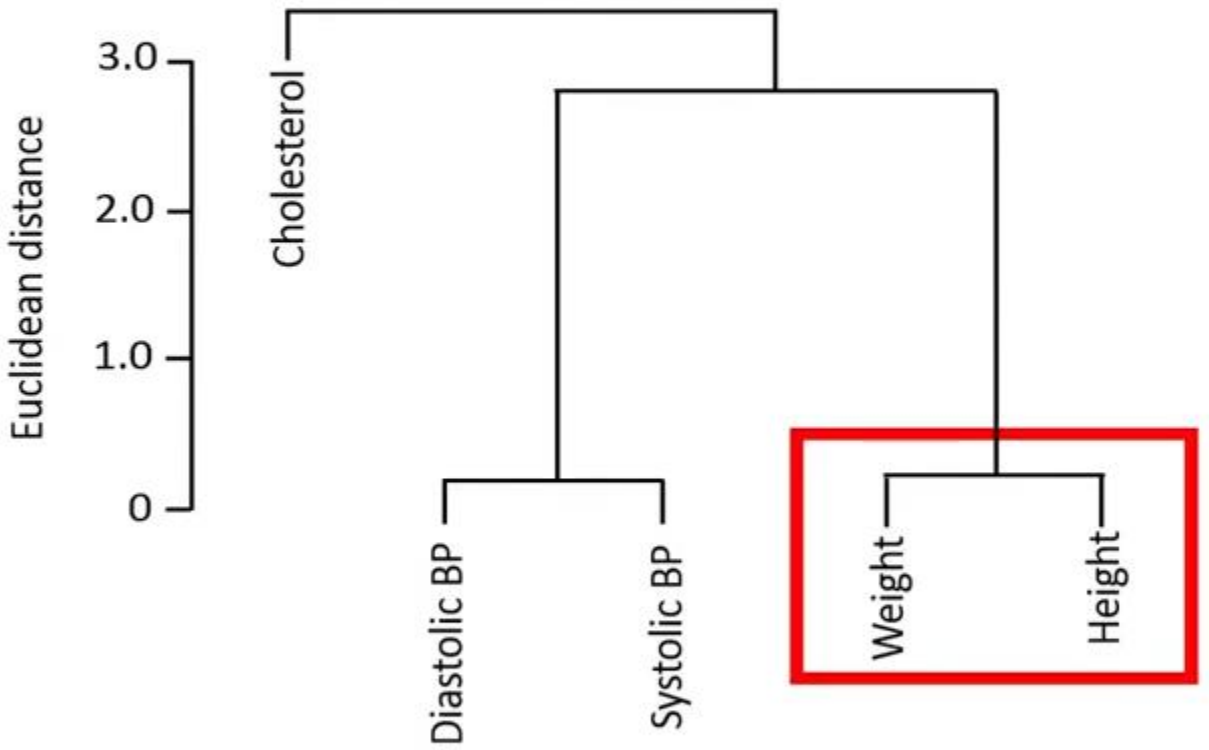
For example, the diastolic and systolic blood pressure variables now group together because they have a similar profile,

Standardize the variables

Distance matrix:

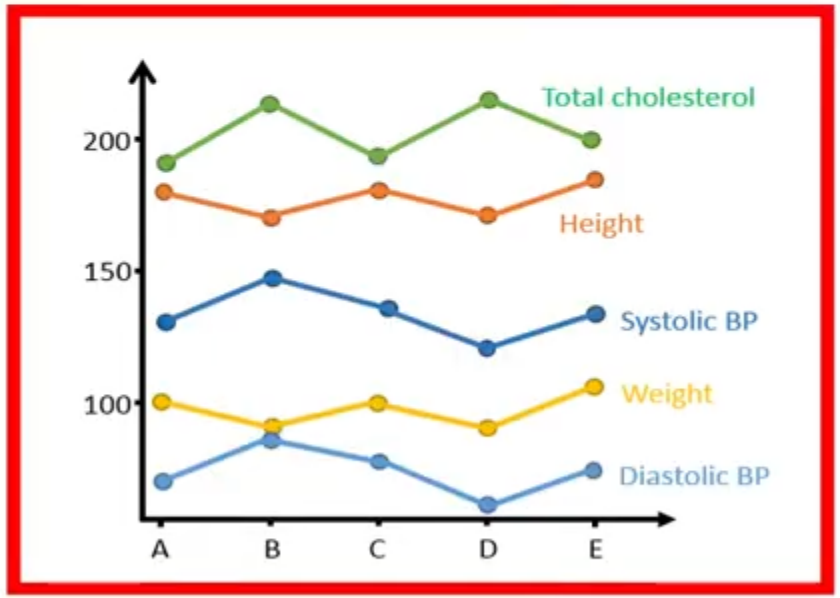
	Diastolic BP	Systolic BP	Weight	Height
Systolic BP	0.21			
Weight	2.70	2.84		
Height	2.76	2.90	0.21	
Total cholesterol	2.89	2.77	3.78	3.82





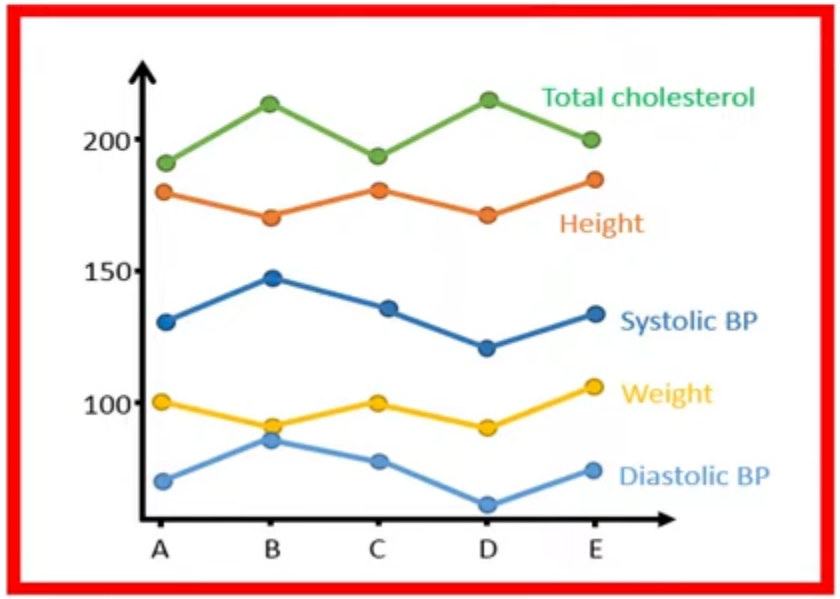
which is also true for the variables weight and height.

Correlation matrix



Another way to group variables with a similar pattern is to use a distance metric based on the correlation coefficient.

Correlation matrix

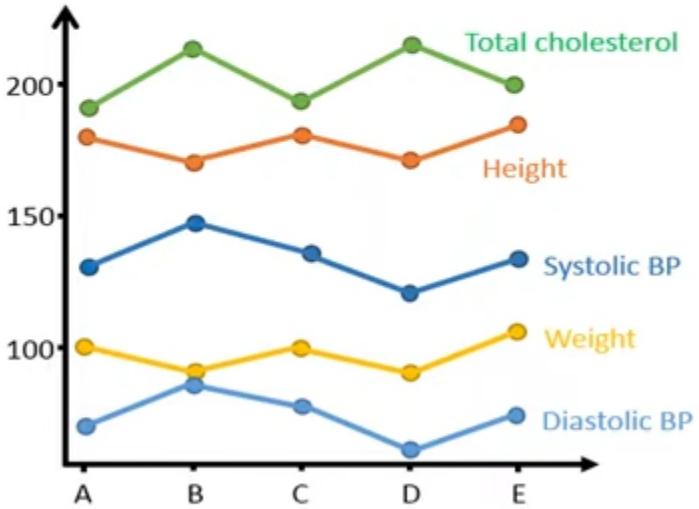


For example, we can calculate the Pearson correlation between each pair of variables,

Correlation matrix

Correlation matrix:

	Diastolic BP	Systolic BP	Weight	Height
Systolic BP	0.99			
Weight	0.09	-0.01		
Height	0.05	-0.05	0.99	
Total cholesterol	-0.04	0.04	-0.78	-0.82

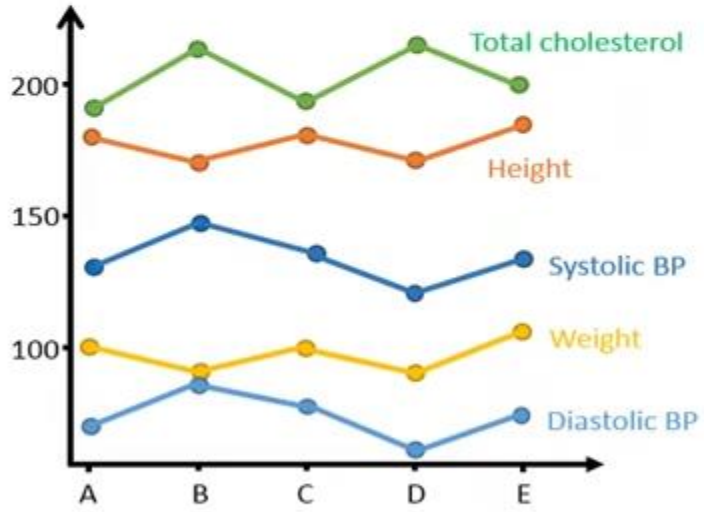


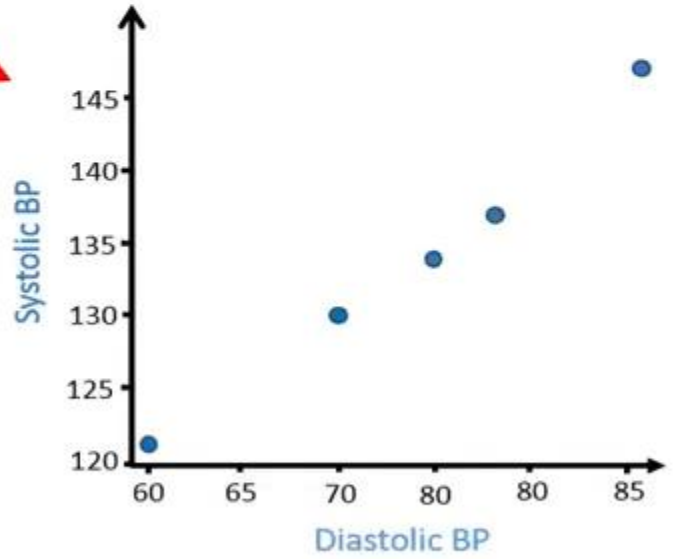
and generate a correlation matrix.

Correlation matrix

Correlation matrix:

	Diastolic BP	Systolic BP	Weight	Height
Systolic BP	0.99			
Weight	0.09	-0.01		
Height	0.05	-0.05	0.99	
Total cholesterol	-0.04	0.04	-0.78	-0.82





We see that the Pearson correlation coefficient between the two blood pressure variables is 0.99.

Correlation matrix

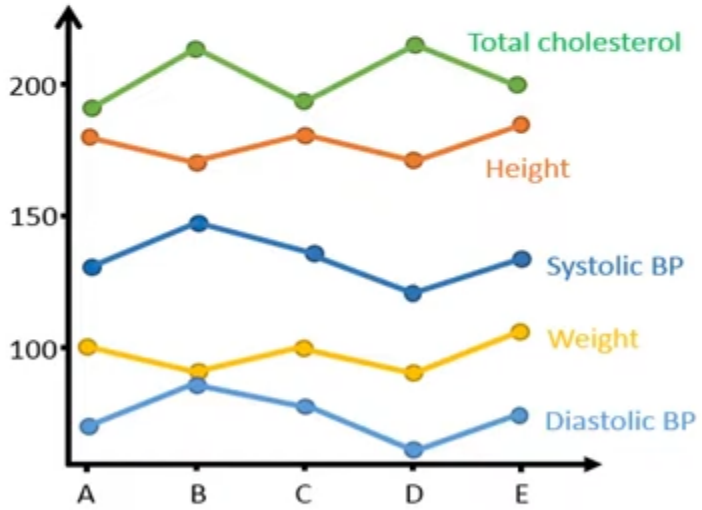
Correlation matrix:

	Diastolic BP	Systolic BP	Weight	Height
Systolic BP	0.99			
Weight	0.09	-0.01		
Height	0.05	-0.05	0.99	
Total cholesterol	-0.04	0.04	-0.78	-0.82

$$d = 1 - cor$$

Distance matrix:

	Diastolic BP	Systolic BP	Weight	Height
Systolic BP	0.01			
Weight	0.91	1.01		
Height	0.95	1.05	0.01	
Total cholesterol	1.04	0.96	1.78	1.82



If we, for example, take one minus the correlation coefficients we can create the following distance matrix.

Correlation matrix

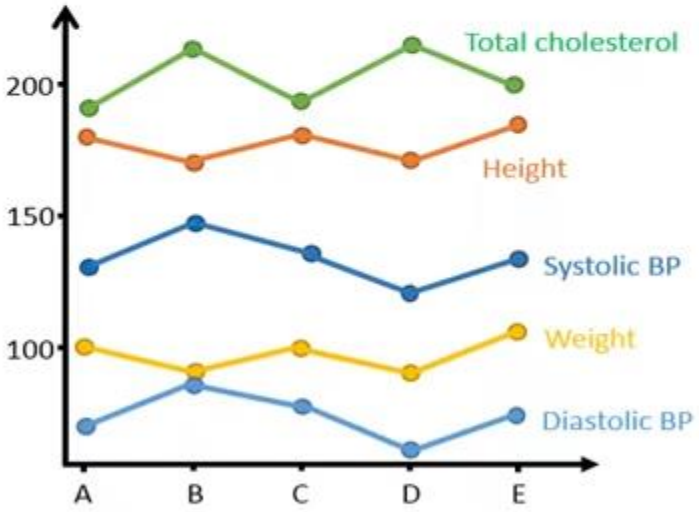
Correlation matrix:

	Diastolic BP	Systolic BP	Weight	Height
Systolic BP	0.99			
Weight	0.09	-0.01		
Height	0.05	-0.05	0.99	
Total cholesterol	-0.04	0.04	-0.78	-0.82

$$d = 1 - cor$$

Distance matrix:

	Diastolic BP	Systolic BP	Weight	Height
Systolic BP	0.01			
Weight	0.91	1.01		
Height	0.95	1.05	0.01	
Total cholesterol	1.04	0.96	1.78	1.82

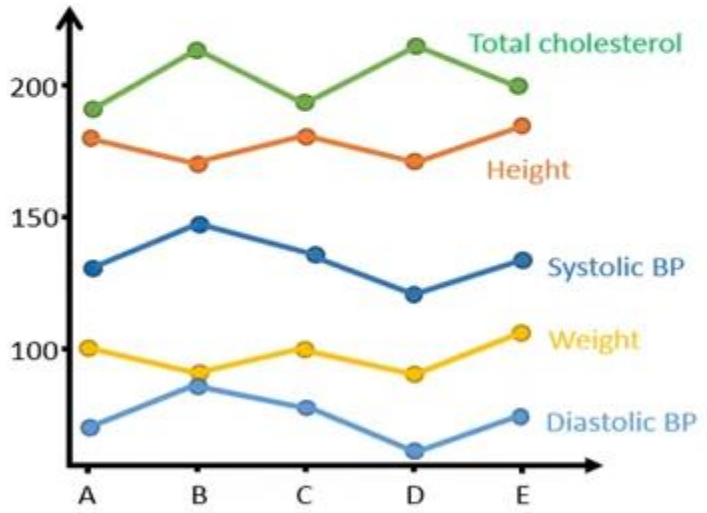


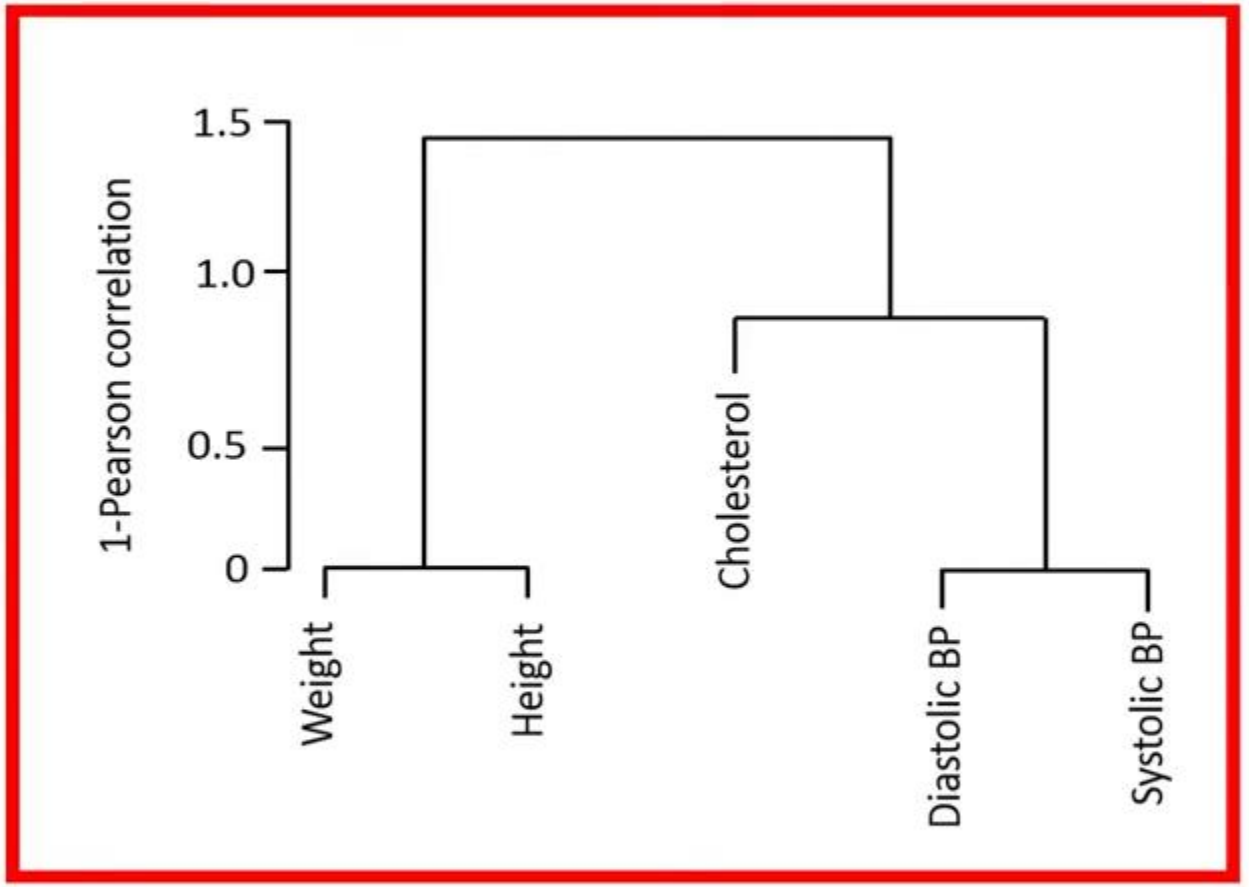
One minus 0.99 is 0.01, which tells us that there is a very short distance between the two blood pressure variables.

Correlation matrix

Distance matrix:

	Diastolic BP	Systolic BP	Weight	Height
Systolic BP	0.01			
Weight	0.91	1.01		
Height	0.95	1.05	0.01	
Total cholesterol	1.04	0.96	1.78	1.82



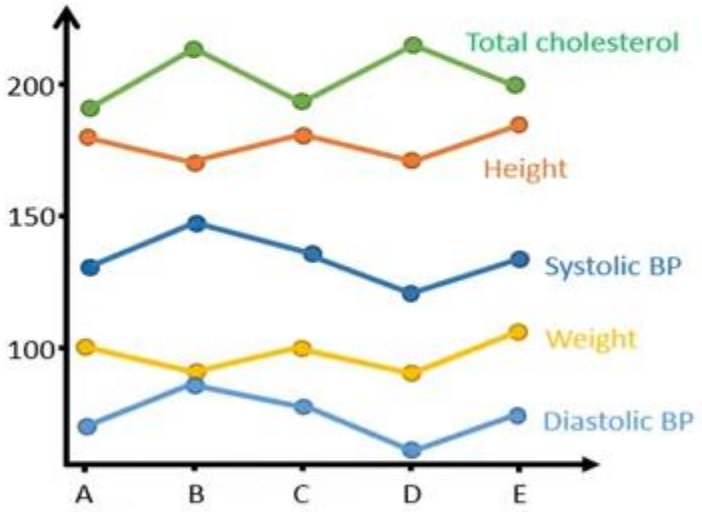


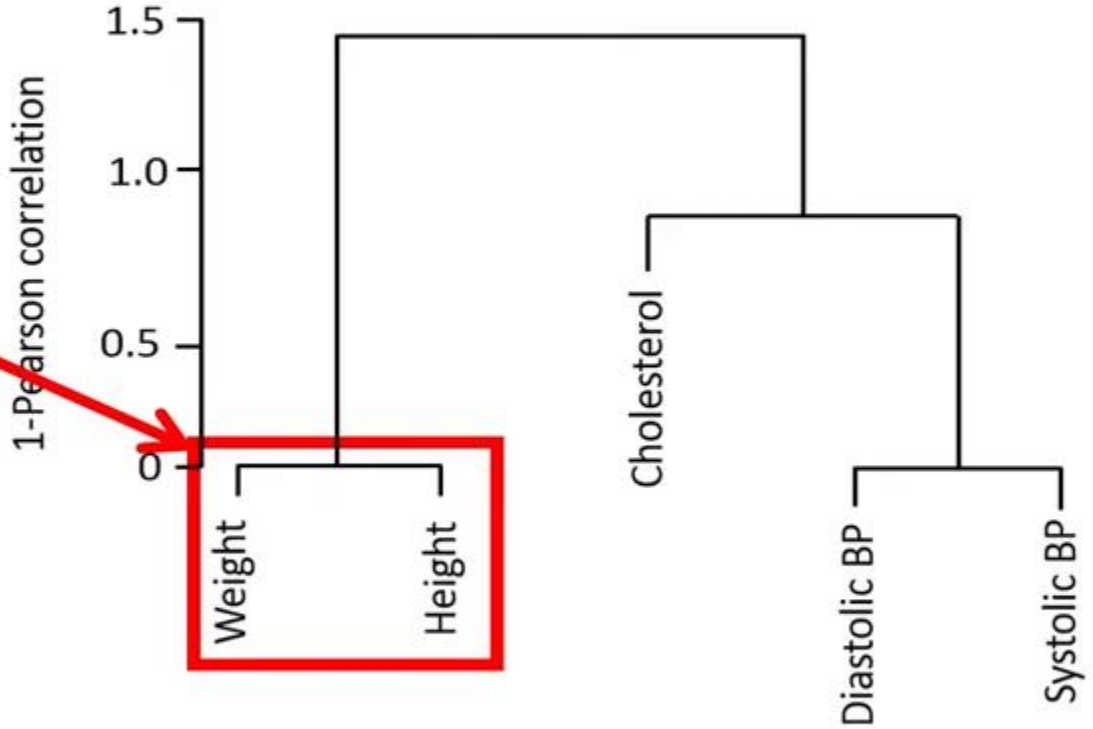
Based on the distance matrix, we can generate the following dendrogram, where the height of this dendrogram now represents one minus the Pearson correlation coefficient.

Correlation matrix

Distance matrix:

	Diastolic BP	Systolic BP	Weight	Height
Systolic BP	0.01			
Weight	0.91	1.01		
Height	0.95	1.05	0.01	
Total cholesterol	1.04	0.96	1.78	1.82



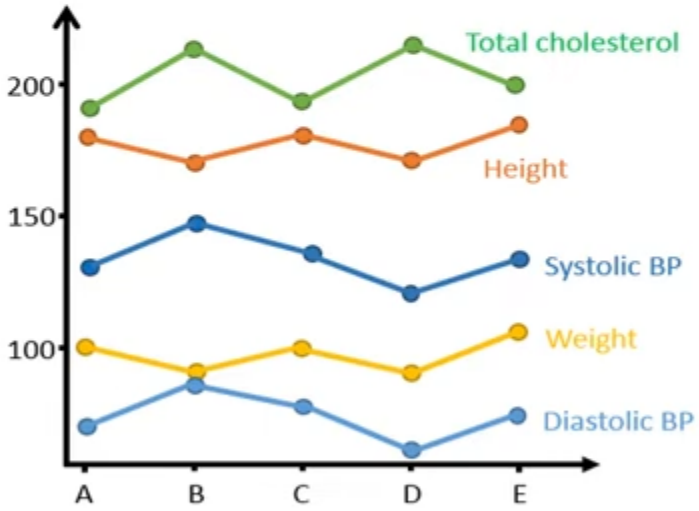


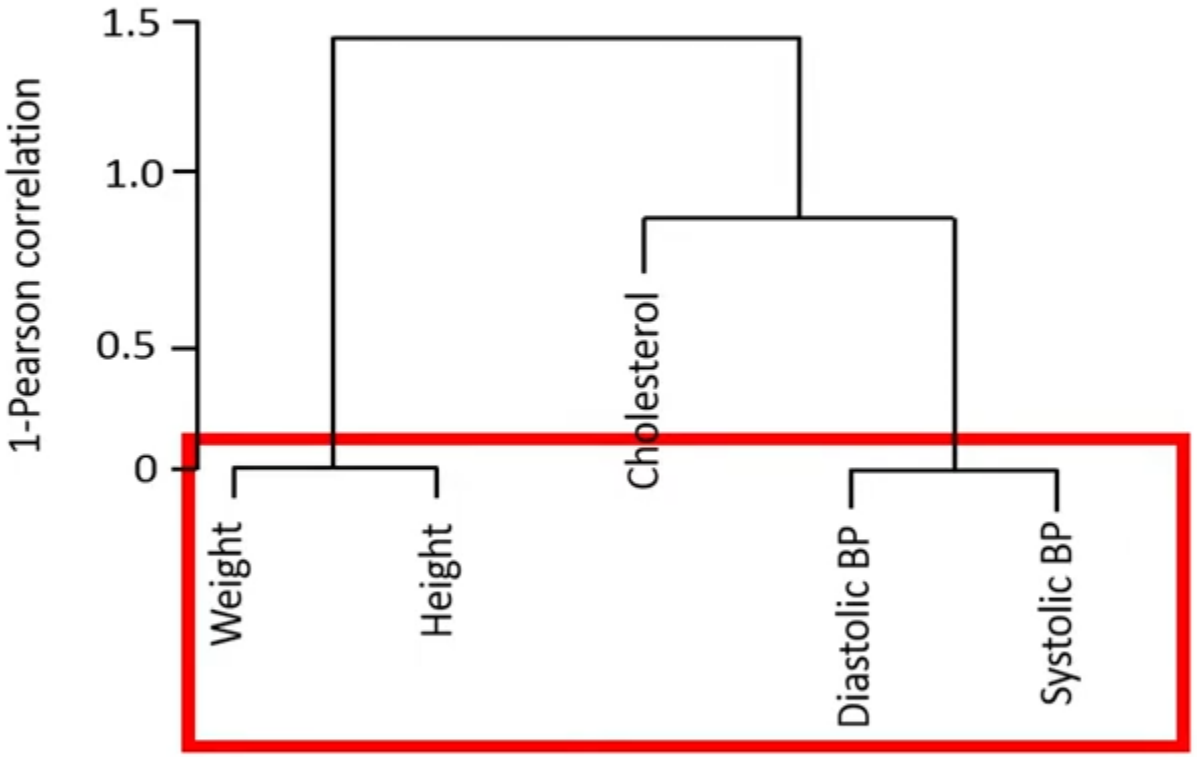
For example, since the correlation coefficient between the variables weight and height was equal to 0.99, the distance is here equal to 0.01.

Correlation matrix

Distance matrix:

	Diastolic BP	Systolic BP	Weight	Height
Systolic BP	0.01			
Weight	0.91	1.01		
Height	0.95	1.05	0.01	
Total cholesterol	1.04	0.96	1.78	1.82





Clusters at the bottom of this dendrogram therefore include variables that have a strong positive correlation to each other.

Transpose the data

	A	B	C	D	E
Diastolic BP (mmHg)	70	86	78	60	75
Systolic BP (mmHg)	130	147	137	121	134
Weight (kg)	100	90	100	90	106
Height (cm)	180	170	181	171	185
Total cholesterol (mg/dl)	190	215	192	215	200

Finally, we will see how we can cluster the five individuals instead of the five variables.

Transpose the data

	A	B	C	D	E
Diastolic BP (mmHg)	70	86	78	60	75
Systolic BP (mmHg)	130	147	137	121	134
Weight (kg)	100	90	100	90	106
Height (cm)	180	170	181	171	185
Total cholesterol (mg/dl)	190	215	192	215	200

If your statistical software clusters based on the rows in the matrix, or on columns, we can first transpose the matrix,

Transpose the data

	A	B	C	D	E
Diastolic BP (mmHg)	70	86	78	60	75
Systolic BP (mmHg)	130	147	137	121	134
Weight (kg)	100	90	100	90	106
Height (cm)	180	170	181	171	185
Total cholesterol (mg/dl)	190	215	192	215	200

	Diastolic BP	Systolic BP	Weight	Height	Total cholesterol
A	70	130	100	180	190
B	86	147	90	170	215
C	78	137	100	181	192
D	60	121	90	171	215
E	75	134	106	185	200

so that the variables now are on the columns instead.

Transpose the data

	A	B	C	D	E
Diastolic BP (mmHg)	70	86	78	60	75
Systolic BP (mmHg)	130	147	137	121	134
Weight (kg)	100	90	100	90	106
Height (cm)	180	170	181	171	185
Total cholesterol (mg/dl)	190	215	192	215	200

	Diastolic BP	Systolic BP	Weight	Height	Total cholesterol
A	70	130	100	180	190
B	86	147	90	170	215
C	78	137	100	181	192
D	60	121	90	171	215
E	75	134	106	185	200

Note that the first column of the transposed matrix,

Transpose the data

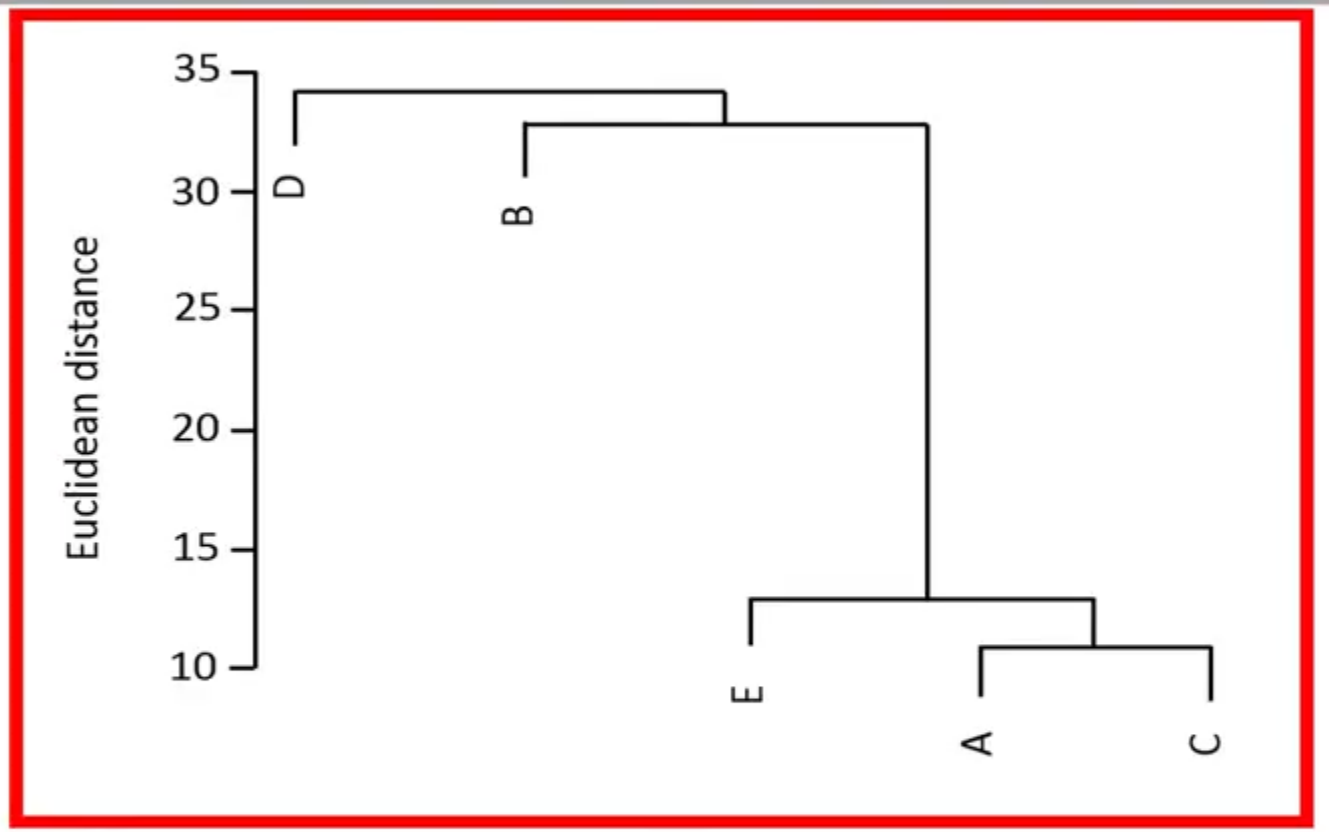
	A	B	C	D	E
Diastolic BP (mmHg)	70	86	78	60	75
Systolic BP (mmHg)	130	147	137	121	134
Weight (kg)	100	90	100	90	106
Height (cm)	180	170	181	171	185
Total cholesterol (mg/dl)	190	215	192	215	200

	Diastolic BP	Systolic BP	Weight	Height	Total cholesterol
A	70	130	100	180	190
B	86	147	90	170	215
C	78	137	100	181	192
D	60	121	90	171	215
E	75	134	106	185	200

is identical to the first row in the original matrix.

Transpose the data

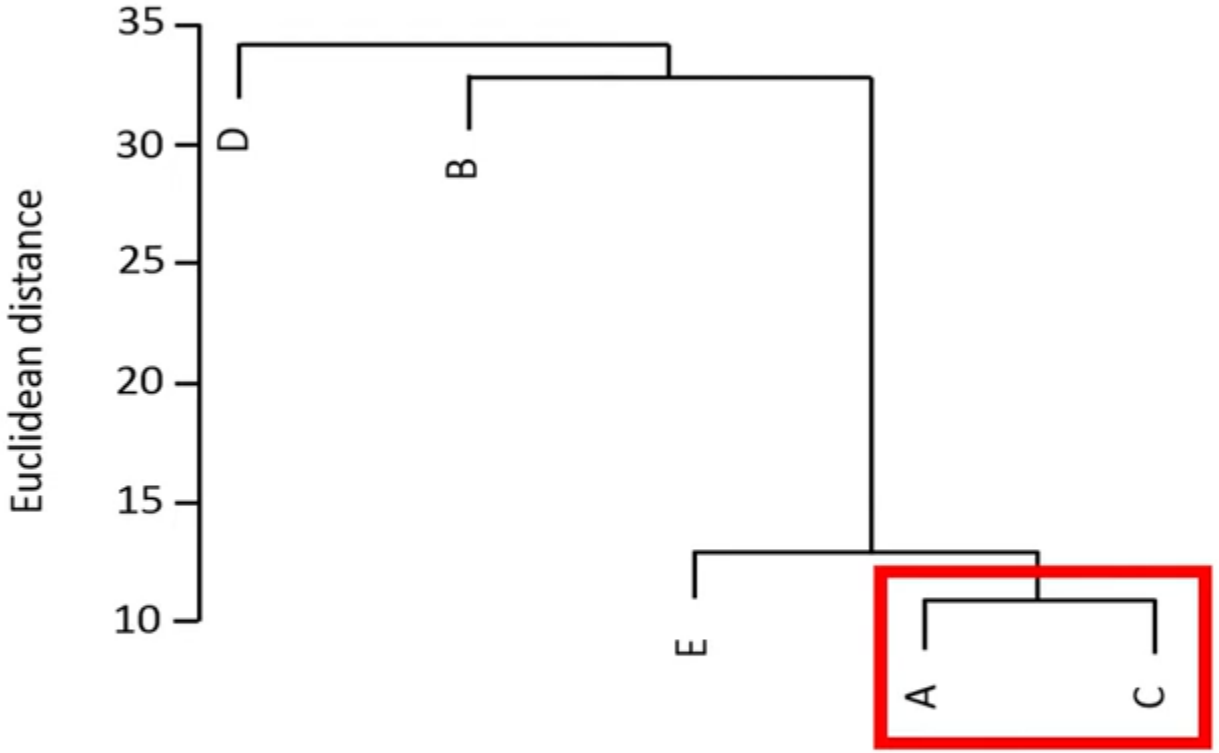
	Diastolic BP	Systolic BP	Weight	Height	Total cholesterol
A	70	130	100	180	190
B	86	147	90	170	215
C	78	137	100	181	192
D	60	121	90	171	215
E	75	134	106	185	200



If we make a dendrogram of the five individuals,

Transpose the data

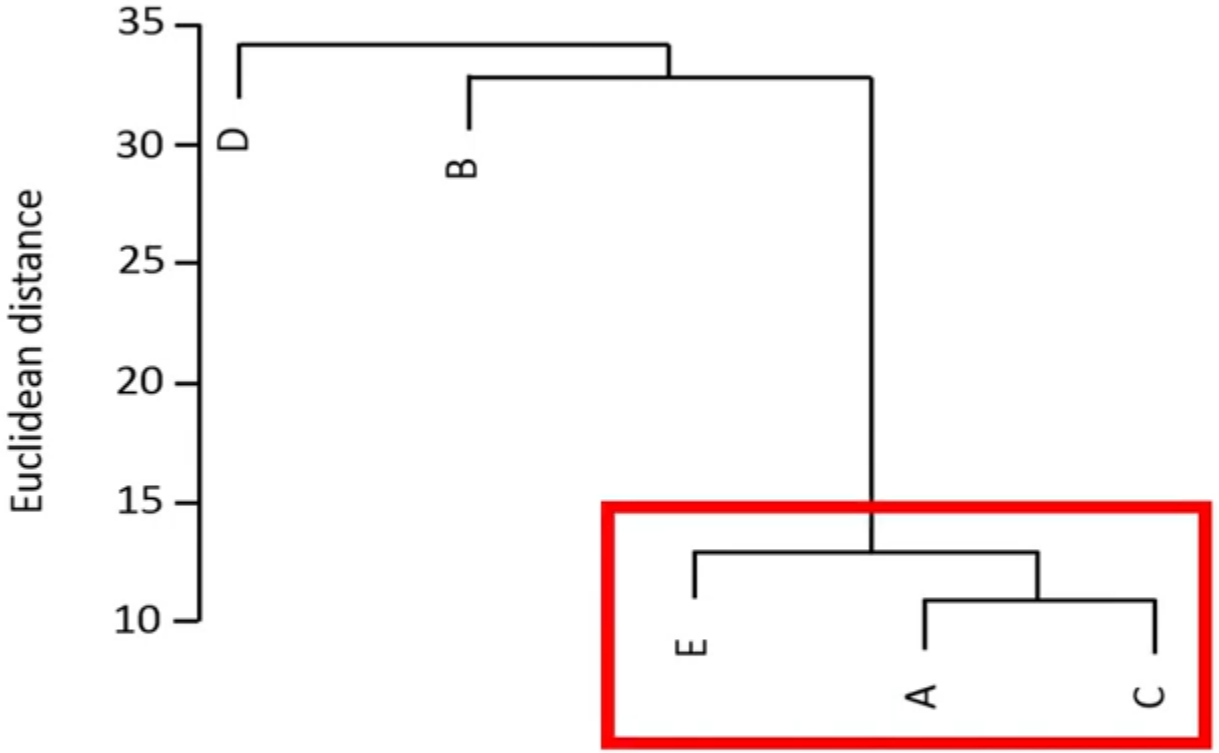
	Diastolic BP	Systolic BP	Weight	Height	Total cholesterol
A	70	130	100	180	190
B	86	147	90	170	215
C	78	137	100	181	192
D	60	121	90	171	215
E	75	134	106	185	200



we see that the individuals A and C have similar values of the measured variables,

Transpose the data

	Diastolic BP	Systolic BP	Weight	Height	Total cholesterol
A	70	130	100	180	190
B	86	147	90	170	215
C	78	137	100	181	192
D	60	121	90	171	215
E	75	134	106	185	200



and that individual E has values that are similar to person A and C.